

Cellular Internet of Things: Optimisation of Random Access and Relaying Communications

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Publication Date:
2020

DOI:
https://doi.org/10.26190/unsworks/21700

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Cellular Internet of Things: Optimisation of Random Access and Relaying Communications

by

Ziqi Chen

A THESIS IN FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY

SCHOOL OF ELECTRICAL ENGINEERING AND
TELECOMMUNICATIONS
FACULTY OF ENGINEERING
THE UNIVERSITY OF
NEW SOUTH WALES



SYDNEY · AUSTRALIA

February 20, 2020

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Faculty	:	**Engineering**
School	:	**Electrical Engineering and Telecommunications**
Thesis Title	:	**Cellular Internet of Things: Optimisation of Random Access and Relaying Communications**

Abstract

The evolution of wireless communications, sensor technology and processing capability has recently created the field of cellular internet-of-things (CIoT). From smart metering to vehicle-to-everything (V2X), CIoT is a technology vital to providing efficient and reliable communications. However, the massive number of devices connecting to the network and their heterogeneous quality-of-service (QoS) requirements pose big challenges in the deployment of future CIoT. In this context, three important problems for CIoT, addressing the optimisation of random access and relaying for both downlink and uplink communications, are solved in this thesis.

We first solve the optimisation of the random access channel (RACH) for machine-type communication devices (MTCDs) in LTE. We construct a framework to analyse random access efficiency in RACH, and then design an access class barring and back-off control algorithm, employing deep reinforcement learning, which is deployed at base stations (BSs). In this methodology, a BS broadcasts control parameters to MTCDs to achieve maximum random access efficiency, under burst activation of devices, while meeting heterogeneous QoS requirements.

We next solve the problem of minimising delay for relaying across the CIoT uplink. In the proposed scheme each MTCD s end-to-end delay is minimised, and overall network uplink efficiency is improved, for millimetre wave (mmWave) networks. We analytically derive the expected end-to-end delay and outage performance of the scheme. Furthermore, we compare 2-hop and 3-hop relaying, demonstrating the scheme's benefits. Theoretical analysis is validated by simulations according to a real-world urban map, with improved average delay and outage probability demonstrated across various network configurations.

The third problem solved is throughput and fairness oriented optimisation for relaying in CIoT. For a full-duplex system, we present a distributed multi-hop relaying and resource allocation scheme by user association for uplink data aggregation. We demonstrate that the data rate of each MTCD and sum data rate across MTCDs are optimised, while maintaining fairness and load balancing across MTCDs. Following from this, for mmWave downlink scheduling, we design an iterative scheduling utilising links between access points, ensuring all UEs are served. Therefore, fairness among MTCDs is significantly improved, while maintaining good spectral efficiency for CIoT downlink scheduling.

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Abstract

The evolution of wireless communications, sensor technology and processing capability has recently created the field of cellular internet-of-things (CIoT). From smart metering to vehicle-to-everything (V2X), CIoT is a technology vital to providing efficient and reliable communications. However, the massive number of devices connecting to the network and their heterogeneous quality-of-service (QoS) requirements pose big challenges in the deployment of future CIoT. In this context, three important problems for CIoT, addressing the optimisation of random access and relaying for both downlink and uplink communications, are solved in this thesis.

We first solve the optimisation of the random access channel (RACH) for machine-type communication devices (MTCDs) in LTE. We construct a framework to analyse random access efficiency in RACH, and then design an access class barring and back-off control algorithm, employing deep reinforcement learning, which is deployed at base stations (BSs). In this methodology, a BS broadcasts control parameters to MTCDs to achieve maximum random access efficiency, under burst activation of devices, while meeting heterogeneous QoS requirements.

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works. We analytically derive the expected end-to-end delay and outage performance of the scheme. Furthermore, we compare 2-hop and 3-hop relaying, demonstrating the scheme's benefits. Theoretical analysis is validated by simulations according to a real-world urban map, with improved average delay and outage probability demonstrated across various network configurations.

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Acknowledgments

First and foremost, my gratitude must go to my two Ph.D. advisors, Dr. David Smith and Professor Aruna Seneviratne, for their constant guidance throughout my journey to Ph.D. I am greatly indebted to Dr. David. He sacrificed hundreds of hours in working with me and taught me how to identify problems, find solutions and present the work. He is a very kind and patient mentor, whose high standards and farsighted instructions directed me all the way to become a genuine researcher. His expertise has greatly helped me overcome obstacles in my research.I also owe my sincere gratitude to Professor Aruna, for his continuous support and encouragement throughout the candidature. His in-depth opinion and valuable feedback have made an immeasurable contribution to my research.

I am thankful for being able to work closely with Dr. Samaneh Movassaghi at the early stage of my study. I would like to give appreciation to Dr. Guohao Lan and Dong Ma for sharing their experience in related fields. I would like to give appreciation to the review panel members of my thesis, Prf. Vijay Sivaraman, Dr. Tim Moors, Dr. Wing Kwan Ng and Dr. Hassan Habibi Gharakheili, for their invaluable comments and feedback to assure my works good progress.

My gratitude must go to my family for their unlimited support and sacrifice. I would like to thank my parents for providing me the support throughout the journey. Their love is most precious in my life. I am more than lucky to have my

wife Jie Tao along the way, without whom I could not finish the candidature. She inspired and encouraged me in my hardest times. Her endless support and love is most valuable in my life.

Last but not least, I would like to acknowledge the support from UNSW and Data61 CSIRO.

Ziqi Chen

Sydney, Australia

September 2019

To *my dearest wife, my parents for their continuous love.*

Scientia Potestas Est

- Francis Bacon

List of Publications

Journal Articles

1. **Ziqi Chen**, David B. Smith and Aruna Seneviratne "*Heterogeneous Massive M2M Communications: Random Access Optimization by Deep Reinforcement Learning*," Submitted in September 2019 to IEEE Transactions on Communications.

2. **Ziqi Chen** and David B. Smith, "*How can multi-hop relaying M2M networks be designed for low latency mmWave communications?*", Submitted in September 2019 to IEEE Transactions on Wireless Communications.

3. **Ziqi Chen** and David B. Smith, "*Spectrally-Efficient scheduling in multi-tier mmWave Networks*", In Preparation for Submission to IEEE Transactions on Vehicular Technology.

Conference Proceedings

1. **Ziqi Chen** and David B. Smith, "*Socially optimal distributed user association for multi-hop machine-to-machine communications*," In Proceedings of IEEE International Conference on Communications (ICC), 2018.

2. **Ziqi Chen** and David B. Smith, "*Heterogeneous machine-type communications in cellular networks: Random access optimization by deep reinforcement learning*," In Proceedings of IEEE International Conference on Communications (ICC), 2018.

3. **Ziqi Chen** and David B. Smith, "*How Multi-Hop Relaying in mmWave Communications Improves Uplink Network Latency*," to Appear In Proceedings of IEEE Vehicular Technology Conference (VTC-Fall), 2019.

4. **Ziqi Chen** and David B. Smith, "*Multi-tier mmWave Networks: Improving Fairness for Spectrally-Efficient Downlink Scheduling*," to Appear In Proceedings of IEEE Vehicular Technology Conference (VTC-Fall), 2019.

Poster Presentation

1. **Ziqi Chen** and David B. Smith, "Heterogeneous Random Access Optimization by Deep Reinforcement Learning," In Australian Communications Theory Workshop (AusCTW), 2019.

List of Publications Contributing to the Thesis

1. **Ziqi Chen**, David B. Smith and Aruna Seneviratne "*Heterogeneous Massive M2M Communications: Random Access Optimization by Deep Reinforcement Learning*," Submitted in September 2019 to IEEE Transactions on Communications.

   This work incorporated as the basis for Chapter 3. The Candidate's contribution to the work:

   - System Design and Implementation (90%)
   - Evaluation and Results Analysis (90%)
   - Paper Writing (90%)

2. **Ziqi Chen** and David B. Smith, "*How can multi-hop relaying M2M networks be designed for low latency mmWave communications?*", Submitted in September 2019 to IEEE Transactions on Wireless Communications.

   This work incorporated as the basis for Chapter 4. The Candidate's contribution to the work:

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   - Evaluation and Results Analysis (90%)

- Paper Writing (90%)

3. **Ziqi Chen** and David B. Smith, "*Socially optimal distributed user association for multi-hop machine-to-machine communications*," In Proceedings of IEEE International Conference on Communications (ICC), 2018.

   This work incorporated as the basis for Chapter 5.1. The Candidate's contribution to the work:

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4. **Ziqi Chen** and David B. Smith, "*Multi-tier mmWave Networks: Improving Fairness for Spectrally-Efficient Downlink Scheduling*," to Appear In Proceedings of IEEE Vehicular Technology Conference (VTC-Fall), 2019.

   This work incorporated as the basis for Chapter 5.2. The Candidate's contribution to the work:

   - System Design and Implementation (90%)
   - Evaluation and Results Analysis (90%)
   - Paper Writing (90%)

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List of Acronyms

AP	Access Point.
BS	Base Station.
CDF	Cumulative Distribution Function.
CIoT	Cellular Internet of Things.
D2D	Device-to-Device.
DRL	Deep Reinforcement Learning.
EED	End-to-End Delay.
eMBB	Enhanced Mobile Broadband.
eNB	Evolved Node B.
H2H	Human-to-Human.
HetNet	Heterogeneous Network.
IIoT	Industrial Internet of Things.
IoT	Internet of Things.
ITS	Intellegent Transportation System.
LOS	Line-of-Sight.
LPWAN	Low Power Wide Area Network.
M2M	Machine-to-Machine.
MIMO	Multiple-in Multiple-out.
mMTC	Massive Machine Type Communication.
mmWave	Millimeter-wave.
MTCD	Machine Type Communication Device.
NE	Nash Equilibrium.
NFC	Near Field Communication.
NLOS	Non-Line-of-Sight.
NR	New Radio.
PDF	Probability Density Function.
PPP	Poisson Point Processes.
QoS	Quality of Service.
RA	Random Access.

RACH	Random Access Channel.
RAN	Radio Access Network.
RFID	Radio-Frequency Identification.
SINR	Signal-to-Interference-and-Noise-Ratio.
UE	User Equipment.
UL	Uplink.
URLLC	Ultra-Reliable Low-Latency-Communication.
V2X	Vehicle-to-Everything.
WANET	Wireless ad hoc Networks.
WBAN	Wireless Body Area Network.
WLAN	Wireless Local Area Network.
WPAN	Wireless Personal Area Network.
WSN	Wireless Sensor Network.

Chapter 1

Introduction

This thesis addresses the problem of random access channel (RACH) optimisation, and multi-hop relaying optimisation, in cellular IoT.

1.1 Background

The term Internet of Things (IoT) first appeared in 1999, and was used in the management of supply chains [5]. Over the last few decades, the definition has become more inclusive, expanding the meaning of 'things' with respect to IoT. In 2015, [6] defined IoT as a global network of machines and devices capable of interacting with each other. *The core concept of IoT is to equip every possible thing with identification, sensing, networking and processing capabilities that allows them to communicate with each other.* There are many IoT applications: labels equipped with near field communication (NFC) capabilities is a typical application of radio-frequency identification (RFID) [7]. Equipping sensors or utility meters with transceivers allows for wireless smart monitoring [8]. Cars sharing information with any or all relevant entities are called vehicle-to-everything (V2X) [9]. From best effort applications such as smart homes and environment monitoring to ultra-

reliable ones such as mission-critical industry IoT and E-healthcare, IoT has a wide and continue growing range of applications, with a key application domain in cellular networks, cellular IoT (CIoT). Overall, both the needs from many unique applications and advances in telecommunication technologies, together enable the fast growth and adoption of IoT.

One of the first IoT technologies is radio-frequency identification (RFID), which enables the automatic identification of anything. Passive RFID tags are not battery powered and they use the power of the reader's interrogation signal to communicate the ID to the reader [7]. This allows for many applications found in, e.g., pharmaceutical production, retailing, supply chain management, bank cards and road toll tags, since the 1980s [10]. Proceeding this, advances in wireless communications and digital electronics have resulted in the development of miniature devices with the ability to sense, process and communicate across short distances. Such interconnected intelligent sensors form a wireless sensor network (WSN). WSNs have had a wide range of applications in environmental monitoring, industrial monitoring, traffic monitoring and so on [8, 11].

Over recent years, thanks to advances in wireless communications, smartphone and sensor technologies, more and more IoT applications have emerged, posing more challenges in their implementation to both academia and industry. The industrial internet of things (IIoT), for example, IoT's application in manufacturing and energy management, allows for a high degree of automation to refine and optimise process controls. It is enabled by cyber-physical systems, edge computing, machine learning and ultra-reliable networking. In the personal sensing domain, a typical example of IoT applications is mobile healthcare [12]. Developments in wearable sensors, micro-electromechanical systems, and wireless body area networks (WBAN) have been integrated to collect health related data, thus

becoming a significant aid in combating growing concerns about healthcare [13]. Vehicle-to-everything (V2X) represents the future of transportation, enabling the realisation of self-driving cars and smart traffic management. The fundamental technology related to V2X is ultra-reliable low-latency-communication (URLLC), which is envisioned as one of three service categories in 5G for cellular networks.

The development of 5G is closely related to the development of IoT, and IoT is considered one essential component of 5G. 5G is the fifth generation cellular network technology, being standardized by the industry association 3GPP [14]. The architecture of 5G networks supports three service categories, which are: enhanced mobile broadband (eMBB); massive machine type communication (mMTC); and URLLC [15]. In them, two service categories, mMTC and URLLC, are specifically designed for future IoT applications. Category mMTC features the supporting of high density of devices (one million connected devices per km^2), long range, low data rate, ultra low cost and long battery life [16]. These are essentially the fundamental requirements of narrowband internet access for sensing, metering and monitoring devices. URLLC features ultra responsive connections with less than 1 ms air interface latency or 5 ms end to end latency between user equipment (UE) and 5G Evolved Node B (eNB), ultra reliable with available 99.9999% of the time, low to medium data rates and high speed mobility [15]. This category serves latency sensitive devices for applications like factory automation, autonomous driving and remote surgery.

To deliver required services, many new technologies have been proposed for cellular networks. Making the cells smaller is a straightforward and extremely effective way to increase network capacity. Cell shrinking has numerous benefits, the most important of which being the reuse of spectrum across a geographic area and the reduction in the number of users in each cell [17]. The effectiveness of this

method has been proved again and again over the generations of cellular networks. More recently, due to global bandwidth shortage, millimeter-wave (mmWave) frequency spectrum has been explored for the 5G new radio (NR) air interface [18]. This new, previously untouched, frequency spectrum provides large chunks of bandwidths, however, it suffers from weak diffraction and high penetration loss through both air and obstacles [19]. To overcome this problem, beamforming has been proposed as a solution, as it is an effective way to concentrate signal power to a directional angle by manipulating phase and relative amplitude of the signals, sent from, or impinging at, an antenna array. This method significantly boosts received power when transmitters and receivers are aligned, as well as eliminating undesired interference from other transmitters.

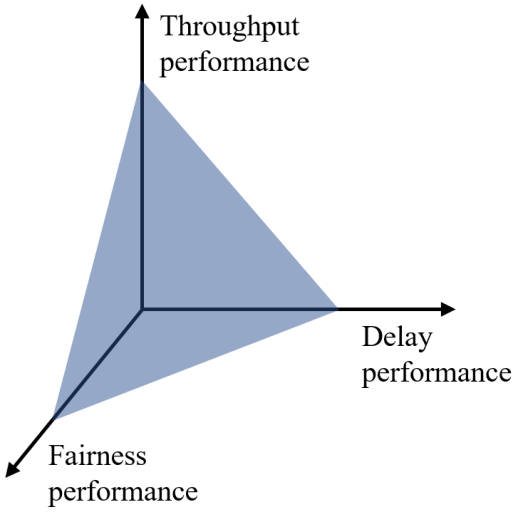
1.2 Motivation

Even though academia and industry have made huge efforts and advances in research into cellular networks to meet future IoT requirements, there is still a fundamental conflict between the massive connections from IoT devices and the constrained resources of IoT [20]. For example, an external event may trigger large amounts of machine-type-communication devices (MTCDs) to become active from an idle state and access the cellular network at once, or a massive number of scheduled MTCDs may periodically request network access to report data at around the same time. Because each activated devices typically carries only a small amount of data, simultaneous transmission attempts to access the eNB require large-scale synchronisation and will result in data traffic flow congestion in the radio access network [21, 22]. Unfortunately, the explosive growth of the cellular IoT has not been well matched by the development of cellular networks themselves.

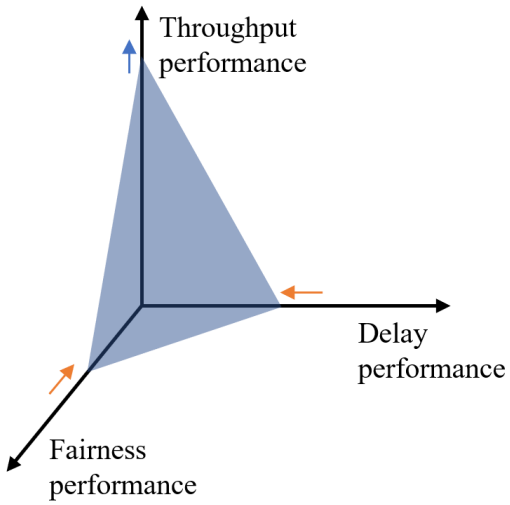
In recent years, relaying in networks has become a common solution to tackle the massiveness problem and non line-of-sight problem brought by the use of mmWave, as the ideas of device-to-device (D2D) communications [23] and wireless backhaul networks [24] mature. The basic idea behind multi-hop relaying in mmWave communications is to allow intermediate relay nodes to aggregate traffic generated in transmitters and forward it to destinations receivers, thus alleviating the access pressure on eNBs [25]. Since the additional routes provided by relay nodes can detour around obstacles, the previously blocked transmission links can be replaced by two-hop line-of-sight (LOS) transmission links. Such improved channel quality will increase mmWave coverage and shorten transmission delay. However, the design of relaying network faces 3 major challenges: throughput, delay and fairness. There exist intrinsic trade-offs between these 3 goals, like pushing down one corner of a floating triangle, the other two corners will inevitably rise up. When we pursue high throughput for the system, the queueing delay at each relay is increased due to higher data arrival rate. On the other hand, when we pursue lower delays, the system can not hold too large volume of data traffic. Moreover, by the pursuing of both goals of delay and throughput, fairness among connected devices is impaired as poor channels are given few, even none resources. If fairness is treated as a first priority, resource allocation and relaying routes should all be designed sub-optimally to complement for devices with poor channels. The relationship between the above mentioned 3 goals is illustrated in Fig. (1.1).

Thus, in this thesis we seek to study these issues and propose solutions to solve the congestion problem in radio access networks and to optimise relaying networks in terms of the tradeoffs of throughput, delay and fairness.

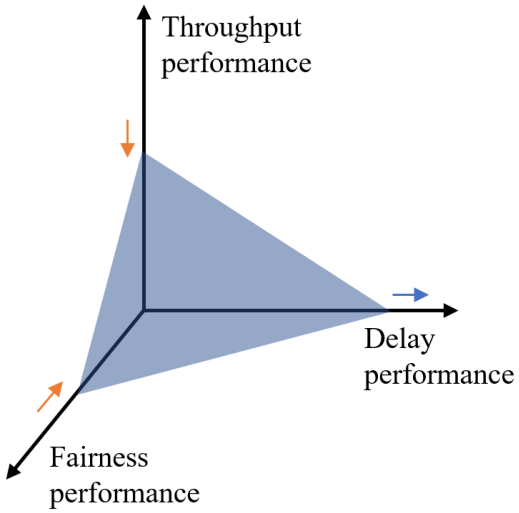
An important key additional motivation, is that although both academia and industry have realised the potential of Cellular IoT and sought to solve related



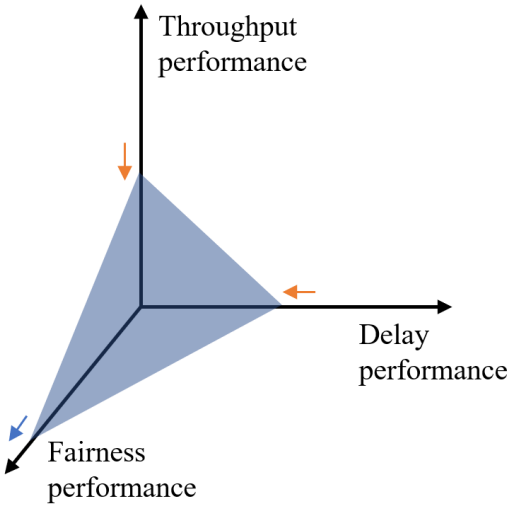
(a) relationship between 3 performances



(b) pursuing throughput



(c) pursuing delay



(d) pursuing fairness

Figure 1.1: Three major challenges in the design of relaying network

problems, one challenge is that the use case of mMTC devices is so vast, that they feature markedly heterogeneous quality of service (QoS) requirements. This is true even if we focus on a single vertical scenario, industrial IoT, where different MTC applications have completely different throughput, delay, fairness and reliability requirements.

These motivations lead to the statement of objectives in the following section.

1.3 Research Objective

Following from our motivation, *the main of objective of this thesis* is to answer: ***How to provide an optimal solution— for the most suitable tradeoff of throughput, delay and fairness in radio access and relaying networks — that complies with heterogeneous QoS requirements?***.

To best address this objective, we decompose it into three individual subtasks with their own specific questions to address:

Task 1: To optimise the RACH for MTC devices in cellular networks: As the RACH is the first step of UEs who want to establish a connection with the eNB and upload data, it is essential that the RACH operates smoothly and efficiently without congestion. Therefore, the optimisation of the RACH for mMTC is of utter importance to meet dense connection requirements, leading to the first two questions to answer:

-Question 1: Is there a way to regulate access requests to the RACH, and how could the regulation be optimised to achieve maximum efficiency?

-Question 2: When facing heterogeneous QoS requirements, how can we regulate different classes of UEs to meet their respective requirements?

Task 2: To reduce latency in multi-hop relaying mmWave networks: even though relaying in mmWave cellular networks alleviates the non-line-of-sight (NLOS) problem and improves coverage range, the latency incurred by each hop could dramatically degrade its QoS performance, especially for delay sensitive applications. Therefore, the next questions to be answered are:

-Question 3: How can we design a relaying network that aims at decreasing latency for the whole system?

-Question 4: How will the hop-number in relaying networks affects delay performance?

Task 3: To Increase throughput and fairness in relaying cellular networks: a relaying network is an efficient solution for data aggregation and data distribution, but traditional relaying algorithms are channel quality oriented. Due to the asymmetric location distribution of UEs, there is likely to be congestion in certain nodes at the uplink, and very low data rates to remote nodes at the downlink, resulting in downgraded system throughput and fairness among connected UEs. Hence, the goal is to design relaying algorithms that optimise overall throughput without compromising fairness for uplink and downlink relaying cellular networks, which leads to the final two questions to answer:

-Question 5: With the aim of optimising both overall system throughput and fairness, how can we design an uplink relaying network that allow UEs to autonomously select relays to connect to?

-Question 6: What is the optimal solution for spectrally-efficient downlink scheduling in a multi-tier relaying network?

In the next section, we will summarize the contributions made by this thesis that directly answer these six questions.

1.4 Research Contributions

The key contributions, answering the questions, can be divided into four major aspects:

1. The design of a deep reinforcement learning (DRL) based access class barring (ACB) and back-off (BO) controlling algorithm that optimises RACH efficiency in a LTE-M cellular network.

2. The design of Latency oriented relaying scheme in mmWave uplink network that is validated through stochastic geometry and queueing theory.

3. The design of socially optimal distributed user association scheme for multi-hop machine-to-machine (M2M) communications which employs game theory.

4. The design of fair spectrally-efficient downlink scheduling scheme based on Hungarian algorithm in a weighted bipartite graph.

Thus, in our contributions we employ a variety of mathematical methods and optimisation techniques. These contributions are described in more detail in the

following subsections.

1.4.1 Deep Reinforcement Learning based Optimisation of Random Access (Chapter 3)

Accommodating a massive number of connected devices in cellular networks is one of the major challenges in cellular IoT. Due to the nature of M2M communications, sporadic uplink requests of a massive number of machine-type communication devices (MTCDs) lead to random access channel (RACH) congestion [26]. Thus we seek to use access class barring and dynamic back-off to regulate the number of simultaneous access requests in a RACH. Furthermore, the scheme is enhanced to provide services to devices with heterogeneous delay constraints.

Hence, in chapter 3, we construct a framework to analyse random access (RA) efficiency for M2M communications in LTE-M with regards to both an ACB scheme and a back-off scheme using a Markov chain, and we explore recent advances in machine learning and its subset, DRL, for usage in optimisation. A DRL agent is then deployed in the eNB to control the ACB factor and back-off window, which are broadcast to MTCDs, to achieve maximum RA efficiency under burst activation of devices. For the case of MTCDs having different QoS requirements, distinct ACB factors and BO windows are applied to devices with different priority classes, each controlled by the DRL agent simultaneously. The key advantage of the proposed algorithm is that the agent can predict the overload situation based on information collected in the process of deploying the random access procedure, then find the best parameter combination for the situation. Results show superior convergence speed and actions taken compared with a traditional RL algorithm, as well as much better success probability and average delay when compared with other state-of-art schemes.

1.4.2 Latency oriented relaying scheme for mmWave uplink (Chapter 4)

For future wireless networks, applications such as industrial IoT are strictly delay-sensitive [27]. Meanwhile, mmWave communications is envisioned as a promising means to provide ultra-high data rate and ultra-low latency services to a massive number of devices [19]. Thus we seek to optimise the delay performance of mmWave uplink data aggregation with relaying in large scale M2M networks.

Hence, in chapter 4, we propose an uplink relaying scheme called "Minimum-Delay" relaying, i.e., a delay-oriented relaying scheme which minimizes each UE's end-to-end delay and improves the overall network uplink efficiency. By applying stochastic geometry and Lagrange optimisation, we derive the expected end-to-end delay performance of the proposed scheme. To answer the question of the effect on performance of the number of hops, we compare the performance of 2-hop and 3-hop configurations of the proposed scheme, finding that using 2-hops performs better under high traffic loads, while using 3-hops has lower delay under low traffic loads. The theoretical analysis is validated by simulations based on a real world urban building map. These results demonstrate that, compared with direct association and a state-of-art max-SINR relaying scheme, our minimum-Delay relaying scheme can improve average delay and outage probability across a variety of network configurations, except under extremely strict delay constraints.

1.4.3 Throughput and fairness oriented relaying scheme (Chapter 5)

A traditional relaying scheme, such as max-SINR relaying, associates with UEs that have the best channel quality [28]. However, such a scheme has a problem of

scalability, since unbalanced locations of devices can lead to partial congestion in the network, especially in the future when total data generated in MTCs is expected to grow exponentially [29]. Thus, we seek to design efficient relaying schemes for both uplink and downlink cellular networks, which maximize overall throughput without compromising fairness. Thanks to uplink-downlink decoupling, we can consider these two networks separately.

In the first part of chapter 5, a relaying scheme for the uplink is considered. Thanks to advances in self-interference cancellation techniques, in-band full-duplex wireless systems are enabled. For full-duplex systems, we present a multi-hop relaying and resource allocation scheme, by user association, for the uplink from MTCDs in a cellular network with the aim to optimise data rate of each MTCD, and sum data rate across MTCDs, while maintaining fairness and load balancing among MTCDs. We define a non-cooperative game for data rate and user association across MTCDs, which has a unique Nash Equilibrium (NE). The game is decomposed into subgames, each consisting of a subset of MTCDs that maximize the sum of utilities for association. A distributed algorithm applying Lagrangian decomposition allows each subgame to reach a NE, and after a small number of iterations the sum utility is maximized such that social optimality — guaranteeing maximum social welfare across all MTCDs — for data rate is achieved, while power consumption is minimized.

In the second part of chapter 5, a relaying scheme for the downlink is considered. Multi-tier mmWave network architectures leverage relay access points (APs) to shorten transmission distances and alleviate blockage, but such architectures pose challenges to the scheduling between each tier. So, we formulate downlink cross-tier scheduling as a nonlinear binary integer programming problem, which maximizes sum throughput from BS to UEs via access points (APs). Then, we de-

sign an iterative scheduling scheme utilizing AP to AP links, which ensures that all UEs are served with polynomial computational complexity. The sum throughput upper bound is derived and validated through simulations, which demonstrates far superior fairness to a state-of-art algorithm, and similar spectral efficiency.

1.5 Thesis Organisation

This thesis is organized into chapters based on contributions. Thus, following from the summary of contributions in chapters 3 to 5 provided in the previous section, here we briefly outline the content of the chapters in this thesis:

Chapter 2 presents a comprehensive literature review. We first introduce the preliminariees of cellular IoT as the research background. Then, we review the related work for both random access channels in MTC cellular networks and multi-hop relaying in MTC cellular networks.

Chapter 3 introduces the design of a DRL-based ACB and back-off controlling algorithm, which both optimises efficiency in the RACH and meets heterogeneous QoS requirements.

Chapter 4 presents the design of our Minimum-Delay uplink relaying scheme, which minimizes each UE's end-to-end delay and improves overall network uplink efficiency.

Chapter 5 introduces the design of throughput and fairness oriented relaying schemes in both uplink and downlink communications.

Chapter 6 concludes the thesis and discusses potential future research directions.

Chapter 2

Literature Review

As made clear in the first chapter, this thesis aims to address the challenges for Cellular Internet of Things (CIoT) in both random access and relaying. Hence, in this chapter, we provide the fundamentals, challenges and potential technologies, design and implementations of CIoT, which serve as the general underlying framework of this thesis. Then, existing works in random access (RA) and multi-hop relaying in CIoT are reviewed, to explain the current state of CIoT with respect to the contributions of this thesis.

2.1 Cellular Internet of Things

As IoT features universal connections of autonomous devices, cellular networks are one of the best choices to provide communications due to their wide coverage and ease of deployment. In this section, we introduce the framework and potential technologies in the CIoT, and then review existing work regarding the design and deployment of CIoT.

2.1.1 Fundamentals of the Cellular Internet of Things

The CIoT features a wide range and large number of autonomous devices communicating without human intervention, and are an essential part of 5G and future wireless systems [30]. Cisco estimates that 12 billion devices will be connected by 2020 [31], while Ericsson goes further in its vision and estimates 18 billion devices to be connected by 2022 [32]. From a point of view of requirements, CIoT is often divided into at least two categories: massive machine-type communication (mMTC) and ultra-reliable low-latency communication (URLLC). In the 5G white paper [33] by the Next Generation Mobile Alliance, the authors describe these two categories in terms of use cases and requirements. Smart wearables and sensor networks belong to mMTC, which feature massive connections, wide coverage, low power, and low costs. URLLC, on the other hand, can be exemplified by high-end applications such as automated driving, industrial automation and eHealth, which require high throughput, high mobility and low latency [34].

To allow for full mechanical automation, the concept machine-to-machine (M2M) communications are developed to achieve comprehensive connections among all smart devices. Three major classes of communications are involved in M2M: Communications between the sensor and the decision maker, Communications among multiple calculation agents within the decision maker, Communications between the decision maker and the action executor [21].

Currently, 3GPP proposed three technologies, named EC-GSM-IoT, NB-IoT and LTE-M, to support IoT communications [35]. These were, to a large extent, designed to serve use cases belonging to the category of mMTC. The work on EC-GSM-IoT and NB-IoT aimed at finding a competitive solution in the low power wide area segment, which was to a large extent defined by unlicensed operation specific for IoT applications. As for the work of LTE-M, the mean objective is to

achieve a 20dB better coverage than existing LTE networks to provide coverage for devices with challenging coverage conditions, such as stationary utility metering devices located in basements. These three technologies all feature long battery life (10 years); massive network capacity (60,000 devices/km2); tolerable latency (10 s for high-priority reports); low throughput (160 bps); and ultra-low device complexity [35].

However, 3GPP cellular technologies are not the only solutions for IoT communications. In recent years, a new set of technologies have emerged in the category of low power wide area networks (LPWAN) [36]. These technologies normally use unlicensed spectrum, corresponding to portions of the public frequency space that are free of licensing costs. They are designed to meet regulatory requirements associated with these frequency bands, but unlike wireless local area networks (WLAN) and wireless personal area networks (WPAN), they provide long-range connectivity, which is the enabler for supporting IoT devices in locations beyond WLAN and WPAN coverage [37].

2.1.2 Challenges and potential technologies in the Cellular Internet of Things

As current standardisation mainly focusses on solving the basic requirement of CIoT: providing ubiquitous internet access service to things, there are still many other challenges waiting to be solved in future cellular networks. The wide range of devices with quite different applications require a new network paradigm – heterogeneous network – to support their distinct quality of service (QoS). To further improve spectral efficiency and solve scaling problem, decentralized communication – device-to-device communication, will play an important role in CIoT. To enable URLLC, high speed transmission in CIoT will likely be provided by a new

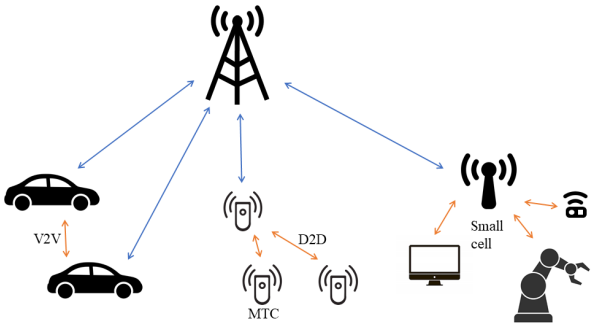


Figure 2.1: Example of CIoT

transmission technology: millimeter wave communication. An example of CIoT is illustrated in Fig. 2.1. Next, we introduce the three technologies in detail.

Heterogeneous Network (HetNet)

As mentioned before, one of the biggest challenges in CIoT is the diverse network requirements from a wide range of devices with quite different applications. Some applications require extremely low power consumption and long battery as long as 10 years, while some others require ultra-low latency as low as 5 ms. Thus, a universal technology accommodating such heterogeneous requirements in CIoT is not well realised. HetNet is a new network paradigm designed to satisfy the on-demand requirements of service-driven IoT in future cellular networks. A Hetnet for IoT is comprised of different cell types and different access technologies. A wide area network can use macrocells, picocells, and/or femtocells to offer wireless coverage ranging from an open outdoor environment to office buildings, homes and underground areas. Small cells are primarily added in hot spots with high user demands, and to fill in areas not covered by the macro-network [38]. A number of Hetnet solutions have been developed, including resource allocation in non-orthogonal mul-

tiple access heterogeneous networks [39], user association for maximising energy efficiency in heterogeneous networks [40], energy-harvesting aware routing algorithm in heterogeneous IoT networks [41].

Femtocells, now more commonly known as "small cells", based infrastructure is a key component in HetNet deployments [42]. Small cells are commonly installed in indoor environments to provide better coverage and capacity as well as to offload traffic from macrocells. Indoor small cells bring receivers and transmitter closer to each other which significantly decrease penetration loss, thus effectively reduce energy consumption and prolong battery life for IoT devices [43]. Indoor small cells, together with other relevant technologies can be used to enhance e-health monitoring systems [44]. On the other hand, small cells can also be deployed outdoors in public transport vehicles, such as buses and trains, to provide service for devices on the move [42].

Device-to-Device (D2D)

Compared with a traditional cellular network, where devices communicate directly with the BS, D2D communication enables data exchange between user devices without the use of the BS, therefore it is a promising technology for optimising spectral efficiency (underlay inband communication) and dealing with the scaling and coverage problem, especially for IoT devices [2]. D2D takes advantage of the closer proximity of communicating devices for best utilization of available resources: improving data rates; reducing latency; and increasing system capacity [45]. D2D is being considered as a key enabling technology in 5G cellular networks due to the inherent need for high data rates, constraints on delay and QoS specific communications. However, there are still many challenges, as it needs efficient discovery mechanisms, intelligent mode selection and association algorithms, complex

resource management and mobility management.

D2D communication has been utilised in unlicensed spectrum for a long time, but it has not been introduced in cellular networks until the fourth generation (4G) of cellular networks, release 12 in 2012 [46]. D2D communications have proved to be useful to integrate into many IoT technologies: including public safety services [47]; vehicular networks [48]; cellular offloading [49]; multi-hop relaying [50], and proximity based services [51]. D2D communications are classified as inband D2D (on cellular spectrum) and outband D2D (on unlicensed spectrum). In the case of inband D2D, there are two ways D2D communications access spectrum resources. In underlay inband communication, D2D user equipments (UEs) compete with cellular UEs and opportunistically access resources occupied by cellular users, thus resulting in improved spectral efficiency [52]. On the other hand, in overlay inband communications, a portion of the cellular spectrum is dedicated for D2D communication, and as a result, radio interference is managed and signal strength is improved for relay assisted networks [53].

Millimeter Wave Communication (mmWave)

Another big challenge for cellular IoT is that many applications require ultra-reliable and low-latency communications (URLLC). For example, V2V and V2X communication require end-to-end delay of milliseconds and extremely high reliability. However, current cellular networks (LTE) can only achieve delay of tens of milliseconds. The reason is that they were originally designed for H2H communications, thus both radio access technology and bandwidth are limited. Thus, in 5G and future cellular networks, the employment of mmWave is inevitable, with 5G New Radio (NR) mmWave bands already incorporated in release 15 from 3GPP [14]. The mmWave bands, roughly corresponding to frequencies from 30

GHz to 300 GHz, have drawn considerable attention because of the huge unused bandwidth [54]. The large chunk of spectrum allows for simultaneous massive connections, each having high data rates due to abundant spectrum resources. Owing to short wavelengths, mmWave devices typically have large antenna arrays packed in a small physical dimension. However, due to much higher carrier frequencies than those used by conventional wireless techniques, mmWave communications have several natural disadvantages, e.g., severe signal attenuation caused by oxygen absorption, they are easily blocked by obstacles due to low penetration and diffraction ability [55]. Therefore, these two factors significantly reduces mmWave communication's transmission range and these links can be disconnected easily. However, beamforming based on mmWave antenna arrays has been widely exploited to concentrate the signal in desired direction and significantly extends transmission range as well as reduces interference [56].

Multi-hop relaying uses the concept of D2D to route wireless data to the destination through other UEs [2]. For mmWave communications, proper relaying algorithms can overcome the blockage problem, increasing the robustness of mmWave links and effectively improving network coverage [57]. Furthermore, a well designed relaying algorithm can minimize the interference among these concurrent links thanks to beamforming enabling directional communications [58].

2.1.3 Design and Implementation of the Cellular Internet of Things

Smart City

One of the first applications of cellular IoT is in smart cities [59]. Unlike smart homes, which largely utilise WPAN (Zigbee) or WLAN (Wi-Fi) to support com-

munications between interconnected smart devices, smart cities rely on large scale cellular networks to gather information and provide remote controls according to residents needs. There are many basic services, such as transportation, metering and environment management, which could be improved in collaboration with CIoT to enhance the quality-of-life of city residents [59, 60]. The most popular applications include environmental monitoring, smart grid [61], traffic management [62] (which includes vehicle communications) and waste management systems. Researchers and industry collaborated in the "Barcelona Intelligent City" project [63] to test the smart city concept with the dual purpose of contributing to the creation of standards and providing an interim solution to cope with the heterogeneity of providers, in particular relating to wireless sensors. The work in [64] presents a framework for the realisation of smart cities through IoT, which covers the complete urban information system, from the sensory level and network support structures to data management and cloud-based integration. However, the processing constraints hinder the IoT devices to offer complex services such as security and QoS. Integrating network services through network function virtualisation can effectively provide security for IoT devices, identifying malicious traffic and ensuring QoS, without introducing significant overhead [65]. Similar to smart homes, communication devices in these applications normally feature low power consumption (10-year 5Wh battery life), but they spread out over large areas and require reliable communications over a much larger distance (maximum coupling loss of 164 dB) than that of smart homes [35]. These applications also feature markedly heterogeneous network requirements, for instance, meter readings for utilities [66] require much less frequent updates than traffic monitoring, but meters can be located deep inside the building requiring much stronger signal coverage [35].

E-health

IoT is expected to be a dominating influence of modern medical and healthcare systems, as it enables the discovery and performance of new diagnostic and treatment techniques [67]. Recent developments in wearable sensors capable of collecting biologically relevant data such as blood pressure, body temperature, cholesterol level, heart rate etc., have enabled new ways to track progress and directly communicate anomalies to a medical professional. Real-time motion metrics to monitor Parkinson's disease can be enabled by placing lightweight wearable sensors in the patient's clothes, and using mobile devices the gathered data can be analysed in a medical data base, enabling the discrimination of relevant threatening conditions, adjusting medications, and facilitating treatments that don't require frequent medical visits [68]. However, the continuous sampling, processing and transmission usually lead to high energy consumption due to limited battery resources. Instead of solely relying on batteries, energy harvesting powered IoT devices can gather energy from the ambient environment to allow for long-term operation without battery replacement or charging [69]. Instead of using GPS and accelerometer, transportation mode detection using voltage signals from kinetic energy harvesting device can further save energy [70]. A Wireless Body Area Network (WBAN) is standardized under IEEE (802.15.6 [71]), solving the data transmission problem for extremely low power devices on, or inside, the human body [72]. Similar to a smart city, these are delay tolerant applications, but there are also low-delay applications, such as real-time emergency response and remote diagnostics.

Industrial IoT

Traditionally, industrial control systems are based on wired networks because of their stability. Nevertheless, replacing wires with radio links has substantial bene-

fits, such as lower deployment costs, higher long-term reliability as wired connections suffer from and tear in motion applications, and higher flexibility in maintenance and upgrading [73]. Industrial IoT (IIoT) represents IoT application in the automation process of manufacturing. In IIoT, the data reliability is extremely stringent, because this data is usually deterministic as it has strict delay constraints [74]. For closed loop control systems, latencies at the millisecond scale are required [75]. URLLC is exactly designed for these kind of applications [76]. The two most critical sources of delay are link establishment (grant acquisition or random access) and packet retransmissions caused by channel errors and congestion. Transmit time interval (TTI), defined as the minimum data block length, is another substantial delay component since it is involved in each link establishment and data transmission/retransmission step [77]. In LTE and LTE-A, a major job of IoT compliant technologies is to accommodate a large number of connections and improve coverage, but the requirements for URLLC is largely neglected. In a recent release of the 5G specification by 3GPP, Release 15, one important goal is enabling URLLC, and multiple approaches have been proposed for mission critical services. Low latency is implemented by using a wide sub-carrier spacing, reducing the number of OFDM symbols used for data assignment, while high reliability is implemented by using the new channel quality indicator (CQI) and a specific modulation and coding scheme (MCS) [78]. However, unlike other domains, the industrial domain typically employs proprietary solutions, which limit interoperability, whereas the other applications domains are usually more open and use standardized protocols more extensively.

Vehicle-to-Everything (V2X)

V2X includes vehicle-to-vehicle (V2V), vehicle-to-pedestrian (V2P), vehicle-to-infrastructure (V2I) and vehicle-to-network (V2N) communications [9]. Through wireless interconnected vehicles, pedestrians and infrastructure, V2X is able to improve road safety, traffic efficiency and the availability of infotainment services. The initial set of LTE technical enablers for V2X communication services, included in release 14, has been substantially enhanced in release 15 [78]. The enhancements are categorized as vehicle platooning, extended sensing, advanced driving and remote driving [79].

V2X support provided by LTE cellular networks makes it a suitable technology for meeting the requirements set by the European Telecommunications standards Institute (ETSI) [80]. ETSI has defined two types of safety messages: cooperative awareness messages (CAMs) and decentralised environmental notification messages (DENMs) [81]. CAMs are periodic messages to interchange vehicles status among vehicles in a close vicinity, while DENMs are used for road hazard warnings to warn road users of dangerous events.

In V2X, both cellular interfaces based on uplink/downlink and D2D interfaces using sidelink will be jointly operated and selected according to V2X service requirements [9]. Intelligent Transportation Systems (ITS) built on V2X consists of four types of entities: vehicles, vulnerable road users, road side units and a central ITS server. All the entities can communicate with each other by cellular based communication or D2D communication. D2D communication provides low latency and short-range communications even out of cellular coverage, while cellular based communication is for wide-area communication with high capacity [82].

2.2 Random Access Channel (RACH)

2.2.1 Random Access Channel Procedures in Cellular Networks

In this section, we introduce the random access (RA) procedure of LTE-M networks. Long-term evolution-M (LTE-M), sometimes referred as Enhanced machine type communication (eMTC) or category M1, is a new type of low-power wide area (LPWA) technology standard under LTE intended for low cost machine type communication devices (MTCDs) for its reduced power consumption and simpler signaling procedure [35]. In LTE-M, UEs perform a RA procedure to request channel resources for uplink transmission to the eNB in this cellular-based network. There are two modes operating in the RA procedure, contention-free mode and contention-based modes [83]. In contention-free mode, a dedicated scheduling request resources is assigned by the eNB to UEs, which provides low latency services to high priority users in situations such as handover or positioning. On the other hand, contention-based mode – the standard mode for network access – is used by regular UEs to synchronize with the enhanced Node B (eNB) for its initial association to the network, recovering from link failure or sending scheduling requests. Thus, we only consider contention-based RA here. The contention-based RACH has four steps, summarized in Fig. 3.2 :

Step 1: Each UE randomly chooses a preamble from a non-dedicated preamble pool known to both UEs and eNB, and transmits the preamble to eNB. In LTE-A, there are 64 available preambles for RACH in each BS, which are generated from a Zadoff-Chu sequence [84]. The transmission of preambles serves as a request for a dedicated time-frequency resource block in the upcoming scheduling transmission.

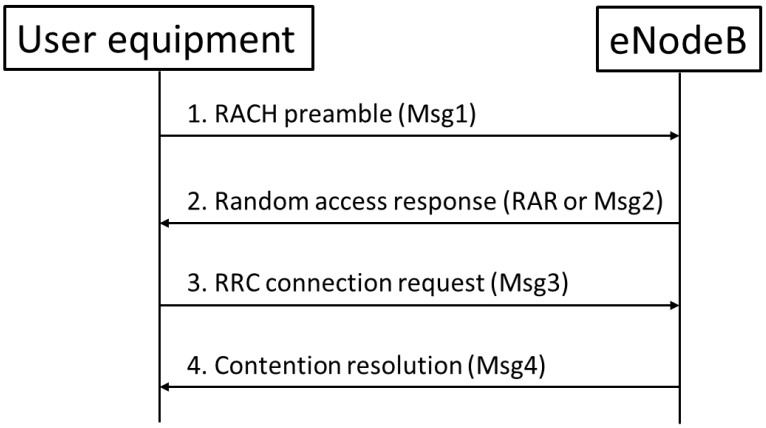


Figure 2.2: Random Access Procedure

Step 2: The eNB decodes the received preambles. A RA response (RAR) will be sent back to acknowledge each decoded preamble, which contains identification of the preamble and uplink grant for step 3 message.

Step 3: The UEs sets a RAR window and waits for the BS to respond with an uplink grant in the RAR. UEs that successfully receive a RAR transmit a radio resource control connection request with identity information to the eNB. When two or more UEs selected the same preamble in step 1, all of them will be granted the same resource block for msg3 uplink transmission, and a collision will happen.

Step 4: eNB broadcasts contention resolution containing IDs of UEs whose msg3 are successfully decoded. There will be no response to the collided msg3 and those UEs selecting the collided preambles are declared failed in the contention resolution.

2.2.2 Enhancements and Variations of the Random Access Channel

Because of the anticipated massive number of MTCDs in LTE-M networks, each carrying only a small amount of data to be transmitted, simultaneous transmission attempts to access the eNB require large-scale synchronization and will result in data traffic flow congestion in the radio access network (RAN) [20, 21]. M2M communications can cause physical random access channel (PRACH) overload in several ways: an external event triggers a large amount of MTCDs to become active from an idle state and access the network at once, or a massive number of scheduled MTCDs periodically request network access to report data. Studies have shown that the actual RA procedure is not efficient for managing massive simultaneous access [26, 85].

Several methods have been proposed to control congestion and provide better RA efficiency. The general strategy of allocating additional random access resources for M2M traffic has been examined from various perspectives [86]. A reporting based resource allocation scheme is proposed in [87], while resource allocation based on delay requirements is considered in [88]. A context aware approach is considered in [89]. Game theory has also been used to optimise resource allocation in random access processes [90]. A Nash equilibrium analysis of M2M arrivals with a game-theoretic perspective of selfish users reaching the RACH has been conducted in [91]. A stochastic coalitional game composed of multiple stages, in which cooperative groups are formed to coordinate the devices' requests for the RACH, is presented in [92]. Stochastic geometry is also applied to analyse the interference between devices for better resource allocation in [93, 94].

Access class barring (ACB) is regarded as an efficient tool to prevent congestion caused by massive concurrent access [95]. ACB is a probability based solution

that limits the number of user equipments (UEs) participating in a RA procedure simultaneously. The optimum ACB factor that best reduces congestion and access delay, when the eNB knows the number of backlogged users, is determined in [96]. The effectiveness of an ACB method in highly congested environments was evaluated according to key performance windows such as delay and energy consumption in [97]. Various methods for controlling ACB have been proposed [98–100], and a Markov-Chain traffic load estimation scheme according to network collision status is proposed in [101]. A dynamic RACH preamble allocation scheme based on the ACB factor is proposed in [102].

A delay estimation based RA scheme was proposed in [103], based on a back-off (BO) scheme, which aims at improving collision detection and resolution capability. A novel dynamic back-off scheme, which takes constrained downlink resources into consideration, is presented in [104]. And to protect RACH from overload, a class-dependent back-off scheme is introduced in [105]

2.3 Multi-hop Relaying in the Cellular Internet of Things

2.3.1 Basics of Wireless Multi-hop Relaying

A relay network is a kind of network topology, where source and destination nodes are interconnected by means of relay nodes because the source and destination nodes can not reliably communicate with each other directly [106]. There are three most commonly used relaying schemes: decode-and-forward, compress-and-forward and amplify-and-forward. In decode-and-forward, the relay node decodes the source message in one block, then re-encodes and forwards the message in the

following block [107]. In compress-and- forward, the relay node can not decode the message sent by the source, but still can help by compressing and forwarding its observation to the destination [108]. In amplify-and-forward, the relay node sends an amplified version of the received signal to destination, thus requiring much less computational power [109].

Wireless sensor networks (WSNs) are composed of a large number of sensor nodes capable of wireless communications, randomly deployed in the area possessing self-organizing capabilities [8]. These features make WSN suitable for various IoT applications in E-healthcare, environmental management and security. The realisation of WSNs require wireless ad hoc networks (WANETs). A WANET does not relay on pre-existing infrastructure, such as routers in wired networks or access points in managed wireless networks. Instead the nodes form a mesh network, participating routing by relaying data for other nodes [110].

In academia, D2D communication was first proposed in [111] to enable multi-hop relaying in cellular networks, which allows for use-cases such as M2M communications, content distribution and data aggregation. As cellular networks are evolving to support IoT applications, relaying plays an essential role in alleviating the access pressure, saving energy consumption, improving system capacity and coverage [112]. Although D2D relaying in cellular networks looks similar to relaying in WANETs, there are still fundamental differences among their architecture. Relaying in cellular networks is expected to be overseen/controlled by a central entity, usually the BS. The availability of a supervising central entity resolves many problems of WANETs such as white space detection, collision avoidance and synchronisation [113].

New technologies for 5G networks are also beneficial to multi-hop relaying [114]. Full-duplex technology allows relay nodes to transmit and receive at the same time

[115]. The extra frequency band in mmWave provides relaying implementations with more time-frequency resource blocks to use, thus largely solves the problem of interference. A UE-based mode can construct new M2M services by the UE becoming a core access node in the IoT network. Novel relay selection algorithms could further improve network capacity and alleviate balanced network loads in relaying cellular networks.

2.3.2 Developments in Relaying for Cellular Networks

D2D communication was first introduced by 3GPP at release 12 as a Proximity-based Service (ProSe) [116], but was mainly intended for public safety use. The same transmission mechanism is also applicable to other scenarios, e.g., local data exchange, M2M communication, UE relaying, and converged heterogeneous networks. In recent years, cooperative transmission based on buffer-aided UE relaying has been proven to be a promising approach to improve transmission efficiency and extend cellular transmission range [117,118]. Uplink aggregation for M2M has been comprehensively investigated in [25]. Data aggregation and resource scheduling schemes for massive machine type communication (mMTC) networks are presented using additional aggregators as intermediate relays [119], or, alternatively, by allowing multi-tier relaying to maximize transmission rate and balance uneven loads [120]. However, there is only limited research on D2D relay-assisted mmWave cellular communications [121,122].

QoS of multi-hop relaying schemes for different physical layer designs is considered in [123]. Average end-to-end delay and maximum achievable per-node throughput in multihop ad hoc networks is derived in [124]. In recent years, ultra-reliable and low-latency communications (URLLC) have drawn much attention because of the more stringent requirement of reliability and delay for future appli-

cations [125]. The problem of URLLC in mmWave enabled multiple-in multiple-out (MIMO) networks was studied in [126] by applying a Lyapunov technique which adapts to channel variations and queue dynamics. For systems with buffers at transceivers, the probabilistic delay for point-to-point mmWave communications is analysed in [127], where the network layer delay bound in the queuing system is derived based on network calculus theory. Multi-hop relaying under a sum power budget is investigated in [128], where a bound on probabilistic delay and effective capacity is derived through stochastic network calculus.

2.4 Concluding Remarks

In this chapter, we introduced the fundamental background of CIoT, followed by an overview of challenges and potential technologies for CIoT including HetNet, D2D and mmWave communications. These new technologies allow for more demanding CIoT services such as heterogeneous QoS, massive connections and URLLC. Recent design and implementations of CIoT were reviewed, with applications across smart cities, E-health, IIoT and V2X. Then, a literature review of the two challenges we aim to address in this thesis: RACH and multi-hop relaying in CIoT; was given in detail in order to give a background of research in these areas. In the next chapter, we will present our research in optimising RACH in cellular networks in detail.

Chapter 3

Deep Reinforcement Learning Based Optimisation of Random Access

As discussed in chapter 1, a major challenge in cellular IoT is accommodating a massive number of connected devices while still providing satisfactory services. In this chapter, we consider the problem of random access channel (RACH) optimisation in cellular networks. Access class barring (ACB) and back-off (BO) schemes are two approaches to improve RACH efficiency through regulating simultaneous access numbers in burst traffic situations. Thus, we present a deep reinforcement learning (DRL) based ACB factor and BO window controlling algorithm, which optimises the RACH for machine type communication devices (MTCDs) with various quality of service (QoS) requirements.

3.1 Introduction

3.1.1 Background

Machine-to-machine (M2M) communications, also known as Machine-Type Communications (MTC), feature a wide range and large amount of autonomous machines communicating with each other, without, or with little, human intervention. It is an essential part of the IoT, and future wireless systems [129–131]. To accommodate versatile M2M traffic characteristics, from best effort applications such as smart metering and environment monitoring, to ultra-reliable ones such as healthcare, public safety and mission-critical industry, a cellular network is a promising means to satisfy such heterogeneous requirements due to their high scalability, diversity and low cost of additional infrastructure deployments [132].

Enhanced machine type communication (eMTC), sometimes referred as long-term evolution-M (LTE-M) or category M1, is a new type of low-power wide area (LPWA) technology standard under LTE intended for low cost MTCDs for its reduced power consumption and simpler signaling procedure. Because of the anticipated massive number of MTCDs, each carrying only a small amount of data to be transmitted, simultaneous transmission attempts to access eNB require large-scale synchronization and will result in data traffic flow congestion in the radio access network (RAN) [20, 21]. M2M communication can cause physical random access channel (PRACH) overload in several ways: an external event triggers large amount of MTCDs to become active from an idle state and access the network at once, or a massive number of scheduled MTCDs periodically request network access to report data.

Devices with eMTC technologies perform a random access (RA) procedure to request channel resources for uplink transmission in cellular networks. The most

important quality of service (QoS) requirement regarding random access is each device's completion time. As delays experienced in random access procedure has a large effect on total uplink latency as well as overall random access capacity. In order to improve QoS, an efficient RA procedure is required to enhance RA performance. A user equipments (UE) performs either contention-free RA or contention-based RA to request uplink scheduling. Due to the characteristic of MTCDs, a contention-based RA is considered as the main access technology to request channel resources in the uplink transmission [85] [83]. The contention-based RA is based on ALOHA-type access, where each UE selects a random preamble from available orthogonal preamble pool, transmitting it to eNB. UEs choosing the same preamble will collide, thus none of them will succeed in the RA procedure, and these devices need to start the RA procedure in the next available RA opportunity (RAO). The RAOs are represented by the repeated PRACHs, and they are reserved in the uplink channel defined by PRACH configuration index. Studies have shown that the actual RA procedure is not efficient for managing massive simultaneous access [85] [26].

In this chapter, we utilize access class barring and a back-off scheme to solve the problem of random access channel congestion. More specifically, we significantly extend our previous work in [133], and introduce a deep reinforcement learning based algorithm, for controlling both access class barring factors and back-off windows, which optimises the random access channel for a massive number of MTCDs with various QoS requirements.

Many Methods have been proposed to control congestion and provide better RA efficiency [86–94]. ACB is regarded as an efficient tool to prevent congestion. It is a probability based solution that limits the number of user equipments (UEs) participating in a RA procedure. Many solutions have been proposed for optimising

ACB [96–102]. Various BO schemes are also proposed to improve RACH efficiency [103–105]. A complete overview of related work is given in chapter 2.2.2.

As MTCDs have a diverse range of applications, QoS requirements in M2M communications are highly variable. Thus, meeting the growing range of various QoS requirements for MTCDs is an urgent area for research. As MTCDs can have various QoS constraints, a multiple access class barring (MACB) scheme is proposed in [1], which assigns distinct access probabilities to different classes of MTCDs requiring different service level. Similar mechanisms are presented in [105, 134, 135], but in those works the parameters are adjusted according to estimated traffic analysed with partial information of the network. [136] presented a dynamic resource allocation scheme that allocates distinguishing number of preambles to delay-sensitive and delay-tolerant MTCDs according to estimated access attempt number.

Reinforcement learning (RL) is a type of machine learning technique that mimics the fundamental way in which humans learn. A reinforcement learning agent evolves by interacting with the environment through observing it, taking actions and receiving immediate reward feedback, while the goal of the agent is to select actions that maximize cumulative future rewards. Deep reinforcement learning (DRL) is an enhanced version of RL that uses a deep neural network (DNN) to approximate the cumulative future discounted reward for each action, so that the agent can make accurate predictions and decisions that lead to a good overall performance for more complicated problems [137, 138]. DRL can be used to solve RACH overload problems because the DRL agent uses network metrics directly to explore network conditions and make adequate control decisions dynamically for optimal long-term performance [133]. RL is also been employed to develop channel access schemes for cognitive radios [139] and wireless sensor networks [140]. As far

as the authors are aware, no prior work studies RACH optimisation based on RL, and we leverage the recent advances in DRL in this chapter to exploit the potential of RACH cellular networks. Since this chapter focuses on RACH optimisation that make use of DRL, we have limited our review of work to RACH only. Readers are referred to [141,142] for a general survey of applications of DRL in communications and networking.

3.1.2 Contribution

In this chapter, *we design a deep reinforcement learning based access class barring factor and back-off window controlling algorithm, which meets diverse QoS requirements for MTCDs and reduces overall delay.* In this context, our main contributions are:

- For the first time, we construct a framework to analyse RA efficiency for M2M communications in a cellular network using Markov Chains. We consider an ACB scheme and a back-off scheme together in the optimisation of RACH. Expected RA throughput and optimal combination of ACB factor and back-off window are given in a saturated traffic model.

- For the first time, we exploit DRL in the design of random access optimisation for M2M communications. Our DRL-based algorithm is formulated to achieve maximum RA throughput by controlling the ACB factor and back-off window. Extensive simulation results in both saturated traffic model and burst traffic model show that the algorithm can maximize RA success number without prior knowledge of MTCDs in the area.

- For the case of MTCDs having different QoS requirements, distinct ACB factors and BO windows are applied to devices with different priority classes,

each controlled by the DRL agent simultaneously. We define the state space, action space and reward function for the DRL agent. The reward function is designed to ensure the highest access probability of delay-sensitive devices while maximizing the RA efficiency at the same time.

- We demonstrate the advantages of DRL over traditional RL for heterogeneous wireless networking. In particular, we show that, compared with traditional RL, DRL can converge to an optimal solution faster and is more precise in terms of action selection, two essential properties for practical deployment in real wireless networks.

- We simulate the RA process under different traffic settings. The results prove that the proposed algorithm achieves maximum RA efficiency in both uniform and heterogeneous QoS networks in terms of success access number. The success probability and average delay of high priority MTCDs are improved compared with another state-of-art scheme.

3.2 System Model and Analysis

In this section, we summarise the random access procedure of eMTC networks. A complete introduction to random access procedure is given in chapter 2.2.1.

UE perform RA procedure to request channel resources for uplink transmission to the eNB. There are two mode of RA: contention-free mode and contention-based mode. The standard mode – contention-based mode – is used by regular UEs to synchronize with the eNB for its initial association to the network, recovering from link failure or sending scheduling requests. In contention-based mode, there are four steps. Step 1: each UE Randomly selects a preamble from a 64 orthogonal preamble pool, then transmit it to eNB. Step 2: eNB sends back RA response (RAR) to

Table 3.1: Summary of main notations

p_{acb}	ACB factor
W	back-off window
p_c	collision probability
U	maximum number of retransmissions
$p_{i,j}$	transition probability from state i to j
B_i	ith state a node enters in steady state
p_i	relative frequency a node enter state i in steady state
$\bar{d}_i$	average time a node stays in state i in steady state
S_i	probability an arbitrary node is in state i
$s_{i,k}$	probability a node is in state i and back-off timer has value k
$s_{i,trans}$	probability of a node sending a random preamble in state i
p_t	probability a given node will transmit in an arbitrary RAO
n	number of active UEs
M	number of orthogonal preambles
C_m	number of node selecting preamble m
N_i	number of active nodes in RAO i
N_i^{trans}	number of nodes sending a random preamble in RAO i
K_i	number of success preamble transmission

acknowledge each preamble. Step 3: UEs successfully receive its RAR transmit a radio resource control (RRC) connection request with identity information to the eNB. When two or more UEs select the same preamble in step 1, they will collide in step 3. Step 4: eNB broadcast contention resolution to UEs, and there will be no response to collided UEs. The RA procedure is summarised in Fig. 3.1.

In our work, we only consider how MTCDs as UEs compete for uplink grants amongst themselves. Therefore we suppose separate resources are allocated to M2M traffic and human-to-human traffic. Table 3.1 gives a summary of main notations appear in system model.

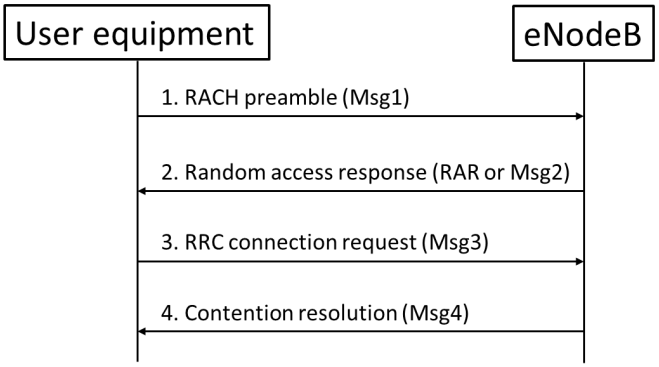


Figure 3.1: LTE-A Contention-based random access (RA) Procedure

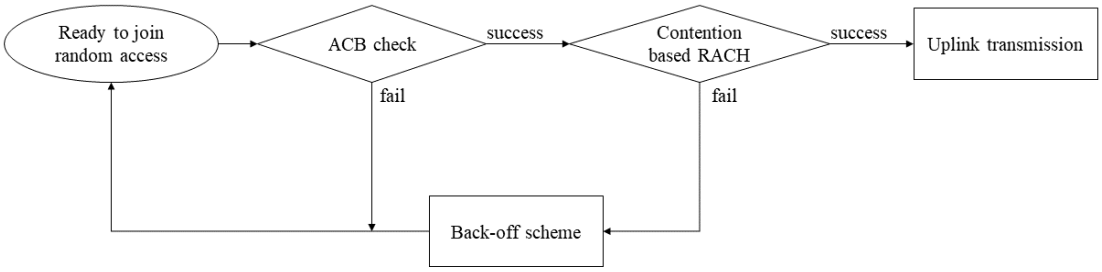


Figure 3.2: Access class barring and Back-off procedure.

3.2.1 Access Class Barring and Back-off scheme

The RACH described above happens in each random access opportunity (RAO). The eNB broadcasts system parameters at the beginning of each RAO, including ACB and BO factors, and then the devices execute the RACH according to the system information in each RAO, and the eNB will be updated at the end of each RAO. The process of access class barring and back-off scheme is illustrated in Fig. 3.2.

Access Class Barring (ACB) is a method to redistribute the network access requests of UEs in the time domain to alleviate RACH pressure. Before each Random Access Opportunity (RAO), the eNB broadcast an ACB factor $p_{ACB} \in (0, 1]$ as part of system information according to the network condition. After receiving ACB factor p_{ACB}, each active UE that wants to participate in RA draws

a random number $q \in (0, 1]$. If $q < p_{ACB}$, then the UE is allowed to transmit the preamble. Otherwise, the UE is blocked from access, stays silent and waits for the next RAO. This process is repeated until the UE generates q greater than the ACB factor. ACB scheme is a basic congestion control method that reduces the RA attempts to avoid massive simultaneous access requests that lead to most preambles being collided. A suitable ACB factor can keep the allowable access within a reasonable number, and helps keep high RA efficiency when the network is overloaded.

A back-off scheme is another congestion control method that defers the UE's re-attempt of participating in a RA procedure for a certain time to alleviate the access number. When a UE fails a RA procedure due to preamble collision, it automatically defers the RACH re-attempt and waits for k RAOs, where back-off time k is randomly drawn between 0 and BO window $k = \mathrm{U}(0, W)$, until it tries again. In the LTE-M standard, back-off window W is broadcasted by eNB before each RAO. However, an exponential back-off scheme deployed in each UE is proven to improve RA efficiency as waiting time increases exponentially with the number of fail times [143].

3.2.2 ACB and Back-off Analysis in saturation mode

In some networks, QoS of MTCDs is not a major concern, or broadcasting multiple ACB factors for multiple classes of UEs is impractical, and a single ACB factor applying to all UEs in the cell is favorable. As all MTCDs in the cell are uniformly treated, the optimisation problem is simplified to maximizing the total number of MTCDs successful in RA over a particular period.

We assume a saturated traffic condition, meaning each node always has packets to upload and always requests uplink resource after each success in RACH. We

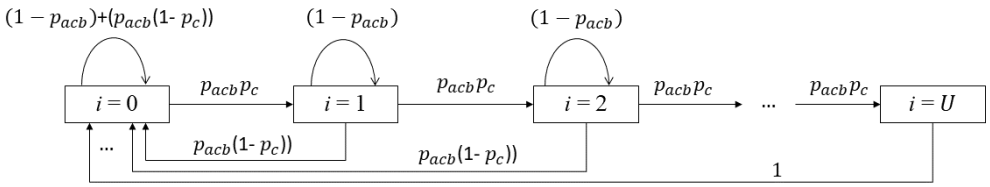


Figure 3.3: State transition diagram of a node in steady state

model the operation of the RACH at an individual node using the state diagram in Fig. 3.3. There is a limit, U, to the maximum attempts per random access request and after reaching state U, a node goes to state 0 with a new random access request even if the preamble transmission was unsuccessful. Denote by B_k the kth state that a node enters in steady state, then B_k is a Markov chain with the transition probability $p_{i,j}$ given as follows:

$$\begin{aligned}
p_{i,0} &= \Pr\{B_{k+1}=0|B_k=i\} = p_{acb}(1-p_c), \quad i=0,1,...,U-1 \\
p_{i,i+1} &= \Pr\{B_{k+1}=i+1|B_k=i\} = p_{acb}p_c, \quad i=0,1,...,U-1 \\
p_{i,i} &= \Pr\{B_{k+1}=i|B_k=i\} = 1-p_{acb}, \quad i=0,1,...,U-1 \\
p_{i,j} &= 0, \quad j\neq 0, \quad j\neq i+1
\end{aligned} \tag{3.1}$$

Let P_i be a probability defined as

$$P_i = \lim_{k\to\infty} Pr\{B_k=i\}, \quad i=0,1,...,U \tag{3.2}$$

then P_i is the relative frequency that a not will enter state i at steady state. Since $\sum_{i=0}^{U} P_i = 1$, according to the Markov chain described in Fig. 3.3, P_i can be obtained as follows:

$$P_i = \frac{1 - P_{acb}P_c}{1 - (P_{acb}P_c)^{U+1}}(P_{acb}P_c)^i, \quad i = 0, 1, ..., U \tag{3.3}$$

The main performance measure in evaluating a network is its throughput. We analyse the saturation throughput in steady state by calculating the number of successful preamble transmissions in a RAO. The probability P_i given in 3.3 is the relative frequency that a node enters state i. But the average time that a node stays in a state depends on back-off window of the state. If a node enters state $i, i \neq 0$, then let integer random variable D_i be from a uniform distribution from $\{0, 1, ..., W-1\}$. After waiting for D_i RAOs. the node will get an ACB check again, which determines whether it can retransmit a random preamble or wait until the next RAO. Note that the node will stay in state i for $D_i + 1$ time slots until the node moves to the next state. On average, a node will stay in state i for

$$\begin{aligned} \bar{d_0} &= \frac{1}{p_{acb}} \\ \bar{d_i} &= \mathrm{E}[D_i + 1] = \frac{W+1}{2} + \frac{1}{p_{acb}}, \quad i = 1, 2, ..., U-1 \end{aligned} \tag{3.4}$$

Therefore, the probability that an arbitrary node is in state i is

$$S_i = \frac{P_i \bar{d_i}}{\sum_{j=0}^{U-1} P_j \bar{d_j}} \tag{3.5}$$

For large M, (3.5) can be approximated as

$$\begin{aligned} S_i &\approx \frac{2 - 2p_{acb}p_c}{2 + (W+1)p_{acb}^2 p_c}, \quad i = 0 \\ &\approx \frac{((W+1)p_{acb} + 2)(1 - p_{acb}p_c)(p_{acb}p_c)^i)}{2 + (W+1)p_{acb}^2 p_c}, \quad i = 1, 2..., U-1 \end{aligned} \tag{3.6}$$

We define $s_{i,k}$ the probability that the node is in state i and the back-off timer has value k

$$S_i = \sum_{k=0}^{W} s_{i,k} \tag{3.7}$$

Since the back-off counter is decreased by one at the elapse of each RAO, when the counter reaches zero, it has the probability of p_{acb} to retransmit a new preamble, and the probability of $1 - p_{acb}$ to hold on and wait for next RAO where it will get another ACB check. Thus the probability of back-off counter being 0 is given as follows:

$$\begin{aligned} s_{0,0} &= 1 \\ s_{i,0} &= \frac{2S_i}{p_{acb}(W + \frac{2}{p_{acb}} - 1)} \end{aligned} \tag{3.8}$$

and the probability of sending a random preamble in state i is thus

$$\begin{aligned} s_{i,trans} = s_{i,0}p_{acb} = p_{acb}, \quad & i = 0, \\ = \frac{2S_i}{W + \frac{2}{p_{acb}} - 1}, \quad & i = 1, 2..., U - 1 \end{aligned} \tag{3.9}$$

Let p_t be the probability that a given node will transmit in an arbitrary RAO. Then, since $s_{i,trans}, i = 0, 1, ..., U$ are the probabilities of mutually exclusive events,

$$p_t = \sum_{i=0}^{U} s_{i,trans} \tag{3.10}$$

For large U, (3.10) can be approximated as:

$$\begin{aligned} p_t \approx & \frac{(2 - 2p_{acb}p_c)p_{acb}}{2 + (W + 1)p_{acb}^2 p_c} + \\ & \frac{2((W + 1)p_{acb} + 2)(p_{acb}p_c)}{(2 + (W + 1)p_{acb}^2 p_c)(W + \frac{2}{p_{acb}} - 1)} \end{aligned} \tag{3.11}$$

Next we analyse the relationship between transmit probability in an arbitrary RAO p_t and collision probability p_c. Denote by n the total number of active

nodes, and N_i^{trans} the number of nodes to transmit in the RAO i. Each of N_i^{trans} nodes selects one preamble from M orthogonal preambles in the pool with equal probability $\frac{1}{M}$. Let's consider preamble m, and let $C_m = 0, 1, ..., c$ denote the number of UEs that select preamble m, $m = 1, 2, ..., M$. The probability that exactly one user selects preamble m is

$$\Pr\{C_m = 1 | N_i^{trans} = j\} = \binom{np_t}{1} \frac{1}{M} \left(1 - \frac{1}{M}\right)^{j-1} \tag{3.12}$$

As each preamble is independent of others, the expected number of successful transmissions is

$$\begin{aligned} \mathrm{E}[K_i | N_i^{trans} = j] &= \sum_{m=1}^{M} P(C_m = 1 | N_i^{trans} = j) \\ &= M \binom{j}{1} \frac{1}{M} \left(1 - \frac{1}{M}\right)^{j-1} \end{aligned} \tag{3.13}$$

Therefore,

$$\begin{aligned} &\mathrm{E}[K_i | N_i = n] \\ &\quad = \sum_{j=1}^{n} P(N_i^{trans} = j | N_i = n) \binom{j}{1} \frac{1}{M} \left(1 - \frac{1}{M}\right)^{j-1} \\ &\quad = \sum_{j=1}^{n} \binom{j}{n} p_t^j (1 - p_t)^{n-j} \binom{j}{1} \left(1 - \frac{1}{M}\right)^{j-1} \\ &\quad = np_t \left(1 - \frac{p_t}{M}\right)^{n-1} \end{aligned} \tag{3.14}$$

By taking the derivative of (3.14) with respect to P_t, we have

$$\frac{\mathrm{d}}{\mathrm{d}p_t} \mathrm{E}[K_i] = n \left(1 - \frac{p_t}{M}\right)^{n-2} (1 - \frac{np_t}{M}) \tag{3.15}$$

When $M \geqslant n$, $\frac{\mathrm{d}}{\mathrm{d}p_t} \mathrm{E}[K_i] \geqslant 0$, meaning that the expected number of successful preamble transmissions is maximized when $p_t = 1$. In other words, no ACB check should be performed and back-off window W should be set to a minimum. Alternatively, if $M < n$, we let the derivative $\frac{\mathrm{d}}{\mathrm{d}p_t} \mathrm{E}[K_i] = 0$, this gives $p_t = \frac{M}{n}$. Thus, we

have

$$p_t^* = \min\left(1, \frac{M}{n}\right) \tag{3.16}$$

where P_t^* denote the optimal transmission probability under the condition. By substituting p_t, we have expected maximum number of success preamble transmission

$$\mathrm{E}^*[K_i] = \begin{cases} n(1-\frac{1}{M})^{n-1}, & \text{where} \quad M \geqslant n \\ M(1-\frac{1}{n})^{n-1}, & \text{where} \quad M < n \end{cases} \tag{3.17}$$

and corresponding collision probability

$$p_c^* = \begin{cases} 1-(1-\frac{1}{M})^{n-1}, & \text{where} \quad M \geqslant n \\ 1-(1-\frac{1}{n})^{n-1}, & \text{where} \quad M < n \end{cases} \tag{3.18}$$

By substituting (3.16) and (3.18) into (3.11), we obtain the optimal p_{acb} and W combination satisfying

$$\begin{aligned} &\frac{(2-2p_{acb}(1-(1-\frac{1}{M})^{n-1}))p_{acb}}{2+(W+1)p_{acb}^2(1-(1-\frac{1}{M})^{n-1})}+ \\ &\frac{2((W+1)p_{acb}+2)(p_{acb}(1-(1-\frac{1}{M})^{n-1}))}{(2+(W+1)p_{acb}^2(1-(1-\frac{1}{M})^{n-1}))(W+\frac{2}{p_{acb}}-1)} = 1, \quad M \geqslant n \\ &\frac{(2-2p_{acb}(1-(1-\frac{1}{n})^{n-1}))p_{acb}}{2+(W+1)p_{acb}^2(1-(1-\frac{1}{n})^{n-1})}+ \\ &\frac{2((W+1)p_{acb}+2)(p_{acb}(1-(1-\frac{1}{n})^{n-1}))}{(2+(W+1)p_{acb}^2(1-(1-\frac{1}{n})^{n-1}))(W+\frac{2}{p_{acb}}-1)} = \frac{M}{n}, \quad M < n \end{aligned} \tag{3.19}$$

Note that the back-off window $W \geqslant 1$, and when $W = 1$, it can be easily derived that $p_t = p_{acb}$. Therefore, for the existence of (3.19), $p_{acb} > \frac{M}{n}$, or otherwise the system can not achieve maximum successful preamble transmissions. Equation (3.19) shows the relationship between p_{acb} and W, in which case the random access channel is optimised under different device active numbers. In practice, a saturated condition rarely exists in M2M networks. However, studying the characteristics of

p_{acb} and W can help us better understand the RACH. In section 3.4, the theoretical optimal p_{acb} and W combination under the saturation traffic model is compared with the action taken by the DRL-based controlling algorithm.

3.2.3 Burst Traffic and Device Priority Configuration

To evaluate congestion solution proposals, 3GPP TR 37.868 [95] defines two more practical traffic models (see table 3.2) for the evaluation of network performance with M2M communications. This gives a guidance of how a network may experience in normal or extreme situations. Traffic model 1 can be viewed as a typical scenario where MTCDs access the network uniformly over a period of time, i.e. in a non-synchronized manner, while traffic model 2 gives an extreme scenario, where a large amount of MTCDs access the network in a highly synchronized manner, i.e., after an application alarm that activates them.

In traffic model 2, all UEs activate at time $0 \leqslant t \leqslant T$. The random access intensity is described by the distribution $p(t)$ and the total number of UEs in the cell is N, and then the number of arrivals in the i_{th} RAO is given by:

$$g(i) = [N \int_{t_i}^{t_i+1} p(t)dt]_I \tag{3.20}$$

where t_i is the time of the i_{th} RAO and the distribution $p(t)$ follows the Beta distribution:

$$p(t) = \frac{t^{\alpha-1}(T-t)^{\beta-1}}{T^{(\alpha+\beta-1)}\mathbf{B}(\alpha,\beta)} \tag{3.21}$$

$\mathbf{B}(\alpha,\beta)$ is the Beta function with parameters α and β.

In the RA procedure, the eNB does not know UEs' identifiers until their respective msg3 are successfully received. Therefore, to assign distinct RA resources with respect to distinct QoS requirements, we classify UEs into 2 categories: high

Table 3.2: 3GPP Traffic models

Characteristics	Traffic Model 1	Traffic Model 2
Number of MTCDs	1000,3000,5000,10000,30000	1000,3000,5000,10000,30000
Arrival dist.	Uniform dist. over T	Beta$(3, 4)$ dist. over T
Dist. period (T)	60 seconds	10 seconds

Table 3.3: Classification of MTC traffic

Class name	Application example	RA QoS requirement
High priority	Seismic alarm/E-care	<400 ms
Low priority	Consumer electronics	<1000 ms

priority UEs and low priority UEs, as shown in table 3.3, and different ACB factors and back-off windows are assigned to each category.

The high priority category contains public safety devices and healthcare applications etc., whose QoS is stringent. Such traffic features a low frequency of occurrence, extremely short delay constraints and high channel access rate requirements. This category of UEs are best represented by traffic mode 2, in which an unpredicted event may trigger thousands of MTCDs activating in a short period of time.

The low priority category includes consumer electronics, factory management sensors and delay tolerance devices such as smart meters. They feature looser delay constraints and lower channel access success rate requirements. Therefore, this category of UEs can be represented by traffic model 1, in which devices are activated uniformly without burst.

We evaluate network performance in a single cell environment, where high priority and low priority UEs co-exist. The network is subject to different access intensities: for high priority UEs, we consider that in each event triggered burst traffic, N_{high} UEs are activated and require access following a Beta$(3, 4)$ distribu-

tion in 10 seconds; low priority UEs' activation follows uniform distribution for the period with an activation rate of N_{low} in 10 seconds. In the network, high priority burst traffic seldom happens, while low priority UEs are activated all the time.

Following from [95], PRACH configuration index 6 is used in our work, which has 1 RAO every 10 ms with M = 54 out of the 64 available preambles are used for contention based RA. Under these conditions, the system offers 100 RAOs per second. Before each RAO, eNB broadcasts ACB factors, and back-off window as part of the system information. In each RAO, a UE, which is activated but not connected to the network, execute ACB check and RACH described in Fig. 3.2. When the total waiting time of a UE exceeds its RA QoS requirement, e.g., 400 ms for high priority and 1000 ms for low priority, the network is declared unavailable by the UE and the problem is reported to an upper layer.

We are interested in estimating the time it takes for the eNB to collect each UE's data. If a preamble is successfully decoded, the actual user data will then be transmitted without contention on PUSCH via scheduled transmissions that take a fixed time. Thus, the time to success in the RA procedure dominates the total uplink delay. We define delay D_i as the time between UE i being first activated and when it is successfully connected to the network after the RA procedure.

3.3 Algorithm Design

3.3.1 Deep Reinforcement Learning

In this section, we introduce reinforcement learning firstly, then deep Q-learning will be presented in detail.

Reinforcement Learning

Reinforcement learning is an important area of machine learning, concerned with how a software agent makes interaction with an environment, trying to take actions in the environment in order to reach optimal states that receive maximal rewards. In reinforcement learning, the environment is typically formulated as a Markov Decision Process (MDP). However, state space, explicit transition probability and reward function are not necessarily required. Thus, reinforcement learning can handle difficult situations that mimic real-world complexity.

Denote by $X = \{x_1, x_2, ..., x_n\}$ the state space, and $A = \{a_1, a_2, ..., a_m\}$ the action set. Based on the current state $x_t \in X$, the agent takes action $a_t \in A$ to the environment and observes new state $x_{t+1} \in X$ according to the transition probability $P_{x_t x_{t+1}}(a)$, as well as, the reward $r(x_t, a_t)$.

In order to act near optimally, the agent must reason about its long term consequences of its actions, thus the agent should not only consider about the immediate rewards, but also the future rewards. The future rewards are discounted with a discount factor $0 < \epsilon < 1$, representing the agent considers current reward more than future rewards. The aim of the reinforcement learning is to find an optimal policy $a^* = \pi^*(x) \in A$ for each state x that maximizes the cumulative discounted rewards over infinite time horizon. The cumulative discounted reward at state x is represented by the state value function:

$$V^{\pi}(x) = E\left[\sum_{t=0}^{\infty} \epsilon^t r(x_t, a_t) | x_t = x\right] \tag{3.22}$$

Because of the Markov property, the next state is only determined by current state and action, irrelevant to history state, the state value function following optimal policy π^* can be written as

$$V^{\pi^*}(x) = \max_{a \in A}\left[R(x, a) + \epsilon \sum_{x' \in X} P_{xx'}(a) V^{\pi^*}(x')\right] \tag{3.23}$$

where $R(x, \pi(x))$ is the mean value of immediate reward $r(x, \pi(x))$, and $P_{xx'}$ is the transition probability from x to x', when action is taken following optimal policy $\pi^*(x)$

Given the reward R and transition probability P, optimal policy π^* can be obtained.

Deep Q-Learning

When reward R and transition probability P are both unknown to the agent, Q-learning is a widely used method to determine the best policy π^*. The Q-value function $Q(s, a)$ is given as

$$Q(x, a) = R(x, a) + \epsilon \sum_{x' \in X} P_{xx'}(a) V^{\pi}(x') \tag{3.24}$$

which represents the cumulative disounted reward when action a is taken at state x, and follows policy π for subsequent states.

Then the cumulative state value function can be written as

$$V^{\pi^*}(x) = \max_{a \in A} \left[Q^{\pi^*}(x, a) \right] \tag{3.25}$$

From (3.25), we can see that by determining an accurate Q-value, we can find the best policy. To determine accurate Q-value, we need observations to train the function approximator which learns Q-value through recursive training. An experience is of format (x, a, r, x'), i.e., state x, action a, immediate reward r and next state x'. The Q-value function of each (x, a) is updated as

$$Q_{t+1}(x, a) = Q_t(x, a) + \alpha \left(r + \epsilon \left[\max_{a'} Q_t(x', a') \right] - Q_t(x, a) \right) \tag{3.26}$$

where α is the learning rate. The Q-value is usually estimated using a function approximator. In deep Q-learning, a deep neural network — deep Q-network (DQN) — is used as a function approximator to estimate Q-value based on the (x, a) pair.

The neural network $Q(x, a; \theta)$ has weights parameters θ adjusted at each iteration to reduce the error.

To train a Q-network, the input needs to be the current state x, and it needs to output the Q-value for each action a in that state. The output should approach, as training progresses, the real Q-value that accurately describes the environment. Therefore, the loss function is

$$Loss(\theta) = \Big(r + \epsilon \max_{a'} Q'(x', a') - Q(x, a)\Big)^2 \tag{3.27}$$

The Q-network is trained towards the target value $r + \epsilon \max_{a'} Q'(x', a')$ by minimizing the loss function $Loss(\theta)$ after each new experience (x, a, r, x') is fed into the network.

Training the Deep Q-Network

One of the challenges for DQN is that the Q-network used in the algorithm tends to forget the previous experiences as it overwrites them with new experiences. Thus, at each time t, the agent stores its experience (x, a, r, x') into a replay memory D. Then it will randomly sample experiences of the replay memories of size L to train the Q-network's weight parameters θ rather than using the consecutive samples as in Q-learning.

We apply an $\varepsilon - greedy$ policy to take actions. The agent will randomly select its action at first by a certain percentage, ε, called exploration rate. This is because at first, it is better for the agent to try all kinds of things before it starts to see the patterns. When it is not deciding the action randomly, the agent will predict the reward value based on the current state and pick the action that will give the highest Q-value. At the start of the training process the ε value is often initialized to a large probability, to encourage exploration in the face of knowing little about the environment. The value is then annealed down to a small constant, as the

agent is assumed to learn most of what it needs about the environment.

$$a_t = \begin{cases} \arg\max_a Q(x, a, \theta), & \text{with probability} \quad 1 - \varepsilon \\ \text{random action}, & \text{with probability} \quad \varepsilon \end{cases}$$

Next we apply the Boltzmann softmax approach to take actions [144]. Instead of always taking the optimal action, or taking a random action as in the $\varepsilon - greedy$ method, this approach involves choosing an action with weighted probabilities. To accomplish this we use a softmax function over the networks to estimate Q-value for each action. In this case the action, which the agent estimates to be optimal, is most likely (but is not guaranteed) to be chosen. The biggest advantage over the ε-greedy method is that information about the likely value of the other actions can also be taken into consideration. We use an additional parameter τ to represent temperature, which is annealed over time. This parameter controls the spread of the softmax distribution, such that all actions are considered equally at the start of training, and actions are sparsely distributed by the end of training.

$$P_t(a) = \frac{\exp(q_t(a)/\tau)}{\sum_n^{i=1} \exp(q_t(i)/\tau)} \tag{3.28}$$

The agent updates the target value every several steps, instead of every time step. The weights of Q-network θ is updated every N time steps. Such modification can make the learning process more stable. As deep reinforcement learning employs a neural network as a function approximator for the Q-value, it is able to deal with a large number of states, or even a continuous state space, which is very suitable for management of dynamic ACB under different situations.

3.3.2 Problem Formulation - Uniform Priority

In this section, we first formulate the problem of assigning a single dynamic ACB factor and a single BO window for every UE in the cell. According to (3.19), the system can easily derive the optimal p_{acb} and W based on the total number of active UEs requesting access n at each RAO. However, in practice, the eNB can not acquire the exact number of active UEs requesting access in the cell. The information it could obtain is limited to the number of successful preamble transmissions, the number of preambles collided and the time it took each UE to succeed in RA procedure when examining those newly connected UEs. Therefore, the eNB can only estimate the upcoming traffic based on such limited information. However, the obtained information indirectly indicates how crowded the random access channel is. If the success number is low while collided number is high, it normally indicates that too many UEs are contenting simultaneously, and the ACB is set too hight or BO window is set too low. On the other hand, when the success number is low while collided number is low, but the delay of successful UEs is considerably high, it's likely because too few UEs are granted access, usually a result of too low an ACB factor or too high a BO window. These are just two examples of typical situations, however, there are many more situations in the operation of RACH that need to be taken into consideration. Furthermore, access class barring and back-off scheme works together to optimise the efficiency of an RA procedure, but which combination gives best result in various situations remains a challenging problem. Mismatching of ACB factor and BO window can cause such a severe downgrade of performance.

In Ethernet and wireless LAN standards, Random access schemes use exponential back-off, which lengthens waiting times exponentially as the number of collisions increase. However, such mechanisms have a drawback. The UE has to

collide many times before it realises the system is overcrowded and lengthen the back-off time to a suitable value to the congestion condition. Those collisions are essentially a waste of RA resources in heavily loaded networks.

By contrast, if a dynamically adjusted back-off window could accurately reflect the current situation, all UEs would wait for a suitable time regardless of their history of collisions.

In our system, we assume all UEs in the cell have the same QoS requirement and they are uniformly treated by an eNB, consistent with section II-E. The eNB has two parameters that it can adjust to optimise the random access channel. ACB factor P_{ACB} determines the probability that a UE is allowed to participate in RA procedure. Backoff window W determines the maximum back-off time if a UE fails in RA procedure. The back-off time is uniformly distributed $T_{BO} = U(0, W)$.

The optimisation objective of the algorithm is to find the ACB factor and BO window pair (P_{ACB}, W) that maximize the number of UEs successfully accessed in the network through a RA procedure in each RAO.

Here, a central agent is deployed inside the eNB, responsible for acquiring RACH information, feeding the collected information to the Q-network. The Q-network outputs the Q-value of taking each action under the current state, and the optimal action $\arg\max_a Q(x, a)$ is taken. After obtaining an action, the agent adjust P_{ACB} and W and broadcast them before the next RAO. Then system will then transfer to a new state after next RAO, and the rewards can be obtained according to the reward function.

Inside the Q-network, the replay memory D stores the agent's experiences. After taking an action, the agent observes new state and reward, then stores the experience (x_t, a_t, r_t, x_{t+1}) into replay memory D. The agent updates weight parameters θ of the Q-network with N_B samples from the replay memory D every T

Algorithm 1: Deep Reinforcement Learning based ACB and BO controlling algorithm

Initialize replay memory D and network parameter θ
for episode $e = 1, ...E$ **do**:
    Initialize the beginning state x.
    for time $t = 1, ...T$ **do**:
        apply ε-greedy or boltzmann softmax approach to determine action a_t
        Execute action a_t, observe the reward r_t and the new state x_{t+1}
        Store the experience (x_t, a_t, r_t, x_{t+1}) in D
        Get mini-batch of samples of size N_B from replay memory D
        train the Q-network to fit with the sample of experience
        update target Q-network parameter θ^- every N time iterations.
    End for
End for

iterations to avoid oscillation. The $\varepsilon - greedy$ policy is utilized to balance the exploration and exploitation — trying random actions to obtain unknown knowledge versus taking the action that maximizes long-term rewards. The implementation of deep Q-Learning is described in **Algorithm 1**.

To obtain the optimal policy, state space, action space and reward function should be properly designed, which is illustrated in the following subsections

State space

The state space of the agent consists of 3 components, the number of UEs successfully accessed in the network through a RA procedure during the las RAO N_s, the number of preambles being detected as collided during last RAO N_c, and the average delay D_{avg} of successfully accessed UEs during last RAO.

$$x = \{N_s, N_c, D_{avg}\} \tag{3.29}$$

The number of possible states is very large. Traditional reinforcement learning is incapable of handling such a problem. However, a deep Q network is able to learn directly from high-dimensional inputs, thus it is proper to be used in our work.

Action space

The action space of the agent contains 2 components. The first one is an adjustment to the ACB factor. The second one is an adjustment to the BO window.

$$a = \{P_{ACB}, W\} \tag{3.30}$$

Reward

The reward needs to represent the objective of the algorithm, which is maximizing the total number of UEs successful in RA procedures under a delay constraint in each RAO. Thus, we define the immediate reward that the agent receives as the number of successful access under a delay constraint after performing action a in the upcoming RAO,

$$r = N_s' \tag{3.31}$$

3.3.3 Optimisation for Heterogeneous Priority

In this section, we present a deep Q-learning agent with respect to heterogeneous priority classes, as described in Section 3.2.3. In a network with high priority and low priority, different ACB factors and back-off windows should be applied to each of them, in order to meet their respective delay constraints.

The goal of the optimisation problem is to maximize the sum of weighted number of successful access in RA procedure of high and low priority.

$$\{P_{ACB}^{high}, W^{high}, P_{ACB}^{low}, W^{low}\} = \arg\max_{a} \left(5 \cdot N_s^{high} + N_s^{low}\right) \tag{3.32}$$

where N_s^{high} represents the number of high priority UEs that successfully complete RA procedure within a high priority delay constraint. N_s^{low} denotes the number of low priority UEs that successfully complete RA procedure within a low priority delay constraint.

The state space, action space and reward function for jointly optimising multiple classes of priorities are defined in the following subsections.

State space

The state space of a deep Q-learning agent consists of 4 components: the number of UEs of each category successfully accessed to the network through RA procedure during last RAO N_s^high and N_s^low; the number of preambles being detected as collided during last RAO N_C; the average delay of successful accessed UEs of each category D_{avg}^{high} and D_{avg}^{low}

$$x = \{N_s^{high}, N_s^{low}, N_c, D_{avg}^{high}, D_{avg}^{low}\} \tag{3.33}$$

Action space

We let the agent decide on the ACB factors and BO windows assigned to each category simultaneously. Specifically, the action of the agent is the adjustment to 4 parameters: ACB factor for high priority P_{ACB}^{high} and low priority W^{high}, BO window for high priority P_{ACB}^{low} and low priority W^{low}.

$$a = \{P_{ACB}^{high}, W^{high}, P_{ACB}^{low}, W^{low}\} \tag{3.34}$$

Reward function

The rewards need to represent the objective of the algorithm, which is maximizing the sum of weighted success number of high and low priority UEs. After exploring different weight parameters, we define the immediate reward that the agent receives as

$$r = 5 \cdot N_s^{high} + N_s^{low} \tag{3.35}$$

Then, with new state space, action space and reward function, the deep reinforcement learning based algorithm presented in **Algorithm 1** is modified to adjust 2 ACB factors and 2 BO windows simultaneously.

3.4 System Evaluation

We present numerical simulations to demonstrate the performance of proposed uniform priority and heterogeneous priority RACH optimisation algorithm. We use TensorFlow in our simulation as the backend to implement deep reinforcement learning. In this section, we give an introduction to TensorFlow, followed by simulation settings. Lastly, simulation results of both uniform priority of heterogeneous priority are discussed.

3.4.1 Simulation Settings

TensorFlow

TensorFlow is an open source software library for expressing machine learning algorithm and implementing such algorithms. TensorFlow has a variety of applications, including speech recognition and computer vision etc. Originally developed by Google Brain team, TensorFlow was later released under the Apache 2.0 open source license in 2015.

TensorFlow mainly provides Python API and it is able to run on multiple CPUs and GPUs for faster speed. TensorFlow can be implemented on a wide range of hardware platforms, from mobile devices to personal computers to large-scale systems running hundreds or thousands of GPUs.

The following functions available in TensorFlow are particularly of use here:

1. Experience Replay: It allows the Q-network to train itself using stored experiences (x_t, a_t, r_t, x_{t+1}) in the memory. which enables the Q-network to learn from more varied sets of past experiences. 2. A Second Target Network: A second network is used to compute the target Q-value during the training period. This avoids feedback loops between the target and estimated Q-value, which destabilizes the system.

Simulation Settings

In our simulation, the detailed parameters in the proposed deep reinforcement learning algorithm are listed in Table 4.1. The deep Q-network consists of 2 hidden layers. The first hidden layer has 24 hyperbolic tangent activation Unit, and the second layer has 48 hyperbolic tangent activation Unit [145]. In the simulation, we set the available preamble number $M = 54$. In DRL, the ACB factor can be adjusted in the range $\{0.01, 0.02, 0.03, 0.05, 0.05, 0.1, 0.15, ..., 1\}$, and BO window can

Table 3.4: Deep reinforcement learning algorithm parameters

Parameter	Value	Description
Hidden layer	2	Number of hidden layers
First hidden layer	24	Number of nodes in the first hidden layer
Second hidden layer	48	Number of nodes in the second hidden layer
Activation	tanh	Node activation: hyperbolic tangent activation function
Mini-batch size	32	How many experiences are used for each training step
Experience replay buffer size	2,000	Training experiences are randomly sampled from this number of recently experiences
Total training episodes	1000	How many episodes are used to train the network model
Training time per episode	1000	How many time steps are used to train the network model in each episode
Discount rate	0.95	Discount rate applied to future reward
Learning rate	0.001	The learning rate used by Adam Optimiser [146]
Initial exploration	1	The starting chance of random action in the $\epsilon - greedy$ exploration
Decay rate	0.9995	The speed of reduction of exploration rate
Final exploration	0.01	The final chance of random action in the $\epsilon - greedy$ exploration
Initial temperature	10	The starting temperature in the Boltzmann softmax exploration
Decay rate	0.9995	The speed of reduction of temperature in the Boltzmann softmax exploration
Final exploration	0.5	The final chance of random action in the Boltzmann softmax exploration

be assigned from the set $\{1, 3, 5, ..., 45\}$ (RAOs). We assume all the messages including preambles are transmitted without error, thus PHY layer problems, such as SINR outage, are ignored. Section 3.4.2 gives simulation results in saturation condition, while Section 3.4.3 tests the algorithm under burst traffic mode. In Section 3.4.4, we assume a heterogeneous priority scenario, and discuss the performance of proposed algorithm.

3.4.2 Saturation Traffic Model Simulation Results

In this section, we assume the network contains a fixed number of nodes in saturation condition, which means each node always has a packet to transmit. Each training episode contains 10 seconds of consecutive uplink resource requests from various number of UEs in the network. The simulation time remains to be 10s,

the same to traffic model defined in [95]. We present the simulation results of the DRL-based algorithm that assigns single ACB factor and BO window to all UEs with uniform priority.

Fig. 3.4 compares the convergence performance of the proposed scheme using both ε-greedy and Boltzmann softmax exploration methods in the deep reinforcement learning algorithm, and traditional tabular reinforcement learning. Boltzmann softmax exploration method involves choosing an action with weighted probability, and is introduced in Section 3.1.1. From Fig. 3.4, we can see that the total reward per episode is low at the beginning. As the learning process continues, more actions are taken and more experiences are observed and stored. With an increase in episode number, the total reward increases until it reaches a relatively stable value, which is around 19.795. We can also observe in Fig. 3.4 that the Boltzmann softmax exploration method has advantages over ε-greedy in terms of convergence speed. It can also be seen that both DRL methods converge to the optimal throughput much faster than tabular RL, and have a much higher success number when they converge. As a result, in practice, DRL is more favorable than traditional tabular RL for its faster convergence speed and higher throughput upon convergence, indicating more precise actions are taken.

Fig. 3.5 shows the plot of W as a function of p_{acb} given in (3.19) for various number of nodes. Each curve represents the potential (p_{acb},W) pairs that optimise the RA efficiency under a given number of nodes. After the DRL-based algorithm is applied to the RACH in saturation mode, the throughput converges and a steady state is reached, in which the actions (p_{acb},W) taken by the DRL agent are shown in Fig. 3.5. From it, we can see that DRL-based algorithm chooses the point with $p_{acb} = 1$ in the theoretical optimal curve after converge in steady state. We can draw the conclusion that our proposed algorithm achieves optimal efficiency in the

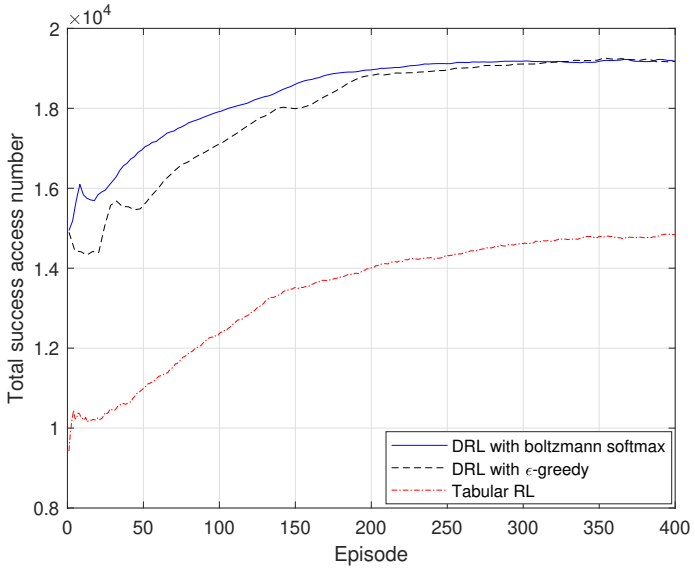


Figure 3.4: Convergence speeds of different learning methods when active number is 100

RACH in steady state, and is still robust under a massive number of active UEs.

3.4.3 Burst Traffic Mode Simulation Results

In Section 3.4.2, we have proved that the proposed DRL-based ACB factor and Back-off controlling algorithm can achieve optimal RA success number per RAO in saturation conditions. In this section, we set the system in practical environments and simulate for the burst traffic model [95] given in Table 3.2. During the simulation period of 10 seconds, UEs are activated according to a Beta$(3, 4)$ distribution, with total number of 20,000.

Fig. 3.6 shows the number of UEs that are successfully connected to the eNB through a RA procedure in each random access opportunity (RAO). Our proposed DRL algorithm controlling both ACB factor and BO window is compared with the scheme where the ACB factor is controlled by a DRL agent while the BO window is either fixed to 1 or binary exponentially increased, and the scheme that ACB is

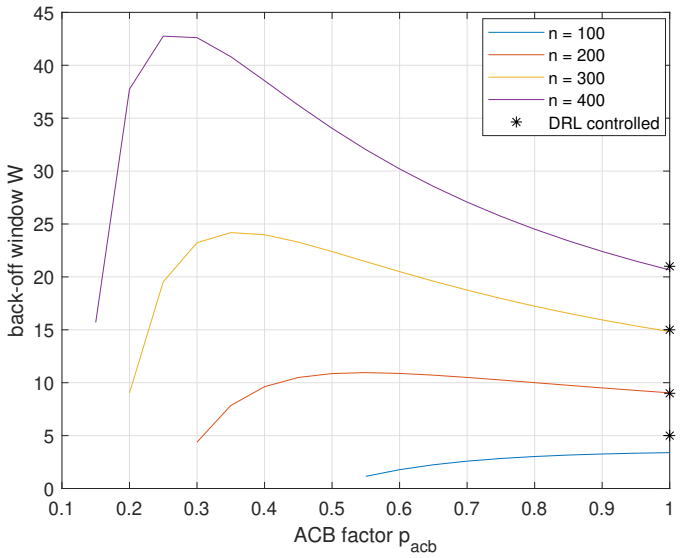


Figure 3.5: Theoretical optimal (p_{acb},W) curve and action taken by DRL algorithm in steady state

not used but exponential BO window is applied. Note that exponential back-off is used in Ethernet and wireless LAN standards, but is not currently supported in LTE-M. We see exponential back-off as a state-of-art back-off scheme that our proposed algorithm can compare with.

The theoretical maximum number of success number per RAO derived from (3.17) is also shown. It can be seen that the proposed algorithm achieves the highest success number over the simulation period, very close to the theoretical maximum. It is because as the learning is converged, the agent is able to find optimal (p_{acb}, W) pair for every state, which consequently gives optimal RA throughput close to the results obtained in saturation mode – 20 success access per RAO on average. The By contrast, when the ACB factor is controlled by the DRL agent while the back-off window is fixed, the success number is less than that of the proposed scheme as the agent may not be able to find the most suitable ACB factor for the fixed window to ensure an optimal RA.

In the second scheme, access class barring is applied and the ACB factor is controlled using deep reinforcement learning. As a result, the congestion during the peak activation period was alleviated through regulating the number of accesses in each RAO. However, in the early phase of massive activation, the scheme handles the situation poorly, as collided UEs tend to re-attempt the RA procedure very swiftly under exponential BO. Therefore, a large amount of UEs still overload the RACH.

The scheme only using an exponential BO performs worst, as expected, due to the fact that when a large number of UEs wish to access the eNB, all of them are allowed to participate in the RA procedure. As a result, very few of them can actually transmit a non-collided preamble. The exponential BO helps alleviate the congestion as the BO window is increased exponentially. But the scheme only helps when obvious congestion has already taken place.

In our proposed algorithm, both the ACB factor and BO window are controlled by the deep reinforcement learning agent. The algorithm is able to assign an appropriate BO window in all phases of congestion to avoid re-attempt wastage. The ACB factor is also adjusted jointly with the BO window, to keep the efficiency of the RA procedure very close to the theoretical maximum.

Fig. 3.7 shows the average delay of successful accessed UEs per random access opportunity (RAO), under different schemes. The delay is calculated as the time between the device is first activated and the time it successfully finishes the RA procedure. It can be seen from the figure that scheme of DRL controlled ACB with back-off window fixed to 1 has highest delay in peak traffic periods. It is shown that the delay under the DRL controlled ACB and BO scheme is larger than the other two schemes. The reason is that the BO window is set much higher than the other two schemes — the BO window can be set directly up to 45 RAOs, while the

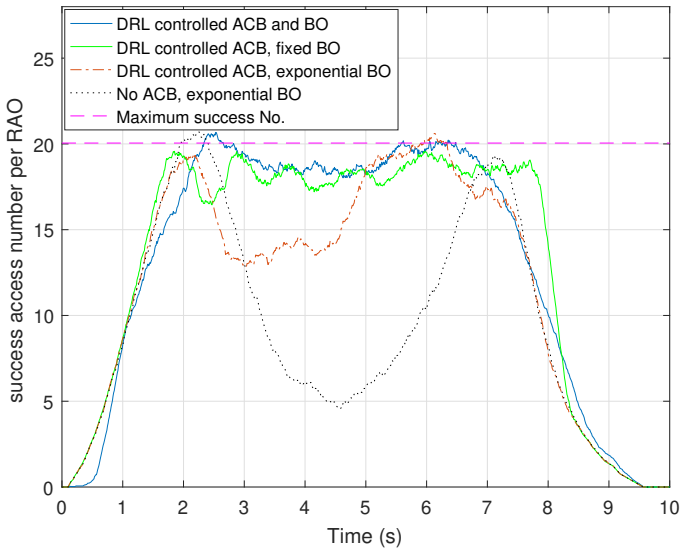


Figure 3.6: Comparison of success access per random access opportunity (RAO) with different control methods

exponential BO window starts at 1 RAOs. However, the benefit of increased delay is higher RA efficiency and more UEs being able to succeed in the RA procedure under a delay constraint.

Figure. 3.8 compares the total number of UEs successes in the RA procedure of different schemes during the 10 second period under different UE activation intensity. In [101], the author propose a UE arrival number prediction method using markov chain, and adjusts ACB factor accordingly. As the total activation number increases, the success number of the proposed algorithm using DRL controlled ACB and BO approaches the maximum. We can see that even the activation number is far beyond the system's capability, and there are no downgrades to the system RA's efficiency. However, the scheme with fixed BO perform a little worse than the scheme with DRL controlled BO because as simultaneous active number increases, it is getting harder to find the suitable ACB factor for BO window fixed to 1. The scheme in [101] is least reliable and the performance downgrades when active

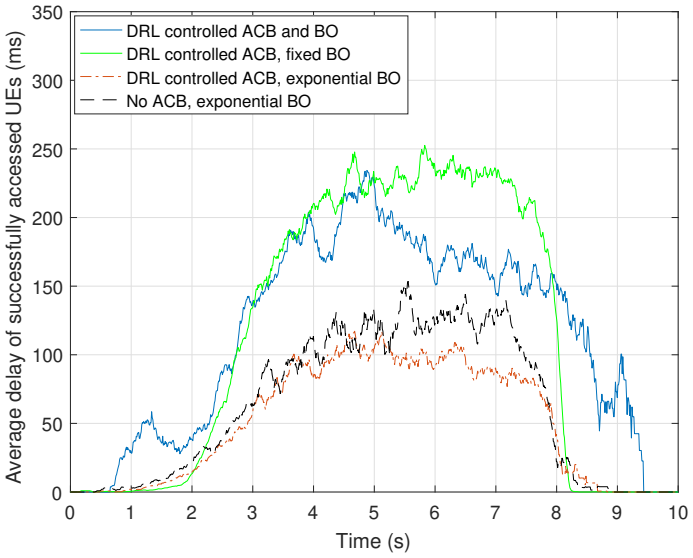


Figure 3.7: Comparison of average delay per random access opportunity (RAO) with different control methods

number get larger, due to reason that the prediction gets less accurate at larger arrival numbers. Besides, even the arrival number is accurately predicted, the ACB factor adjustment algorithm is not designed to achieve optimal RA efficiency, especially at large arrival numbers.

3.4.4 Heterogeneous Priority Simulation Results

In this section, we present the simulation results of assigning different ACB factors and BO windows to different classes of UEs according to their respective delay constraints and congestion status in burst traffic mode. In the simulation, low-priority and high-priority class MTCDs co-exist in the coverage area of the eNB. During the simulation period of 10 seconds, 10 low priority UEs are activated at each RAO, hence 10,000 in total. High priority UEs are activated according to a Beta$(3, 4)$ distribution, with total number of 15,000.

Fig. 3.9 shows the number of the two priority classes of UEs that successfully

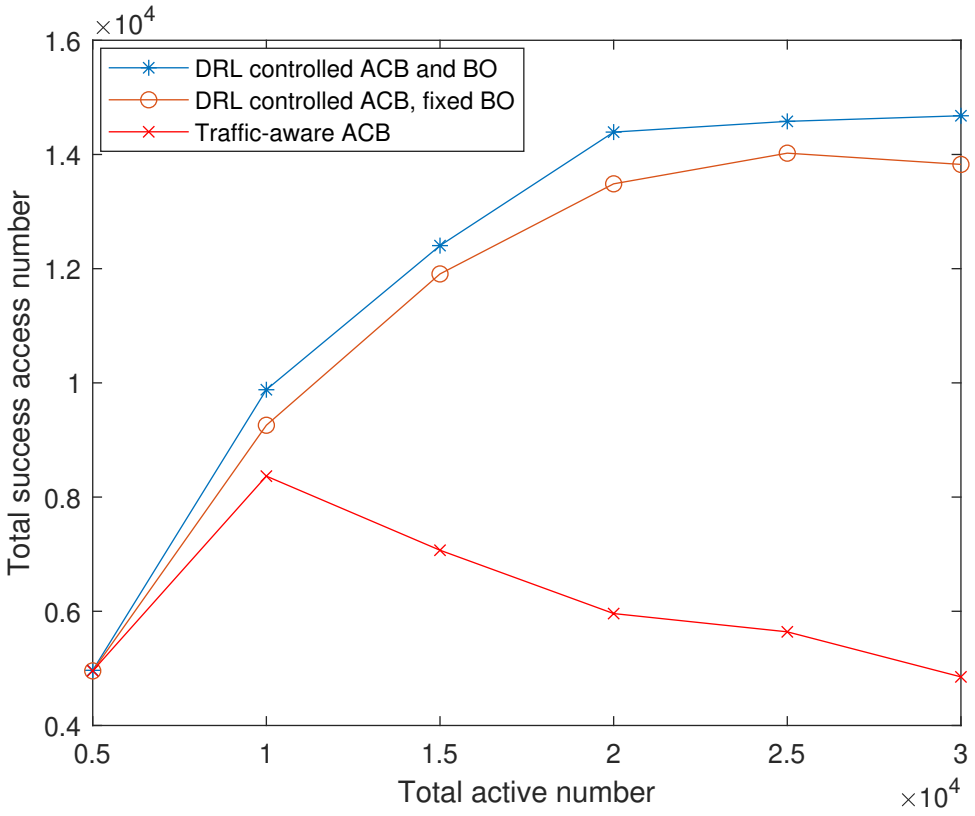


Figure 3.8: Comparison of total success access number with different control methods

connected to the eNB through a RA procedure in each RAO, using DRL-based ACB and BO controlling algorithm. The theoretical maximum number of successes per RAO derived from (3.17) is also shown. It is shown from Fig. 3.9 that when burst activation of high priority UEs happens, they receive a higher ACB factor and lower BO window, allowing them to succeed faster in the RA procedure. At this time, many low priority UEs are rejected due to a low ACB factor assigned to them, ensuring better performance of high priority UEs. Although the access of high and low priority classes are controlled separately, their combined success number stays very close to the maximum value, which ensures RACH's efficiency is maximised. This indicates that the best effort services are given to low priority devices as long

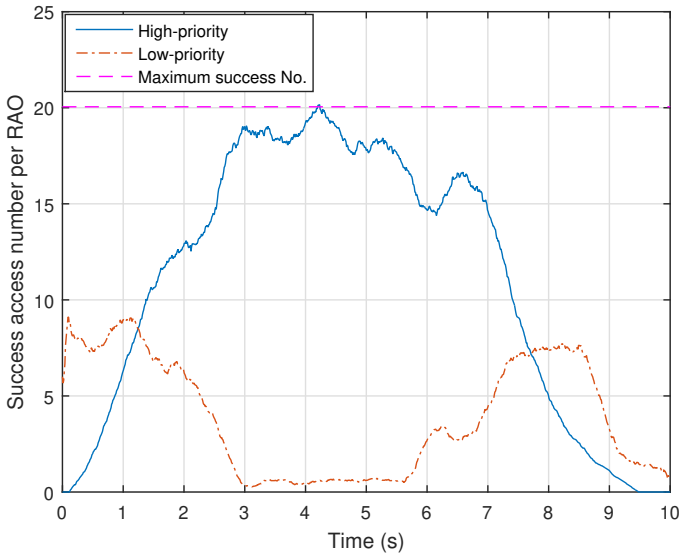


Figure 3.9: Comparison of success access per random access opportunity (RAO) of high and low priority class

as high priority services are not impeded. Fig. 3.10 compares the average delay of successfully accessed UEs at each RAO of two priority categories. We can see that the delay of the high priority classed was maintained as extremely low even in burst traffic. However, the delay of the low priority class is purposely sacrificed due to a lower ACB factor and a longer BO window.

Then, we run the simulation under different activation intensity. Of the total number of activations, 2/5 of them are low-priority UEs uniformly activated in 10s, and 3/5 of them are high-priority UEs activated following a Beta$(3, 4)$ distribution. Fig. 3.11 shows the number of successful access per RAO of high and low priority classes. Our proposed DRL-based algorithm is compared with the Multiple Access Class Barring scheme in [1]. Under a very low activation intensity, e.g., 5000 and 10000 devices are activated in 10s, both high and low priority UEs have close to 100% success probability. As activation intensity grows, the success probability of the low priority class drops firstly. At intensity above 20000 activations in

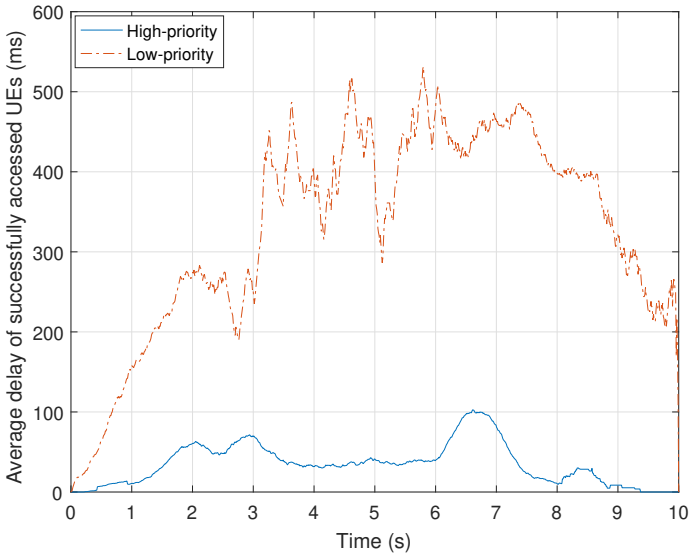


Figure 3.10: Comparison of average delay per random access opportunity (RAO) of high and low priority class

10s, the RACH at peak traffic is saturated, and the success probability of high-priority is downgraded. Compared with the MACB scheme, our proposed DRL-based algorithm achieves a higher success probability for high priority UEs under high intensity activation.

Fig. 3.12 compares our DRL-based algorithm and the MACB scheme in terms of the average delay of high and low priority users. For our proposed algorithm, we can see the average delay of high-priority UEs slowly increases as activation intensity grows. However, the delay of low-priority UEs is largely affected by the activation intensity due to the fast decrease in ACB factor and the fast increase in BO window. The average delay of the low priority class is up to 5 times more than that of the high-priority class in extreme cases where 30,000 UEs are activated in 10s. On the other hand, in the MACB scheme, the average delay of high priority UEs increases significantly with the increase of activation intensity, resulting in up to 2 times that of our proposed DRL-based algorithm.

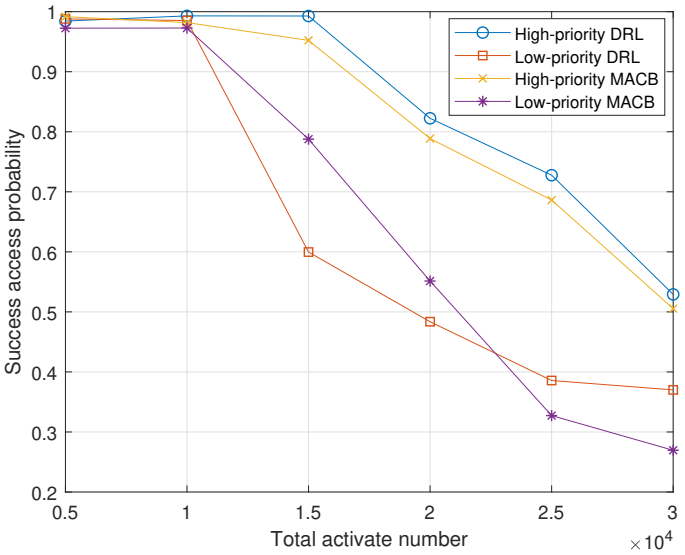


Figure 3.11: Comparison of successful access per random access opportunity (RAO) of high and low priority class under different activation intensity (comparison with MACB [1])

3.4.5 Discussion regarding distributed deployment

In the proposed algorithm, the DRL agent is deployed at the eNB, controlling the ACB factor and BO window to be broadcast to UEs. However, the idea of deploying a machine learning agent at UEs has emerged [147]. At the current stage, training DRL agents directly inside MTCDs so that they can choose parameters individually is not, as yet, a good option, because: Firstly, the cost of deploying machine learning at each UE is huge, as capable processing units are needed to perform basic machine learning, and the algorithm is designed for massive MTC thus the cost of the distributed deployment could be 100s to 1000s times of centralized deployment of machine learning. Secondly, due to the characteristics of machine learning, more information leads to better performance. The learning agent at UEs does not have global information about the system, nor is there an efficient way to

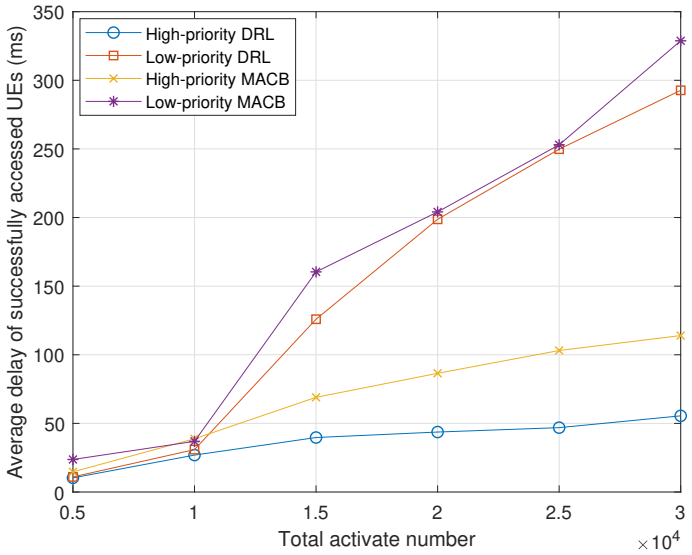


Figure 3.12: Comparison of average delay per random access opportunity (RAO) of high and low priority class under different activation intensity (comparison with MACB [1])

exchange the information among UEs. Therefore, training UEs individually is of very low efficiency. Finally, as network standards keep evolving, maintenance and upgrading of protocols and algorithms are much more convenient for eNBs rather than for a massive number of UEs/machine type communication devices. Thus, we believe at this stage deploying the DRL agent at the eNB is better than deploying at UEs.

3.5 Concluding Remarks

In this chapter, the random access network overload issue in cellular networks has been addressed. A deep reinforcement learning based access class barring (ACB) & back-off (BO) controlling algorithm for solving random access channel (RACH) congestion problem in an LTE-M network has been developed, which is likely to be

applicable to future cellular networks, in particular for massive machine-to-machine (M2M) communications. The algorithm allocates ACB factors and BO windows to each different priority classes of machine-type communication devices (MTCDs) in order to satisfy their respective QoS requirements. Through simulation of burst MTCD activation, it has been shown that the proposed algorithm assigns ACB factors and BO windows that achieves maximum RA efficiency in both uniform and heterogeneous priority networks. Furthermore, it achieves higher success probability and shorter delay for delay-sensitive MTCDs, compared with other schemes. With careful tuning, DRL-based algorithm could be applied to other key aspects of cellular networks optimisation. As processing power and learning algorithm advances, it could be much easier and cost-efficient to deploy learning agents in mobile devices. In the next chapter, we will investigate relaying in cellular networks to reduce end-to-end delays for data transmission of MTCDs.

Chapter 4

Minimum-Delay Relaying in mmWave Uplink

In chapter 3, we presented a DRL based RACH optimisation algorithm for a massive number of connections. However, as the proposed algorithm solves an accessing and synchronisation problem for cellular IoT networks, there is still a big challenge in optimising data transmission due to the emerging massive amount of data generated from IoT devices. Millimeter Wave (mmWave) communication will play an essential role in future cellular networks as the abundant spectrum available at mmWave band is capable of delivering high transmission speeds and capacity. Because the high attenuation and low penetration capability of mmWave constrains its transmission range, multi-hop relaying is seen as a promising tool to extend network coverage and capacity. In this chapter, we investigate the delay performance in mmWave uplink relaying networks, and present a delay-oriented relaying scheme, which automatically allow user equipments (UEs) to choose relays based on network topology and congestion conditions aiming to minimise estimated delay.

4.1 Introduction

4.1.1 Background

Machine-to-machine (M2M) communications, also known as Machine-Type Communications (MTC), feature a wide range and large number of autonomous Machine Type Communication Devices (MTCDs) communicating with little or no human intervention, and are an essential part of the internet of things (IoT) and future wireless systems. To satisfy versatile massive M2M traffic characteristics, from best effort applications like water/gas metering systems and environmental monitoring, to ultra-reliable ones such as healthcare, public safety and mission-critical industry, enhancing cellular networks is a promising means to accommodate such heterogeneous requirements. 5G aims at using ultra-high frequency - mmWave spectrum - to provide ultra-high data rate, ultra-low latency whilst supporting even more devices. 5G is a promising, high-efficiency solution to future IoT that not only provides internet access to billions of autonomous devices, but also meets key requirements such as low power and large coverage area [148].

Wireless communications in mmWave bands (from 30 GHz to 300 GHz) have attracted much attention from academic community and industry for its potential to fulfil multi-gigabits per second (Gbps) transmissions in future mobile broadband access [149] [150]. Due to its many appealing properties, such as abundant spectrum bands and ultra low delay, 3GPP has included mmWave spectrum in NR operating bands in FR2 for release 15, also known as 5G phase 1. Phase 1 defines a period of research for frequencies under 40 GHz to address the more urgent subset of the commercial needs, while phase 2 (release 16) will focus on frequencies up to 100 GHz.

However, mmWave communication suffers from weak diffraction ability and

high penetration loss through both air and obstacles. For example, the mmWave propagation measurements show that tinted glass has a penetration loss of 40.1 dB, while brick pillars have loss of 28.3 dB, at 28 GHz frequency [151].

One solution to the blockage problem in mmWave communications is to allow intermediate relay nodes to forward traffic generated in transmitters to destination receivers. Since the additional routes provided by relay nodes can detour around obstacles, the previously blocked transmission links can be replaced by two-hop Line-of-Sight (LOS) transmission links. Such improved channel quality will increase mmWave coverage and shorten transmission delay.

Low latency is a crucial metric for evaluating the quality of service in future 5G mobile communications [152]. Maintaining low latency is essential for various delay-sensitive applications such as video conferencing, vehicle-to-everything(V2X), E-healthcare, real-time remote control and especially, industrial M2M communication. The end-to-end delay (EED) mainly consist of 4 components as follows [153]: propagation delay (sending data through medium), transmission delay (pushing the data into medium), queueing delay (the time packet spends in the buffer) and processing delay (period to process the packet and determine routes). In these four components, queueing delay is the dominant part, while other delays are negligible [128]. Therefore, queuing systems are widely utilised to evaluate delays in mmWave networks [126–128, 154]. For buffer-aided relaying systems, a principle task is to minimize the queueing delay.

In recent years, cooperative transmission based on buffer-aided UE relaying has been proven to be a promising approach to improve transmission efficiency and extend cellular coverage [117, 118]. Many uplink aggregation schemes have been proposed, either using additional aggregations or existing UEs as relays [119, 120]. However, there is only limited research on relay-assisted mmWave cellular

communications [121,122]. Various multi-hop relaying scheme have been proposed, with focuses on Quality of service (QoS) [123], URLLC [126] or power budget [128]. A complete review of related works is given in chapter 2.3.2.

Base stations (BSs) are commonly assumed to be arranged in hexagonal or square grids in 3GPP evaluation. However, grid models are often too ideal because BS deployments are constrained by terrain and obstacles, and, furthermore, such models lack tractability. In recent years, stochastic geometry has gained popularity as a powerful tool for modeling/evaluating wireless networks [155, 156]. This is because stochastic geometry leads to a mathematically tractable framework, and the positions of BS, UE and obstacles in real life often resemble point processes [156, 157]. Lin et al. [2] use stochastic geometry to analyse the connectivity of mmWave networks with multi-hop relaying, and they derive the analytical probability that there exists a path between source and destination potentially including relays. It is shown that multi-hop relaying can improve connectivity, but both latency (delay) and effective capacity are not considered. In this chapter, we take steps towards filling the gaps in prior works by analysing latency and delay outage probability in mmWave uplink multi-hop relaying system, and show the improvement of our proposed Minimum-Delay relaying scheme over a state-of-art max-SINR scheme.

4.1.2 Contribution

The main objective of this chapter is to investigate the delay performance on a mmWave uplink multi-hop relaying cellular network. Considering buffer-aided mmWave relaying systems, the delay performance is studied in terms of average delay, outage probability and Jain's fairness, while network efficiency is investigated in terms of effective capacity and spectral efficiency. More precisely, we propose a delay-oriented relaying scheme, named Minimum-Delay relaying, which

automatically allow UEs to choose relays based on network topology and congestion condition aimed to minimise estimated delay. Furthermore, to characterize the delay, we derive corresponding estimated EED by applying stochastic geometry and Lagrange optimisation, and compare against real world PPP-based simulation. Then, Minimum-Delay relaying is compared with a state-of-art max-SINR relaying scheme and a direct association scheme in terms of delay performance and network efficiency. The main contributions are:

- To the best of our knowledge, mmWave uplink data aggregation delay performance has not been investigated previously, and thus the QoS of applying mmWave in large scale M2M is not clear up till now. In this chapter, we comprehensively investigate the delay performance of a state-of-art max-SINR relaying scheme. We derive expected end-to-end delay performance of this relaying scheme by applying stochastic geometry.

- Furthermore, we propose a uplink relaying scheme named "Minimum-Delay" relaying, i.e., a delay-oriented relaying scheme which minimizes each UE's EED and improves the overall network uplink efficiency. By applying lagrange optimisation, we derive the expected EED performance of proposed scheme.

- We extend our Minimum-Delay relaying scheme to a network architecture with three tiers, allowing a maximum of 3-hop transmission. We derive the expected EED of each tier devices, and compare the analytical results with numerical simulations. The results show a 3-tier architecture has a distinct advantage over 2-tier architecture under low data arrival rate, but performs worse under higher traffic load.

- We investigate the delay performance of both noise-limited and noise+inteference mmWave channels. According to numerical results, the

interference can be ignored for high obstacle densities.

- We show that our Minimum-Delay relaying scheme performs better than a direct association scheme and the max-SINR relaying scheme in the four key metrics of average delay, delay outage probability, effective capacity and spectral efficiency.

- We validate our analytical results against simulations based on 3GPP network evaluation scenarios and channel models, and a real world urban building map. We then explore the effect of various parameters, such as average data arrival rate and BS density, on the performance of relay-assisted mmWave networks.

4.2 System Model

4.2.1 Geometric Assumption

Poisson Point Processes

In this chapter, we model the spatial location of MTCDs and obstacles as 2D homogeneous Poisson Point Processes (PPPs) [155]. A PPP defined in $\mathbb{R}$ is a random process with number of points Φ in a bounded Borel set $\mathcal{B} \subset \mathbb{R}^2$

$$\mathbb{P}(\Phi(\mathcal{B}) = k) = \frac{\Lambda^k}{k!} e^{-\Lambda}, k = 0, 1, 2, \tag{4.1}$$

where $\Lambda = \lambda v_2(\mathcal{B})$ is the expectation of the Poisson random variable for some intensity λ and $v_2(\cdot)$ denotes the Lebesgue measure in $\mathbb{R}^2$. If λ is constant, the PPP is said to be homogeneous.

Obstacles

In general, obstacles have arbitrary shapes. In outdoor urban environments, obstacles are commonly buildings, which can be approximately modeled in 2D as rectangles or polygons. To enable tractability, here we model the obstacles as cylinders distributed in a 2D homogeneous PPP Φ_o with intensity λ_o.

Machine Type Communication Devices (MTCDs)

We model the locations of BSs as a 2D homogeneous PPP Φ_b with density λ_b . We also model the locations of MTCDs as the distribution of 2D homogeneous PPP Φ_m with intensity λ_m, and all MTCDs in the cell are candidate relays, which have the ability to relay other MTCDs' packets. Since the candidate relays and obstacles are homogeneously distributed PPPs, we focus our analysis on a typical BS or MTCD, which we assume is located at origin o. This is permissible in a homogeneous PPP by slivnyak's theorem [158].

4.2.2 mmWave Beamforming

In M2M communications, mmWave systems leverage directional beam forming to compensate for high path loss and minimize inter-cell interference. Here, we consider a simple sectored antenna array model for both mmWave transmitters and receivers [159]. The main lobe gain and side lobe gain of the transmitter antenna are denoted by G_m and G_s, respectively. For a link, if the receiver is located in the main lobe of the signal pattern, the beamforming gain at the transmitter side is G_{m_t}; otherwise, if the receiver is covered by the side lobe, the beamforming gain of transmitter is G_{s_t}. Similarly, we define G_{m_r} and G_{s_r} as the main lobe gain and side lobe gain at the receiver side. With an assumption of an $N \times N$ uniform planar square antenna array with half-wavelength antenna spacing, the main lobe gain

and side lobe gain are given as [160]:

$$\theta = 1.732/N, \quad G_m = N^2 \quad and \quad G_s = 1/\sin^2\left(\frac{3\pi}{2N}\right), \tag{4.2}$$

where θ denotes the half-power beamwidth of the main lobe and is determined by antenna array property.

The antenna array gain g from an arbitrary transmitter to an arbitrary receiver is

$$g = \begin{cases} G_{m_t}G_{m_r}, & \text{with} \quad \text{probability} \quad \frac{\theta_t}{2\pi}\frac{\theta_r}{2\pi} \\ G_{m_t}G_{s_r}, & \text{with} \quad \text{probability} \quad \frac{\theta_t}{2\pi}(1-\frac{\theta_r}{2\pi}) \\ G_{s_t}G_{m_r}, & \text{with} \quad \text{probability} \quad (1-\frac{\theta_t}{2\pi})\frac{\theta_r}{2\pi} \\ G_{s_t}G_{s_r}, & \text{with} \quad \text{probability} \quad (1-\frac{\theta_t}{2\pi})(1-\frac{\theta_r}{2\pi}). \end{cases} \tag{4.3}$$

Note that the beam forming gain on the desired cellular link is always the on-boresight gain $g = G_{m_t}G_{m_r}$. For tractable reasons, we assume that the interferer's antenna boresight is uniformly distributed over $(0, 2\pi]$.

4.2.3 Line-of-sight (LOS) Probability

Let the random variable $H \in [h_{min}, h_{max}]$ denote the obstacle height with probability density function (PDF) $f_H(h)$, and let h_{T_x} and h_{R_x} denote the transmitter and receiver heights, respectively. According to [2], the LOS probability between two communicating nodes separated by planar distance d is

$$p_L(d) = qe^{-\beta d}, \tag{4.4}$$

where $q = e^{-\eta\lambda_o\pi E[R^2]}$ and $\beta = 2\eta\lambda_o E[R]$, random variable R denotes the obstacle radius and $\eta = 1 - \int_0^1 \int_{h_{min}}^{sh_{R_x}+(1-s)h_{T_x}} f_H(h)\mathrm{d}h\mathrm{d}s$ is the obstacle thinning factor, λ_o denotes the obstacle intensity. Note that η may be different for direct links to BS and relay links because receivers have different antenna heights. We denote these different parameters as η_b and η_m, respectively.

4.2.4 Channel Model

In this chapter, we consider the following path loss model for both LOS and none-line-of-Sight (NLOS) links.

$$PL(d) = A_1 + A_2 \log_{10}(d) + A_3 \log(f_c), \tag{4.5}$$

where d(m) is the distance between the transmitter and receiver, f_c(GHz) is the carrier frequency, A_1 is the intercept, A_2 includes the path loss exponent, A_3 describes the path loss frequency dependence. The values of A_1, A_2 and A_3 are specified in the 3GPP New Radio Study work group [161], for both LOS and NLOS channels.

In the mmWave spectrum, due to high attenuation, multi-path effects are negligible. At 60GHz, for example, the channels closely match an Additive White Gaussian Noise (AWGN) channel. Thus, multi-path fading will not be considered in the channel model. The received signal power over a mmWave link of length d is

$$P_{R_x} = P_{T_x} g / PL(d), \tag{4.6}$$

where g denote the antenna array gains at both transmitter and receiver; P_{T_x} denotes the transmit power.

The received mmWave signal will experience interference from nearby inter-cell transmitters operating in the same frequency band.

$$SINR = \frac{P_{R_x}}{\sigma^2 + \sum_{i>0} B_i g_i PL(d_i)}, \tag{4.7}$$

where B_i is a Bernoulli random variable with parameter $PL(d)$, and d_i is the distance between the receiver and respective interferer. The noise power is incorporated in σ^2.

4.2.5 Machine Type Communication Devices

We assume all MTCDs in the area are candidate relay, so that every MTCD can gather data packets from other devices and forward them to either another relay or associated base station. Each MTCD is equipped with a data buffer of finite size. The size of data that are stored in the relay buffer is denoted by $L(L \geqslant 0)$. The buffer is increased by the size of a data message when a message is successfully received by the relay and is decreased by the size of message data it successfully transmitted to another MTCD or base station. When the size of data stored in the relay buffer reaches buffer size, the relay stops accepting new messages and drop any newly received messages until more space is freed up.

4.2.6 End-to-end Delay (EED)

The EED over a path is the summation of delays experienced by all the hops along the path. Here we only consider the delay incurred by transmission and queueing. In order to compute the EED metric over a wireless channel, each MTCD needs to monitor the queueing delays at each relay MTCD, as well as its own transmission time to potential relays. For example, the EED between MTCD n and base station, consists of all queueing delays experienced at each relay over the link and transmission time from MTCD n to its associated relay,

$$EED_n = \mathrm{E}[\text{transmission delay} + \text{total queueing delay}]. \tag{4.8}$$

Here we suppose the transmission delay between relays and between relays and BSs are negligible, because those links are very typically line of sight channels and the transmission rates are normally very high.

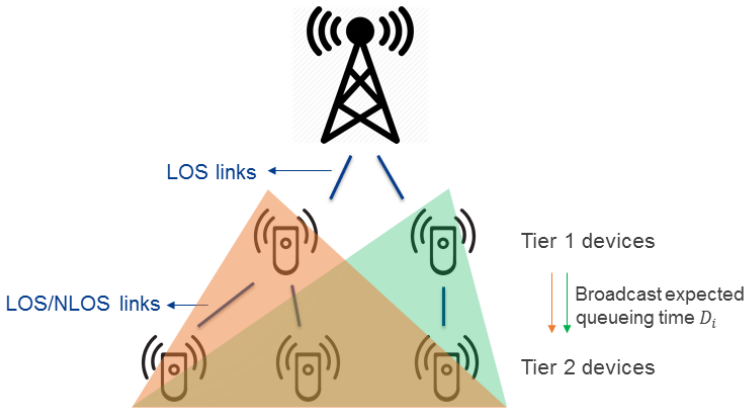


Figure 4.1: System diagram of 2-tier relay selection model.

4.3 Scheme Design and Analysis

4.3.1 Minimum-Delay Relaying Scheme

In this section, we present the multi-hop relaying scheme by optimising EED of each MTCD in the cell. The relay selection model, shown in Fig. 4.1, is as follows:

First Tier Relays

If a MTCD i has LOS communication with the BS, it is considered a candidate relay, and the set of all LOS candidate relays is denoted by $\mathcal{I}$. The relay calculates the expected time to upload all the data in its buffer to the associated BS based on link capacity

$$D_i = \frac{L_i}{c_i}, \tag{4.9}$$

where L_i is the queue length (bits) of MTCD i and c_i is the link capacity (bits/second) from MTCD i to BS (Shannon's capacity),

$$c_i = B \log_2(1 + SINR_i), \tag{4.10}$$

where B is denoted as bandwidth, and $SINR_i$ represents the signal-to-interference-plus-noise ratio measured between i and associated BS. And then, the relay i broadcasts the expected queueing time D_i, as an estimation of minimum waiting time a newly received data packet can be forwarded by this node.

Second Tier MTCDs

If MTCD n receives a broadcast D_i from potential relay MTCD i, it measures the channel between itself and the broadcasting MTCD i. We denote by $D_{n,i}$ the expected transmission time to upload all buffered data between the MTCD and potential relay

$$D_{n,i} = \frac{L_n}{c_{n,i}}, \tag{4.11}$$

where L_n is the queue length of MTCD n, $c_{n,i}$ denotes the expected transmission rate between MTCD n and relay i from (4.10). Then, the estimated EED between MTCD n and base station when associating with relay i is given as

$$EED_{n,i} = D_{n,i} + D_i. \tag{4.12}$$

At every association window, a MTCD receives multiple service times from different potential MTCD relays. Thus, the estimated EED through those candidate relays can be calculated. The MTCD n selects the relay that minimise its estimated EED

$$i = \arg\min_i(EED_{n,i}), \quad i \in \mathcal{I}. \tag{4.13}$$

4.3.2 Minimum-Delay relaying steady-state analysis

The distance between the MTCD and base station is denoted by d. As noted before, denote by $\mathcal{I}$ the set of LOS MTCDs to base station and u the number of LOS

MTCDs to base station, then $\mathcal{I}$ follows an inhomogeneous PPP of $\lambda_{LOS} = \lambda_m \cdot p_L(d)$, with distance d of MTCD to the nearest base station.

Lemma 1 *Based on the system model described in Section III, the expected distance of i_{th} nearest LOS MTCD to BS is given as*

$$\mathrm{E}[d_{LOS}(i)] = \int_0^\infty \frac{2(\pi\lambda_{LOS}(d))^i}{(i-1)!} d^{2i} e^{-\pi\lambda_{LOS}(d)d^2} \mathrm{d}d, i \in \{\mathcal{I}\}, \tag{4.14}$$

where $\lambda_{LOS}(d)$ is the LOS MTCD intensity within distance d

$$\lambda_{LOS}(d) = 2\pi\lambda_m q \left(\left(-\frac{d}{\beta} - \frac{1}{\beta^2} \right) e^{-\beta d} + \frac{1}{\beta^2} \right). \tag{4.15}$$

Proof. The proof is provided in Appendix A □

Then we suppose the interference to mmWave relaying is negligible due to its high attenuation through obstacles. In section VI, we compare the results obtained from two simulation settings, one considering interference and one only taking noise into consideration. It is shown that the numerical results are very close between noise-only and interference+noise, so that in analysis we can ignore interference on mmWave links for simplicity. Therefore the expected transmission rate of the nearest i_{th} LOS relay is Shannon's capacity with $SINR(i) \approx SNR(i)$, the signal-to-noise ratio of the i_{th} LOS relay,

$$SNR(i) = \frac{P_{T_x}}{PL(\mathrm{E}[d_{LOS}(i)])\sigma^2}. \tag{4.16}$$

Consider each relay as a queue. There are two sources of message arrivals at the relay, the messages that are generated at the relay and the messages that are transmitted to the relay by other MTCDs. As Poisson distribution can model the number of occurences of a rare event in a very large population, classic Markov queue systems with Poisson arrivals has been long used for stochastic traffic modeling [162]. We assume the message arrivals occur at rate ω according to a Poisson

process. The message size follows an exponential distribution, thus, to transmit each message, the service time also follows an exponential distribution. We assume the service rate μ in the queue, where $1/\mu$ is the mean time to transmit each message. In queueing theory, such queue is defined as an M/M/1 queue. To avoid the queue growing indefinitely long, we have $\omega < \mu$, where the message arrival rate is lower than message service rate.

The average number of messages in the buffer is $\frac{\lambda}{\mu-\lambda}$. According to Little's law [163], the mean sojourn time is given as

$$D = \frac{1}{\mu - \omega}. \tag{4.17}$$

If queueing delays are the only nonnegligible source of delay in a network, and if each link traffic can be modeled as Poisson message arrivals with independent exponentially distributed length, then link (i, j) delay

$$D_{i,j}(f_{i,j}) = \frac{1}{\mu - \omega} = \frac{s}{c_{i,j} - f_{i,j}}, \tag{4.18}$$

where $c_{i,j}$ is the capacity of link (i, j), $f_{i,j}$ is the data arrival rate (bits/second) of MTCD j, and s is the mean of message size distribution (bits).

EED of first tier relays

First of all, we examine the first tier relays. Since the first tier relays gathers all the data generated in the cell network, and forward those data to base station, in steady state the sum of all first tier MTCDs' transmission rate equals expected sum of all generated data in this second, denoted by Γ.

$$\Gamma = k\rho, \tag{4.19}$$

where k is the total number of MTCDs in the BS's coverage, and ρ is the average data generation rate of each MTCD. Let m be the number of first tier relays per each BS's coverage, and $\mathcal{M}$ be the set of first tier relays.

Theorem 1 *Based on the system model in Section III, under the Minimum-Delay scheme, the expected delay of the i_{th} closest first-tier MTCD relay to BS is given as*

$$D_{i,b}(f_{i,b}) = s/\left(c_{i,b} - \frac{\sqrt{\lambda m}c_{i,b} - \sqrt{c_{i,b}}}{\sqrt{\lambda m}}\right), \quad i \in \mathcal{M}, \tag{4.20}$$

where $c_{i,b}$ is the Shannon's capacity from i_{th} MTCD relay to BS, calculated from (4.10), (4.14), (4.15) and (4.16), m is the number of first tier relays, λ is the lagrange multiplier,

$$\begin{cases} \lambda = \left(\frac{\sum_{i=1}^{m}\sqrt{c_{i,b}}}{\sqrt{m}(\sum_{i=1}^{m} c_{i,b} - \Gamma)}\right)^2, & c_{i,b} \in \{c_{1,b}, c_{2,b} \ldots c_{m,b}\} \\ m = \arg\min_m \lambda, & m = 1, 2 \ldots u. \end{cases} \tag{4.21}$$

The data arrival rate of MTCD i is

$$f_{i,b} = \frac{\sqrt{\lambda m}c_{i,b} - \sqrt{c_{i,b}}}{\sqrt{\lambda m}}, \quad c_{i,b} \in \{c_{1,b}, c_{2,b} \ldots c_{m,b}\}. \tag{4.22}$$

Proof. The proof is provided in Appendix B □

Thus, for the i_{th} closest first tier relay, the average delay per message (waiting time + transmitting time) from (4.18) is given as

$$D_{i,b} = \frac{s}{c_{i,b} - f_i} \quad c_{i,b} \in \{c_{1,b}, c_{2,b} \ldots c_{m,b}\}, \tag{4.23}$$

where s is the average message size. Then the average EED of first tier MTCDs is given

$$E[D_{T1}] = \frac{\sum_{i \in I} D_{i,b}}{m} \quad D_i \in \{D_{1,b}, D_{2,b} \ldots D_{m,b}\}. \tag{4.24}$$

EED of The Second Tier MTCDs

The second tier MTCDs need only transmit total generated data reduced by data generated from first tier relays: $\Gamma_{T_2} = \Gamma - \Gamma_{T_1}$, where $\Gamma_{T_1} = \Gamma \cdot \frac{m}{k}$ and k is the

number of MTCDs in each BS's coverage. Each second tier MTCD j associate with the first tier MTCD, which provides the lowest EED,

$$i = \arg\min_{i} \left(\frac{s}{c_{j,i} - \rho} + D_i \right). \tag{4.25}$$

For a random second tier MTCD, the probability of choosing i_{th} first tier relay i is

$$p(i) = \frac{f_{i,b} - \rho}{\Gamma_{T2}}, \forall i \in \mathcal{M}. \tag{4.26}$$

Therefore, the cumulative distribution function (CDF) of choosing first tier relay with delay lower than t:

$$F_{T1}(t) = \sum_{i=1}^{m} (f_{i,b} - \rho)[D_i > t]/(\Gamma - m\rho), \tag{4.27}$$

where $[\cdot]$ are Iverson brackets, $[P]$ is defined to be 1 if P is true and 0 if it is false.

Thus the CDF of the minimum of EED $F_{T2}(t)$ is given as

$$\begin{aligned} F_{T2}^{EED}(t) = 1 - \prod_{i \in j} \left[1 - F_{T1} \left(t - \frac{s}{c_{LOS,T2,i} - \rho} \right) \right] \\ \prod_{i \in n} \left[1 - F_{T1} \left(t - \frac{s}{c_{NLOS,T2,i} - \rho} \right) \right], \end{aligned} \tag{4.28}$$

where $c_{LOS,T2,i}$ and $c_{NLOS,T2,i}$ are the shannon's capacity of the links between a random second tier MTCD to i_{th} nearest LOS relay and NLOS relay, respectively. j is average number of relays that has LOS link to i in each BS's coverage, $j = \pi d^2 \lambda_{LOS}(d)$, and n is the number of NLOS relays, $n = m - j$.

Therefore, the expected EED of second tier MTCD is

$$\mathrm{E}[EED_{T2}] = \int_0^\infty (1 - F_{T2}^{EED}(t)) \mathrm{d}t. \tag{4.29}$$

Overall, the expected delay for all MTCDs in the area is presented as

$$\mathrm{E}[EED] = \frac{m}{k} D_{T1} + \frac{k - m}{k} EED_{T2}. \tag{4.30}$$

Delay Outage Analysis

In cellular networks, traffic load is a key determinant of delay outage. We define delay outage as the EED exceeding predefined required tolerable delay $EED > \tau$. Thus, the probability of delay outage $p_{D,O}$ represents the probability that a MTCD's expected EED is larger than tolerable delay. Intuitively, larger data generation rate in each MTCD will incur larger queue length in both first tier and second tier MTCDs, resulting in network congestion and longer end-to-end delay experienced by each MTCD.

The probability of delay outage of a first tier MTCD is

$$p_{D,O,T1} = p(D_{T1} > \tau) = \frac{\sum_{i=1}^{m}[D_{i,b} > \tau]}{m}, \ i \in \{\mathcal{M}\} \tag{4.31}$$

where m denotes the number of first tier MTCDs of each BS's coverage.

A second tier MTCD delay outage happens when a first tier relay has been associated with, which provides a minimum expected EED, but the first tier delay plus its transmission duration to associated relay still exceeds delay threshold. Thus the probability of delay outage of a second tier MTCD is

$$p_{D,O,T2} = p(D_{T1} + D_{T2} > \tau) = p(EED_{T2} > \tau) = 1 - F_{T2}^{EED}(\tau), \tag{4.32}$$

where F_{T2}^{EED} is the CDF of minimum EED of second tier MTCDs defined in (4.28).

Theorem 2 *The probability of delay outage for all MTCDs in the network is*

$$p_{D,O} = \frac{k-m}{k} p_{D,O,T2} + \frac{m}{k} p_{D,O,T1}, \tag{4.33}$$

where $p_{D,O,T2}$ and $p_{D,O,T1}$ are defined in (4.31) and (4.32).

Proof. Directly follows from derivations (4.31) and (4.32). □

4.3.3 Max-SINR Relaying (Bechmark) Steady-state Analysis

In the benchmark max-SINR association scheme [2], the second tier MTCD associates with the first tier relays (has LOS conmmunications with BS) that has the highest measured SINR, regardless of the current situation of the corresponding relay. Intuitively, first tier MTCDs that are far away from their associated base station often experience lower uplink transmission rate than nearer ones due to longer transmitting distance. However, the number of associated second tier MTCDs of each relay does not depend on their respective uplink rate. This results in much longer queueing time in those relays with a lower uplink rate. Furthermore, the uneven traffic loads make the situation worse because the relaying scheme is fixed in the max-SINR scheme, losing the diversity gain provided by alternative routes.

Theorem 3 *Based on the system model in Section III, under the max-SINR scheme, the probability of delay of first tier MTCD in max-SINR association scheme higher than* τ *is*

$$p_{D,O,T1}(\tau) = 1 - \int_0^{\tau} f_{T1}(D_{T1})\mathrm{d}D_{T1}, \tag{4.34}$$

where D_{T1} *is the random variable of first tier MTCD delay,* $f_{T1}(D_{T1})$ *denotes the PDF of expected first tier MTCD delay:*

$$f_{T1}(D_{T1}) = 2\pi\lambda\phi e^{-\pi\lambda\phi^2}, \tag{4.35}$$

where

$$\phi = 10^{\left(\left(\frac{P_{t_x}G_{m_t}G_{m_r}}{(2^{cT1/B}-1)\sigma^2} - A_1 - A_3\log_2(f_c)\right)/A_2\right)} \tag{4.36}$$

and

$$c_{T1} = \frac{s}{D_{T1}} + \frac{k\rho}{m}, \tag{4.37}$$

where the second term is the average data transmission rate per each first tier MTCD.

PROOF. From combining (4.4), (4.5) and (4.10) with (4.18), this is a straightforward derivation. □

Theorem 4 *Based on the system model in Section III, under the max-SINR scheme, the probability of delay outage of second tier MTCD in max-SINR scheme is*

$$p(D_{T1} + D_{T2} > \tau) = 1 - \int_0^{\infty} f_{T1}(D_{T1}) F_{T2}(\tau - D_{T1}) \mathrm{d}D_{T1}, \tag{4.38}$$

where $f_{T1}(D_{T1})$ is the PDF of expected first tier MTCD delay given in (4.35), $F_{T2}(t_2)$ is the CDF of second tier MTCD expected transmission time.

$$\begin{aligned} F_{T2}(t_2) = &\int_0^{t_2} 2\pi\lambda_{LOS,T2}\phi e^{\left(-\pi\lambda_{LOS,T2}\phi^2\right)} \mathrm{d}D_{T2} \\ &+ \int_0^{t_2} 2\pi\lambda_{NLOS,T2}\phi e^{\left(-\pi\lambda_{NLOS,T2}\phi^2\right)} \mathrm{d}D_{T2}, \end{aligned} \tag{4.39}$$

where

$$\lambda_{LOS,T2} = \frac{(k-m)}{k}\lambda_m P_L(\phi), \tag{4.40}$$

$$\lambda_{NLOS,T2} = \frac{(k-m)}{k}\lambda_m (1 - P_L(\phi)), \tag{4.41}$$

$$\phi = 10^{\left(\left(\frac{P_{t_x} G_{m_t} G_{m_r}}{(2^{cT2/B}-1)\sigma^2} - A_1 - A_3 \log_2(f_c)\right)/A_2\right)}, \tag{4.42}$$

$$c_{T2} = \frac{s}{D_{T2}} + \rho. \tag{4.43}$$

PROOF. The proof is provided in Appendix C □

As a result the average delay of max-SINR relaying scheme is

$$\mathrm{E}[D_{T2}] = \frac{u}{k}\int_0^\infty f(D_{T1})D_{T1}\mathrm{d}D_{T1} + \frac{k-u}{k}\int_0^\infty (1 - F_{T2}(\tau))\mathrm{d}\tau, \tag{4.44}$$

where u is the number of LOS MTCDs to BS.

4.3.4 3-tier Network Analysis

In this section, we discuss the performance of increasing the number of tiers in the multi-hop relaying network. In previous discussion, we only considered a two-tier structure, where only one tier acts as the relay tier, gathering data from nearby MTCDs and forwarding them to the associated base station though a LOS link. Here, we add another tier — a second tier of relays that gathers data from nearby MTCDs and forwards them to first tier relays through LOS links. We define the three-tier network structure as a three-tier network structure consisting of the first tier MTCDs that associate with base stations with direct LOS links, the second tier MTCDs that associate with any first tier MTCDs with direct LOS links, and third tier MTCDs than only communicate with second tier MTCDs.

We will discuss the average delay in this section. Similar to the two-tier structure, the expected delay of first tier is given in (4.25). To satisfy the data gathering rate of first tier relays in steady state, second tier MTCDs associate with first tier relays in order to feed in the required rates. For each first tier relay i, the data transmission rate equals the data gathering rate plus its own data generating rate ρ.

Then, the first tier MTCD i, $i \in \mathcal{M}$, is associated with a set of second tier MTCDs who have highest LOS transmission rates to MTCD i. Let n_i be the number of second tier relays that associate with first tier MTCD i, and $\mathcal{N}$ be the

set of those n_i second tier relays. To minimize the average uploading delay for all gathered packets,

$$\min \sum \frac{D_{(j,i)}}{n_i} = \min \sum \frac{s/(c_{j,i} - f_{j,i})}{n_i}, j \in \mathcal{N},\ i \in \mathcal{M} \tag{4.45}$$

$$s.t. \sum f(j,i) = f_{i,b} - \rho. \tag{4.46}$$

Similar to the minimisation process of first tier MTCDs, the expected per message queueing delay in second tier MTCD j associated with MTCD i is

$$D_{j,i} = s/\left(c_{j,i} - \frac{\sqrt{\lambda_{min}}c_{j,i} - \sqrt{c_{j,i}}}{\sqrt{\lambda_{min}}}\right), \tag{4.47}$$

where

$$\begin{cases} \lambda = \left(\frac{\sum_{j=1}^{n_i}\sqrt{c_{j,j}}}{\sqrt{n_i}(\sum_{j=1}^{n_i} c_{j,i} - f_{i,b} - \rho)}\right)^2, \quad c_{j,i} \in \{c_{1,i}, c_{2,i} ... c_{n_i,b}\} \\ n_i = \arg\min_{n_i} \lambda. \end{cases} \tag{4.48}$$

Then the average message queueing delay of second tier MTCDs is given

$$E[D_{T2}] = \frac{\sum_{i\in\mathcal{M}} \sum_{j\in\mathcal{N}} D_{j,i}}{\sum_{i=1}^{m} n_i} \tag{4.49}$$

For tractable reasons, we suppose the locations of second tier MTCDs follow a PPP of density $\lambda_{T2} = \frac{\sum_{i\in\mathcal{I}} n_i}{k}\lambda_m$.

For simplicity, we assume each third tier MTCD $r \in \mathcal{R}$ is associated with the second tier MTCD j which has the highest link capacity between k and j.

$$i = \arg\min_{r} \left(\frac{s}{c_{r,j} - \rho}\right), r \in \mathcal{R}. \tag{4.50}$$

The PDF of distance between a third tier MTCD to closest second tier MTCD is

$$f(d_{T2,T3}) = 2\pi\lambda_{T2}(d_{T2,T3}) d e^{-\pi\lambda_{T2}(d_{T2,T3})d^2}. \tag{4.51}$$

The expected queueing delay in third tier MTCDs is

$$E[D_{T3}] = \int_0^\infty f(d_{T2,T3}) \frac{s}{B\log_2(1 + \frac{P_{Tx}g}{PL(d_{T2,T3})\sigma^2}) - \rho} \mathrm{d}d_{T2,T3}. \tag{4.52}$$

Table 4.1: Summary of parameters used in the Simulation.

λ_o	obstacle density	0.42
P_{T_x}	transmit power	0.4mW
A_1	path loss model intercept	28 (LOS)
		13.54 (NLOS)
A_2	path loss exponent	22 (LOS)
		39.08 (NLOS)
A_3	path loss frequency dependence	20 (LOS, NLOS)
f_c	carrier frequency	28GHz
B	bandwidth	10MHz
N_0	noise density	-164 dBm/Hz
F_{noise}	noise factor	9dB
s	average message size	10Kb
ϱ	multiplexing factor	1
L	buffer size	16Mb

Thus, the average EED for all MTCDs in a 3-tier relaying network is

$$E[EED] = \frac{m}{k}E[D_{T1}] + \frac{\sum_{i=1}^{m} n_i}{k}(E[D_{T1}]) + E[D_{T2}]) + \frac{k - m - \sum_{i=1}^{m} n_i}{k}(E[D_{T1}] + E[D_{T2}] + E[D_{T3}]), \tag{4.53}$$

where D_{T1} is calculated from (4.24) with m first-tier MTCDs.

4.4 System Evaluation

In this section, we validate our analytical results with simulations based on the 3GPP network evaluation method, and explore the effect of different parameters on the delay outage and effective capacity.

4.4.1 Simulation Setup

In our PPP-based network models, 9 BSs are deployed in the area and they are placed evenly in road junctions for best LOS probability. UEs are assumed to be dropped according to a 2D homogeneous PPP. If there are k UEs per cell on average, then the density of UEs is $\lambda_r = k\lambda_b$, where λ_b is the BS intensity.

BS and UE Distribution

In 3GPP's grid-based evaluation models, the UMa scenario has a hexagonal cell layout with an inter-site distance (ISD) of 500m, which corresponds to a 2D homogeneous PPP density 4.62×10^{-6} for base stations. In our PPP-based network models, UEs are assumed to be dropped according to a 2D homogeneous PPP. If there are k UEs per cell on average, then the density of UEs is $\lambda_r = k\lambda_b$, where λ_b is the BS intensity.

Obstacle Distributions

We use real building data from urban Chicago area to simulate obstacle distribution. The simulated area is shown in Fig. 4.2. The reason why we choose a real-world building map is that the simulation based on real map gives more realistic and instructive results. We can observe that the building is of various shapes and sizes, and most buildings are constructed in blocks, which is a distinguish difference from other models with randomly distributed obstacles. Note that only 2D map is used in the simulation, and we assume there exist LOS links when there is no building in between the transmitter and receiver, regardless of the relative heights. 2D and 3D distances are different in cellular networks due to the BS and user antenna heights. For tractable reasons, we use 2D distance keeps the simulation relatively simple. We note that the obstacle intensity $\lambda_o = 0.42$ in the

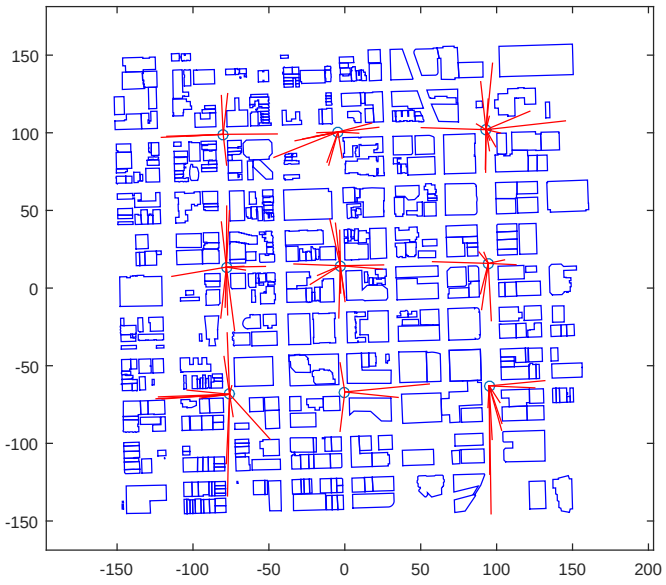


Figure 4.2: Real Building Data from Urban Chicago Area in Simulation.

simulated area.

Antenna Configuration

To model beamforming, we consider that each BS has a 4×4 antenna array. We assume that all MTCDs have 2×2 antenna arrays for both transmitting and receiving. The transmitting power of each device $P_{T_x} = 0.4$mW. Antenna array gains and beamwidth are determined according to (4.2).

Channel Models

We use the path loss model from 3GPP evaluation methodology documents. For mmWave links, the LOS path loss models for the UMa is

$$PL_{LOS} = 28 + 22\log_{10}(d) + 20\log_{10}(f_c), \tag{4.54}$$

where d is the distance in meters and f_c is the carrier frequency in GHz. The NLOS path loss models for the UMa is

$$PL_{NLOS} = 13.54 + 39.08 \log_{10}(d) + 20 \log_{10}(f_c). \tag{4.55}$$

In the simulation, we set the carrier frequency for mmWave links to 28GHz, and the bandwidth to 10MHz. The noise power is set to -164 dBm/Hz, noise factor $F_{noise} = 9dB$ and the average message size is $s = 10$Kb. Note that 3GPP uses 3D distance in its mmWave path loss models. To avoid complicating the analysis in this paper, we use 2D distance for d, which is commonly assumed in stochastic geometry based network performance analysis. The multiplexing factor ϱ is set to be 1, which means we consider a fully loaded uplink network model with orthogonal multiple access such that the BS schedules exactly one transmission link between two MTCDs, or between base station and MTCD, on each uplink sub-channel at a given time within a cell. We set the MTCD buffer size $L = 16$ Mb.

4.4.2 Validation of Analytical Expressions

In this section, we compare our analytical results with the UMa performance evaluation scenario based on urban chicago map simulations.

2-tier structure

In Fig. 4.3, we compare the mmWave 2-tier Minimum-Delay relaying, obtained analytically from (4.30), against urban chicago map based UMa performance simulations. As expected, the results obtained by analysis provide an estimate on the UMa performance simulations. Additionally, we observe that the results obtained by our delay analysis become more accurate as average data arrival rate increases, but the analysis can not provide accurate delay estimation when data arrival rate is

too low. This is because at low data arrival rate, the randomness of data generation cause it very hard to predict the delay. For example, in Fig. 4.3, Minimum-Delay analysis accurately models the estimated delay when the average data arrival rate is higher than 2 Mb/s. We hence conclude that the analytical delay estimation in (4.30) is a reasonable approximation in 3GPP UMa scenarios.

To determine if we can ignore the interference on mmWave links, we compare the two UMa performance simulations results, one considering only noise and the other one considering both interference and noise. We observe that for high BS density networks, the mmWave links demonstrate noise-limited behaviors. The average delay of direct-association scheme (all MTCDs directly associate with BSs) and max-SINR relaying [2] is also shown in Fig. 4.3. We can see that the average EED is significantly lower in Minimum-Delay relaying compared with the direct-association scheme and max-SINR relaying. Moreover, it is clear that the average EED under the max-SINR relaying scheme is even higher than that of the direct-association scheme, especially in a high data arrival rate area.

Number of tiers effect on delay

The average delay for a 3-tier Minimum-Delay relaying scheme is illustrated in Fig. 4.4, with comparison with the 2-tier Minimum-Relaying scheme. In Fig. 4.4, analytical results obtained accurately characterize the slope of simulated results. We notice that the benefits of increasing number of tiers is limited. In low traffic load situations, the 3-tier structure outperforms the 2-tier structure since the additional tier allows for shorter transmission distance and higher probability of LOS links, thus dramatically improving the transmission rate of second tier and last tier MTCDs to upper tiers. However, in high traffic load scenarios, e.g., average data arrival rate higher than 1.5 Mb/s, the 3-tier structure will introduce larger delays

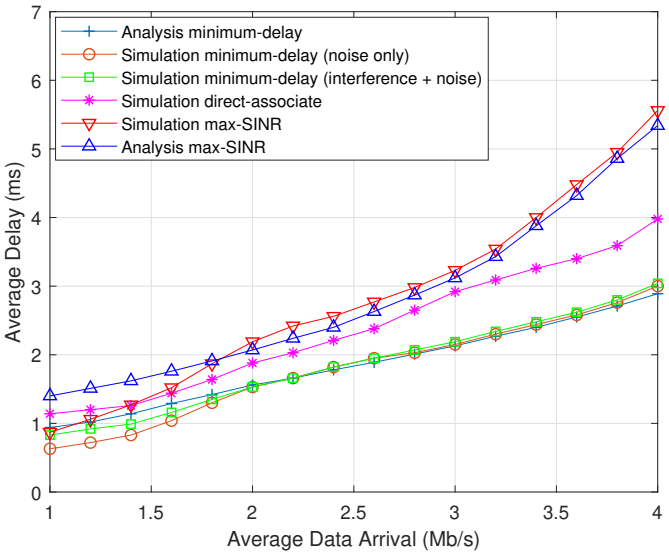


Figure 4.3: Validation of the 2-tier analytical Minimum-Delay relaying average delay in (4.30) with simulation and comparing with other benchmark schemes (max-SINR [2]).

because queueing delays dominate the EED and end users need to queue twice, both at first and second tier relays.

Base station density effect on delay

In Fig. 4.5, we show the average delay obtained analytically from (4.11), with UMa performance simulation with respect to base station density. From Fig. 4.5 we can see that the analytical results accurately predict the average delay experienced by devices under different base station densities. We observe that as base station density increases, the average delay decreases. However, when base station density exceeds a certain threshold, e.g., $p > 300\,\text{per}\,\text{km}^2$, deploying more base stations will not improve average delay significantly. This is because at this density, link distances are short enough to support extremely high data rates compared with the average data arrival rate. It is also shown that there is no sign of downgrade of

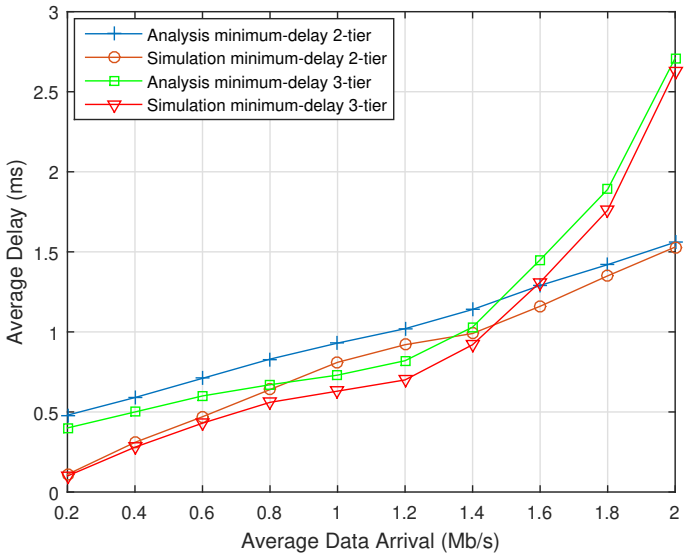


Figure 4.4: Average delay comparison between 2-tier and 3-tier Minimum-Delay relaying.

delay performance at a high BS density deployment. This observation shows that due to blockage and attenuation, interference remains low even in densely deployed BS networks.

4.4.3 Performance Results

Delay Outage Improvement of Min-delay Relaying

The delay outage probability for 2-tier Minimum-Delay relaying and max-SINR relaying cellular network obtained from UMa performance simulation is shown in Fig. 4.6 with respect to delay threshold. Note that we fix base station density to 400 per km^2, and the average data arrival rate is set to 1.8 Mb/s and 2.2 Mb/s. We can see that Minimum-Delay relaying performs better than the max-SINR relaying scheme. At an extremely strict delay threshold, e.g., 2.5 ms, the Minimum-Delay relaying scheme has larger outage probability than the max-SINR scheme. This is

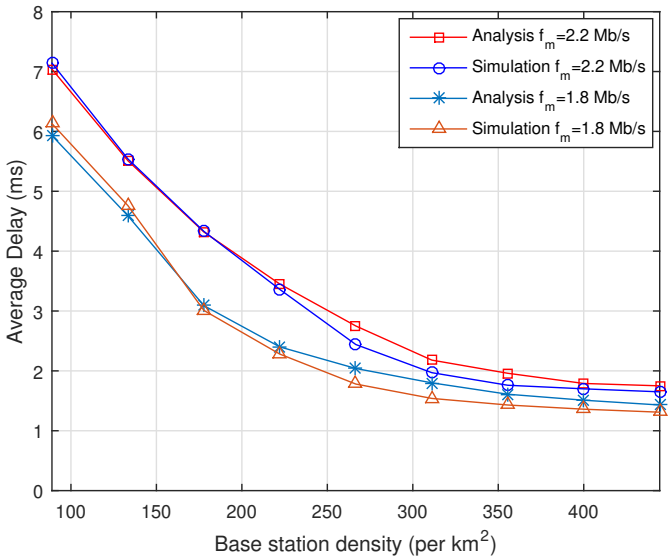


Figure 4.5: Average delay of 2-tier Minimum-Delay relaying with respect to base station density.

because the Minimum-Delay scheme evens out the delays of MTCDs, preventing extremely large delays of certain MTCDs by directing them to transmit through lightly loaded relays. Thus the probability of having extremely low delays is lower than that of max-SINR relaying scheme.

However, the Minimum-Delay scheme can minimise large EEDs, as shown in Fig. 4.6. The outage probability of the Minimum-Delay scheme fast approaches 1% as delay threshold increases, meaning that most of the MTCDs fully utilize resources in each cell. By contrast, the outage probability of max-SINR scheme barely improves as we relax the delay threshold.

Uplink Effective Capacity

For the smooth operation of cellular networks, it is important to ensure that the average data arrival rate is below the service capacity, which means that the service capacity of the network should satisfy potential data traffic. More specifically,

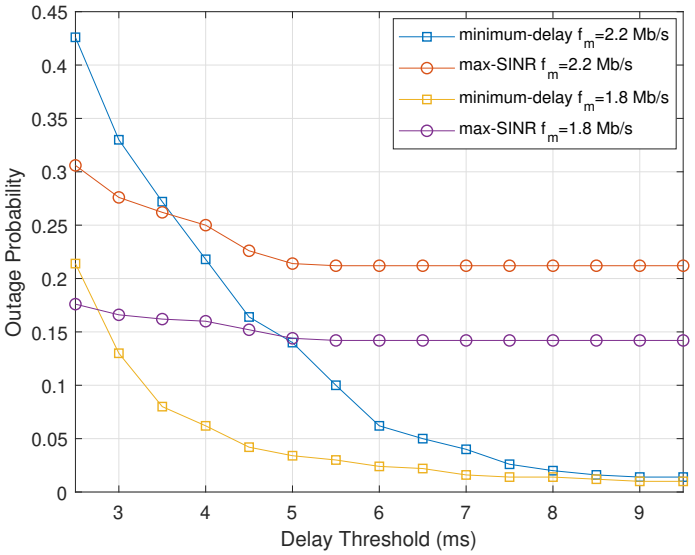


Figure 4.6: Outage improvement of Minimum-Delay relaying compared with max-SINR association [2].

service capacity represents the largest amount of data rate a network can transport without causing significant delay outages. we define the effective capacity as the maximum data arrival rate of UEs that 90% of Users experience EED under outage threshold.

The analytical and simulated effective capacity of Minimum-Delay and max-SINR relaying schemes is illustrated in Fig. 4.7. Clearly, the derived analytical effective capacity accurately describes simulation results obtained from a real world building map. From Fig. 4.7 we find that Minimum-Delay relaying exhibits remarkable advantages when delay outage threshold is high, while the effective capacity of the max-SINR scheme barely increases as threshold increases. These findings indicate that the proposed Minimum-Delay scheme better uses the resources in the cell, allowing for heavier traffic loads at each MTCD and still provide satisfying QoS performance. Besides, the proposed scheme is a more versatile network arrangement since the effective capacity gets higher when delay outage threshold

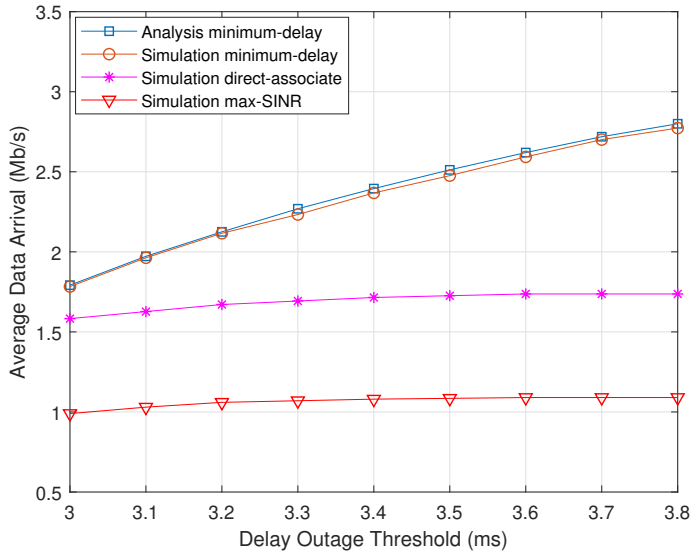


Figure 4.7: Analytical and simulated effective capacity for Minimum-Delay relaying schemes, where outage threshold is 90% and BS density $\xi = 400/\text{km}^2$, compared with max-SINR relaying [2]

is relaxed, compared with the max-SINR scheme where the effective capacity remained unchanged. In this sense, the Minimum-Delay relaying scheme is a better option for heavy traffic scenarios, and this insight is also in line with the results obtained from Fig. 4.6.

Spectral Efficiency

Spectral efficiency (SE) is defined as the bit per unit bandwidth per cell. Generally, the SE is calculated using Shannon's theorem as $\gamma = \log_2(1 + SINR)(\text{bit/s/Hz})$. However, in multi-hop relaying systems, to upload a packet more than one transmitter may be used, which requires more orthogonal spectrum bands to transmit if each transmitter is assigned an orthogonal frequency band in a cell. For example, if a packet is relayed once, the spectrum used to transport this packet is twice as much as that of the direct link. Thus, the SE for this packet is only half of the

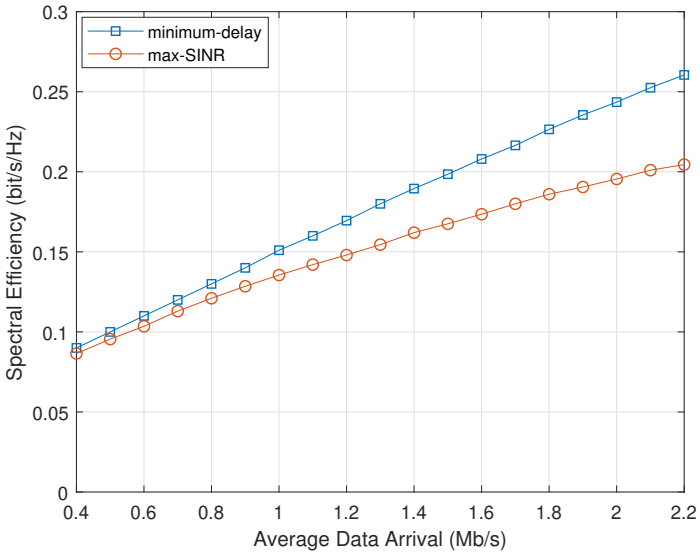


Figure 4.8: Spectral efficiency comparison between Minimen-Delay relaying and max-SINR relaying [2].

original SE. From Fig. 4.8, we can see that the SE of the Minimum-Delay relaying scheme is higher than that of the max-SINR scheme, especially in high data arrival range. This is because the traffic is balanced over the network via the process of seeking the relay that provides the lowest EED, congestion is avoided and message drop due to a full buffer is reduced. As a result, the SE in our Minimum-Delay relaying scheme increases linearly in proportion to average data arrival, while the SE of the max-SINR scheme is lower than that of the Minimum-Delay scheme by a growing margin as the traffic load increases.

Jain's Fairness

Jain's fairness [164] index is a metric for evaluating the fairness of the EED of all MTCDs,

$$J = \frac{(\sum_{i=1}^{k} \delta_i)^2}{k \sum_{i=1}^{k} \delta_i^2}, \tag{4.56}$$

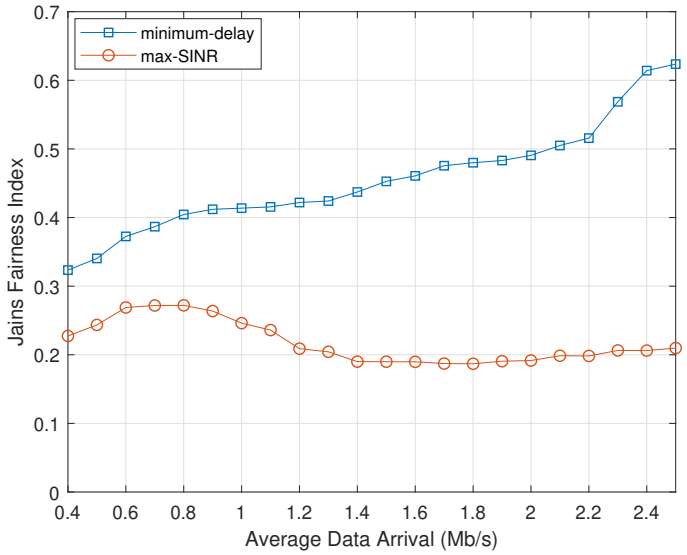


Figure 4.9: Jain's fairness of Miminum-Delay relaying compared with max-SINR relaying [2].

where δ_i denotes the EED of MTCD i in simulations. Fig. 4.9 illustrates Jain's fairness index of both Minimum-Delay and max-SINR relaying schemes. It is shown that Jain's fairness index of Minimum-Delay relaying increases with average data arrival rate, while Jain's fairness index of max-SINR relaying firstly experience a drop, then a gradual slight increase with average data arrival rate. The advantage of fairness of Minimum-Delay relaying over max-SINR gets larger as average data arrival rate increases, as the traffic is balanced over the network via the process of seeking the relay that provides the lowest EED. As the data arrival rate increases, queueing delays in first tier relays dominates EED. The delay advantage of first tier relays (not having to transmit from second tier to first tier) has less effect, gradually, on the overall delay because the commonly shared queueing delay makes up larger and larger portion of the overall delay. On the other hand, max-SINR relaying does not achieve the fairness benefit as traffic grows because queueing delay in each relay varies too much, and the network is not balanced.

4.5 Concluding Remarks

We have proposed a multi-hop relaying scheme for mmWave cellular networks, which optimises uplink latency. Using stochastic geometry, we derived expected end-to-end delay and delay outage probability models for both 2-tier and 3-tier relay-assisted mmWave networks under the proposed Minimum-Delay relaying scheme and the max-SINR relaying scheme, in which each MTCD seeks to minimize its own estimated delay through exchanged information. Our analysis is validated by simulation based on a real world building map, and shows that mmWave links demonstrate noise-limited behaviors, with the proposed scheme achieving shorter delay, lower outage probability and higher effective capacity compared with the state-of-art max-SINR relaying scheme. Finally, Jain's fairness index is measured through simulation, showing that the load is balanced among relays. Now we have solved the optimisation problem of reducing delay for data transmission. Network capacity, one the other hand, is also an important aspect to optimise in CIoT. Hence, in the next chapter, we investigate relaying schemes that improve system capacity and fairness among MTCDs.

Chapter 5

Throughput and Fairness Oriented Relaying

In chapter 4, we presented relaying schemes which focus on reducing end-to-end delay for each machine-type communication device (MTCD). Here, in chapter 5, we switch our focus to optimise throughput of the relaying network with considerations to fairness among devices. In Section 5.1, by using D2D communications, we present a multi-hop relaying and resource allocation scheme in a full-duplex network aiming to optimise data rate for each MTCD. Then, in Section 5.2, through adding access points (AP), we develop a cross-tier scheduling algorithm for a multi-tier downlink mmWave network that optimises the overall downlink throughput while ensuring every user is fairly served.

5.1 Socially Optimal Distributed User Association for multi-hop Machine-Type Communications

5.1.1 Introduction

Machine-to-machine (M2M) communications, also known as Machine-Type Communications (MTC), feature a wide range and large number of autonomous devices communicating without human intervention, and are an essential part of 5G, internet of things (IoT) and future wireless systems [129].

Connection between machine-type communication devices (MTCDs) and MTC servers are provided through base stations (BSs) in cellular networks. However, due to the massive number of MTCDs, each carrying a small amount of data to be transmitted, simultaneous transmission attempts from these MTCDs will require large-scale synchronization and result in congestion of the radio access network (RAN) [20, 21]. Thus, potential solutions to support MTC in cellular networks include modifications of random access channel [165]; base station association schemes for load balancing [166]; and uplink aggregation [25].

One approach to solve the congestion problem is to cluster MTCDs into small groups and relay their data traffic through an MTC gateway (MTCG) or a node with superior processing and transmission capability, e.g., [167]. This hierarchical structure significantly reduces the rate of direct connection, therefore the load on BSs can be alleviated. Besides, such clustering techniques also reduce energy consumption, increasing the scalability and network lifetime. 3GPP support towards group-based management of MTCDs also gives impetus to researchers to study M2M relaying schemes [168]. The clustering scheme proposed in [169] aims at min-

imizing the total energy spent in a wireless sensor network when all sensors report data through the clusterheads to the processing center. A hierarchy of clusterheads is then generated in the algorithm, and observed that the energy savings increase with the number of levels of hierarchy.

Most clustering schemes only allow for one-hop relaying, in which one cluster head is selected to gather M2M data generated in the area. Note that machine generated data often features a relaxed requirement for delay, such as when used for environment monitoring, in comparison with human-to-human (H2H) communication. Therefore, a multi-hop relaying scheme for non-critical MTCDs can benefit from reduced transmission distances, thus achieving a higher data rate with the same transmit power, or lower power consumption while maintaining equal link capacity with relaxed delay constraints. But, multi-hop relaying can lead to inefficient use of system resources as an extra dedicated time, or frequency, slot is usually required for relay transmissions. However, recent advances in self-interference cancellation techniques enable in-band full-duplex wireless systems [170], which transmit and receive simultaneously in the same frequency band. Compared with half-duplex systems in which transmitting and receiving are allocated orthogonal time-frequency resource blocks, multi-hop relaying schemes considerably improve spectral efficiency when incorporating an in-band full-duplex system.

Furthermore with a large number of MTCDs coupled with an increasing amount of data for each transmit, and with more functions are incorporated in MTCD, the total data generated in an M2M network could easily exceed that of H2H communications. But up-to-now, most M2M schemes focus on reducing outage probability and power consumption, and this is rarely done in conjunction with maximizing either data rate or sum data rate.

In this chapter we present a multi-hop relaying and resource allocation scheme,

by user association, for the uplink from MTCDs in a cellular network with the aim to optimise data rate of each MTCD, and sum data rate across MTCDs, while maintaining fairness and load balancing among MTCDs. Minimal extra packet overhead, only 8% at the MAC [171], is required across multiple hops compared with a single-hop scheme. We define a non-cooperative game for data rate and user association across MTCDs, which has a unique Nash Equilibrium (NE). The game is decomposed into subgames, each consisting of a subset of MTCDs that maximize the sum of utilities for association. A distributed algorithm applying lagrangian decomposition allows each subgame to reach NE, and after a small number of iterations the sum utility is maximized such that social optimality for data rate is achieved while power consumption is minimized. In this context load is balanced and fairness is maintained.

5.1.2 System Model

Channel Characteristics

In uplink (UL) cellular networks, the default association scheme is max-signal-to-interference-plus-noise ratio (max-SINR), connecting directly to the BS [172]. But there may be a massive number of MTCDs concurrently associating with the BS, causing regional overload and thus unacceptable packet drop. To overcome this problem, a relaying scheme is proposed, that uses the geometrical characteristics of the network to maximize overall transmission rate, while maintaining fairness and load balancing among relays and BSs. Here we assume that all BSs have full buffers and the same number of resource blocks, while the network is quasi-static, in which the air interface and location of MTCDs changes slowly.

We assume an UL cellular network consisting of K BSs. An example of network with $K = 1$ is illustrated in Fig. 5.1. We denote by B the set of all BSs and D the

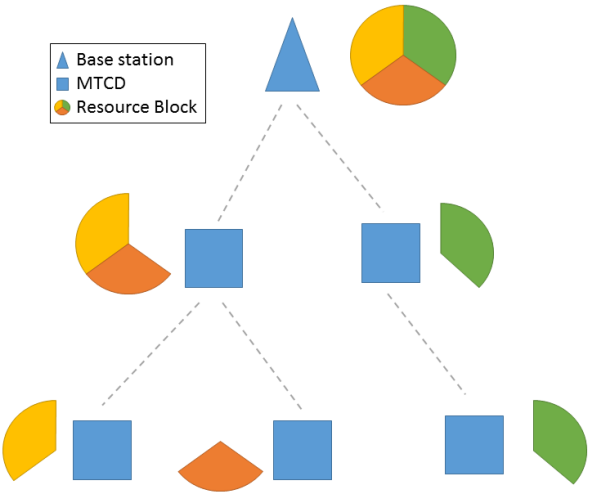


Figure 5.1: Illustration of a three-tier M2M cellular network

set of all MTCDs. The MTCD must be associated with one of the BSs to establish a connection, but may skip the direct link to BSs and establish a connection to the BS via other MTCDs and use them as relays. Relays are MTCDs that act as a bridge to assist other devices transmissions in addition to their own traffic transmission, and the relay set is denoted by R. Note that a relay is also an MTCD so $R \subset D$. Denote by E the set of end devices, that does not assist any other MTCD, $E \subset D$.

During the connection period, we denote by $c_{i,k}$ the achievable rate between MTCD i and BS k as

$$c_{i,k} = \log_2(1 + \gamma_{i,k}), \tag{5.1}$$

where $i \in D$ and $k \in B$, $\gamma_{i,k}$ is the received SINR of the link between device i and BS k. Similarly, we define achievable rate $c_{i,j}$ and $c_{j,k}$ for $i \in D$, $j \in R$, $k \in B$, as the achievable rate of the single link between MTCD i and relay j, and between relay j and BS k, respectively.

We analyse the uplink of an orthogonal frequency division multiple access

(OFDMA) system, such as LTE-A, where the BSs distribute the spectrum in units of resource blocks (RBs). The BSs estimate the required RBs for H2H communications and allocate the remaining RBs for M2M communication. Since in OFDMA system, RBs are all orthogonal to each other, no interference between H2H and M2M or between M2M devices is considered. Thus, in OFDMA, we neglect interference and Signal-to-noise ratio (SNR) γ is calculated

$$\gamma_{i,k} = \frac{P_{i,k} g_{i,k}}{\sigma^2}, \forall i \in D, k \in R \cup B \tag{5.2}$$

where $P_{i,k}$ is the transmit power of device i, and $g_{i,k}$ denotes the channel gain between device i and node k (a relay or a BS), which in general includes path loss, shadowing and antenna gain.

Let real variables $r_{i,k}$, $r_{i,j}$, $r_{j,k}$ be the rate of link between nodes i and k, and i and j, and j and k respectively. Then specify the fraction of resource blocks allocated to serve device i from BS k as $y_{i,k}$. The sum of fraction of RBs allocated to all served MTCDs by each BS equals to 1

$$\sum_{i \in D} y_{i,k} = 1, \forall k \in B \tag{5.3}$$

We suppose that each MTCD can be served by only one BS. Thus, the RBs allocated to MTCD i is $y_i = y_{i,k}, \forall y_{i,k} \neq 0$.

Then, the transmission rate between node i and j is

$$r_{i,j} = c_{i,j}\, y_i, \forall j \in D \cup B \tag{5.4}$$

Note that the actual rate of data generated by MTCD i arriving at target BS is denoted by η_i, then the actual rate is always less or equal to the rate between MTCD i and its connected relay

$$\eta_i \leqslant r_{i,j}, \forall i, j \in D \cup B \tag{5.5}$$

Utility Function of association by rate

We formulate an optimisation problem which involves finding $\{x_{i,j}\}$ corresponding to association (i.e., $x_{i,k} = 1$ if client i is associated with BS k, otherwise $x_{i,k} = 0$. $x_{i,j}x_{j,k} = 1$ if client i is served by BS k via relay j).

Each client i must be associated either directly with BS or connected by a relay.

$$\sum_{k\in B} x_{i,k} + \sum_{j\in R} x_{i,j} = 1, \forall i \in D \tag{5.6}$$

whereas each relay j can assist more than one device, and each end device can setup connection to BS via multiple relays, i.e., relays can assist end devices, and other relays.

Note that each relay, j, transmits data to the next relay or BS k at rate $r_{j,k} = c_{j,k}\, y_j$. The aggregated data rate with which relay j can transmit can not exceed the supported achievable rate for link through relay j

$$\left(\sum_{i\in D} r_{i,j}x_{i,j} + \eta_j\right) \leqslant c_{j,k}\, y_j, \forall j \in R, k \in R \cup B, \tag{5.7}$$

Furthermore, we suppose relays operate in in-band full-duplex mode [170]. More specifically, once a relay assists other devices, it sends both the assisted devices' traffic, and its own traffic, over the same time-frequency-spatial resource block, so the combined fraction of RBs of assisted devices equals the fraction RBs allocated to their associated relay, as illustrated in Fig. 5.1.

$$\sum_{i\in D} x_{i,j}\, y_{i,k} = y_{j,k}, \forall j \in R, k \in B \tag{5.8}$$

The optimisation problem can be formed as maximizing an aggregate logarithmic utility as:

$$\max_{x,r,y} \sum_{i\in D} \log(\eta_i), \tag{5.9}$$

given the constraints of (5.5), (5.6), (5.7), and (5.8). Logarithmic utility function's decreasing slope means that marginal utility diminishes as the rate increases, as opposed to simple linear function, in which the marginal utility keeps unchanged. Thus, logarithmic utility function of data rate encourages to improve the lower rate devices rather than higher rate devices, which naturally satisfies a realistic resource allocation strategy that provides load balancing and fairness amongst devices.

5.1.3 Distributed Algorithm for Association and Resource Allocation

In this section, we propose a distributed algorithm for each device to autonomously determine its association variable and subsequent resource allocation of rate and transmit power. The optimisation problem (5.9) is NP-hard.

When a network is small, the optimal user association can be done in a centralized manner. But solving the problem in large scale, namely for thousands of devices covered by each BS, is impractical as complexity exponentially increases with device number. Besides, a centralized approach requires global network information, introducing greatly excessive signaling overhead. Therefore, a low complexity distributed association algorithm is required. Thus, we propose a non-cooperative game, in which each device acts as a selfish player, to efficiently solve the optimisation problem.

Game Setup and Decomposition

The non-cooperative game solves the problem of association. The fraction of RBs for each device is allocated by either the BS or relay that it is associated with, and the game assumes each MTCD has explicit information about how many RBs it will be allocated for each node with which it associates. Therefore, the game is

given by:

Players: all devices in the network $i \in D$, $i = 1, \ldots, K_{all}$

Action space: each player/device i has the same action set $\boldsymbol{X}$, where pure strategy $x_i \in \boldsymbol{X}$ corresponds to the decision of device i of which relay or BS to associate with.

Utility function: The utility of device i is calculated by:

$$U_i = \log(\eta_i) \tag{5.10}$$

under constraints of (5.5), (5.6), (5.7), and (5.8).

We decompose the game into subgames and divide MTCDs into N tiers. Each subgame either consist of BSs and first tier T_1 MTCDs, or consist of MTCDs in two neighboring tiers T_n and T_{n-1}. In each subgame, lower tier MTCDs act as players, trying to associate with only one of the upper tier MTCDs (or, if in T_1, the BS itself) in order to maximize their utility U_i.

$$x_{i,j} = 0, \forall i \in T_n,\ j \in T_{n'},\ n' - n \neq 1 \tag{5.11}$$

$$\sum_{j \in T_{n-1}} x_{i,j} = 1, \forall i \in T_n \tag{5.12}$$

The set of tier 1 MTCDs are defined as

$$T_1 = \{i \,|\, c_{i,k} > \min(c_{i,j}, c_{j,k})\}, \forall i, j \in D, K \in B \tag{5.13}$$

Each T_1 relay has a higher achievable rate directly to the BS than the maximal achievable rate to the BS through any other device, which means that it will seek to directly connect to BSs rather than transmitting via other MTCDs. Because utility is logarithmic function of actual rate, it can be easily proved that any other link via relay has lower utility than that of direct connection.

The set of rest of tiers are defined one after another as n increases until all MTCDs are assigned to a tier

$$T_n = \{i \mid c_{i,j} > \min(c_{i,j'}, c_{j',j})\}, \forall i, j' \in D, j \in T_{n-1} \tag{5.14}$$

Each T_n MTCD has higher achievable rate to MTCDs in T_{n-1} if directly connected to that higher tier, than the maximal achievable rate through any other device, which means it seeks to directly connect to a relay in T_{n-1} rather than via any other MTCD. With number of network tiers N, then the association of all T_1 MTCDs to BSs, and association from all T_n MTCDs to T_{n-1} MTCDs each represent a subgame, for any $n \leqslant N$.

Resource Allocation

The goal of each MTCD in the relevant subgame is to determine its association for maximal utility, while by properly allocating resource blocks to them, the goal of optimum social welfare in each subgame can also be achieved. The optimisation problem is less complex in each subgame compared to the complete game, so we start from determining an optimal resource allocation in each subgame.

The MTCDS only choose which relay to associate with, while BSs or upper tier relays determine the resource allocation. Note that the optimisation goal in (5.9) is finding maximal sum utility, causing the BS or upper tier relay's to assign RBs to assist MTCDs in the next lower tier to achieve optimum social welfare.

K_j is the number of devices either directly or indirectly assisted by node j, so $K_j = \sum_i x_{i,j} K_i$. And, the overall number of devices is K_{all}. The following lemma provides the optimal resource allocation for each device.

Lemma 1 *Assuming the association variable x is fixed and feasible, then the optimal fraction of resource blocks allocated to each device in each subgame is:*

(1) $y_j = K_j/K_{all}, \forall j \in T_{n-1}$

(2) $y_{i,j} = y_j\, K_i/(K_j+1),\ \forall i \in T_n, j \in T_{n-1}$

PROOF. Considering all assisted MTCDs by relay j, the objective function of subgame for K_j assisted MTCDs is

$$\begin{aligned}\max_y \sum_{i\in D} K_i \log(r_{i,j}/K_i) &= \sum_{i\in D} K_i \log(y_{i,j}c_{i,j}/K_i) \\ &= \sum_{i\in D} \left(K_i \log(c_{i,j}) + K_i \log(y_{i,j}/K_i)\right) \end{aligned} \tag{5.15}$$

where $\sum_{i\in D} \log\, K_i(c_{i,j})$ is constant relative to assisted number of node i and SINR $\gamma_{i,j}$. Then the objective function is equivalent to maximizing the geometric mean:

$$\max_y \sum_{i\in D} K_i \log(y_{i,j}/K_i) \equiv \max_y \frac{1}{K_j} \log\left(\prod_i^{k_j} (y_{i,j}/K_i)^{K_i}\right). \tag{5.16}$$

as the geometric mean is no greater than the arithmetic mean, and then the above equivalence holds if and only if $y_1/K_1 = y_2/K_2 = \cdots = y_n/K_n$. Therefore, optimising (5.15) jointly with relay's own data rate, $y_{i,j} = y_j \cdot K_i/(K_j+1)$. □

Dual Decomposition and Distributed Algorithm

In each subgame, i.e, association between T_n and T_{n-1}, the optimisation problem is decomposed into a local sub-problem after optimal resource block allocation described in **Lemma 1**:

$$\begin{aligned} &\max_x \sum_{i\in T_n} \sum_{j\in T_{n-1}} x_{i,j} k_i \log\left(\frac{\min(c_{i,j}, r_j)}{\sum_{k\in T_n} x_{k,j} K_k + 1}\right) \\ &s.t. \sum_{j\in T_{n-1}} x_{i,j} = i, \forall i \in T_n \\ &\quad x_{i,j} = \{0,1\}, \forall i \in T_n,\ \forall j \in T_{n-1} \end{aligned} \tag{5.17}$$

Where r_j is the transmission rate of node j to upper tier or BS. Denote by $K_j = \sum_i x_{i,j} K_i$ the number of devices either directly or indirectly connected to relay j, the optimisation problem becomes:

$$\max_x \sum_i \sum_j x_{i,j} K_i \log(\min(c_{i,j}, r_j) - \sum_j K_j \log(K_j + 1) \tag{5.18}$$

To solve problem (5.18), we use lagrangian dual decomposition, in which a lagrangian multiplier μ is added to relax the coupled constraint. Thus, the dual relaxation of (5.18) is

$$\min_\mu D(\mu) = f_x(\mu) + g_K(\mu) \tag{5.19}$$

where

$$f_x(\mu) = \max_x \sum_i \sum_j x_{i,j} K_i \log(\min(c_{i,j}, r_j) - \mu_j) \tag{5.20}$$

$$g_K(\mu) = \max_K \sum_j K_j(\mu_j - \log(K_j) + 1) \tag{5.21}$$

Here we propose a distributed algorithm that maximizes social welfare, i.e., sum utility of all players, also ensuring Pareto efficiency across all players.

The two maximizations, $f_x(\mu)$ and $g_K(\mu)$ can be solved separately by nodes in tiers T_n and T_{n-1} in a distributed manner. We propose Algorithm 1 for MTCDs in two neigboring tiers. In each iteration, the part for the upper tier runs with an updated association variable and determines the multiplier μ before the lower tier. This ensures each MTCD player in the lower tier n always gets updated information of their potential relays before commencing the subgame.

The multiplier μ_j act as a price sent by upper tier relays determined by their load situation, which can be either positive or negative. If $\sum_i x_{i,j}(t)$ exceeds $K_j(t)$, μ_j increases, otherwise it decreases. Therefore, when relay j is overloaded, it will increase its price μ_j and fewer devices will associate with it, while other underloaded relays will attract more devices.

Algorithm 1: Distributed Algorithm for MTCDs of neighboring tiers in subgame

1: **MTCDs in lower tier**

(a). Detect uplink SINR by feedback from upper tier node, and receive μ broadcast by each upper tier node.

(b). Determine BS $j*$ satisfying:

$j^* = \arg\max_j(K_i \log(\min(c_{i,j}, r_j)) - \mu_j)$

2: **MTCDs in higher tier or BS**

(a). To obtain the maximizer of (19), we set its gradients to be 0,

$K_j(t+1) = \min\{N_d, e^{(\mu_j(t)-1)}\}$

(b). The Lagrange Multiplier value μ_j is updated as

$\mu_j(t+1) = \mu_j(t) - \delta(t) \cdot (K_j(t) - \sum_i x_{ij}(t))$

Where $\delta(t) > 0$ is the step size.

The algorithm can be viewed as a game with players $i \in T_n$. Each player has action space $\boldsymbol{X}$, which is the decision of choosing MTCD j from $T_n - 1$, or BS j if $n = 1$ to associate with. The goal of any player i is to maximize its utility function:

$$U_i^n = \max_j(K_i \log(\min(c_{i,j}, r_j)) - \mu_j), \forall j \in T_{n-1} \tag{5.22}$$

The association scheme presented in Algorithm 2 requires few iterations to converge to the social optimum.

After a few iterations, and when system convergence is detected, transmit power of MTCDs is adjusted to:

$$p_i = \left(2^{(r_i/W)} - 0.9\right) \cdot E\left[\frac{g_{i,j}}{\sigma^2}\right] \tag{5.23}$$

where W is transmit bandwidth, and $E[\frac{g_{i,j}}{\sigma^2}]$ is estimation of channel gain divided by noise.

Algorithm 2: Distributed Algorithm Iteration for MTCDs

1: $n \leftarrow 1$

2: MTCD j in T_n determines its new tier assignment by (5.13) through updated achievable rate information

$\{c_{j,k}\}$ exchanged between MTCDs and BSs

3: **While** $n \leqslant N$ **do**

4: Lagrange Multiplier μ_j is updated by **Algorithm 1.2** with current association:

$\mu_j \leftarrow \mu_j - \delta \cdot (K_j - \sum_i x_{i,j})$

5: Resource block allocated to j, y_j is updated by **Lemma 1**

6: **if** all MTCDs in T_n finish step 4 and 5 **then**

7: $n \leftarrow n + 1$

8: **end if**

9: MTCD i in T_n determines its new tier assignment by (5.14) through updated achievable rate information $\{c_{j,j'}\}$ exchanged between MTCDs

10: MTCD i in T_n determine its association variable

by **Algorithm 1.1**:

$x_{i,j} \leftarrow \arg\max_j (K_i \log(\min_{i,j}, r_j)) - \mu_j)$

11: Assisted number is updated: $K_j = \sum_i x_{i,j} K_i$

12: **end while**

Nash Equilibrium and Social Welfare

A subgame n is defined as the game of MTCDs $i \in T_n$ as players to determine association with MTCDs $j \in T_{n-1}$ or $j \in B$ if $n = 1$.

Theorem 1 *There exists a pure strategy Nash Equilibrium (NE) in each association subgame, which after a few iterations will converge to achieve maximal sum utility in the subgame.*

PROOF. Suppose ${x_i^n}^*(\mathbf{x_{-i}^n}^*)$ is the best response of player i in subgame n to the actions of all other players such that ${x_i^n}^*(\mathbf{x_{-i}^n}^*) = \arg\max_{x_i} U_i(x_i^n, \boldsymbol{X_{-i}^n})$

The action space of each player is convex and compact, while best response ${x_i^n}^*(\mathbf{x_{-i}^n}^*)$ of each player is nonempty and convex, and U_i^n is continuous and compact,which proves the existence of NE in the subgame.

Sum utility optimisation in each subgame is solved by performing lagrange dual decomposition giving a NE best-response utility for an MTCD incorporating μ ,the data rate offered by the upper node, and the demanded rate each MTCD wants to transmit at, which rapidly converges to the optimal solution of the lagrangian function, and sum utility is maximized. □

Lemma 2 *The complete game has a subgame perfect equilibrium (SPE).*

PROOF. Because every finite extensive form game has a subgame perfect equilibrium [173], the NE reached in each subgame is part of the NE in complete game. The association variables $\{x_{i,j}\}$ determined by each subgame form the complete games association, independent of the history. □

Note that for each subgame, $U_{i,j}^n = (K_i \log r_j - \mu_j)$. For each device, $\frac{\partial U}{\partial r_j} = \frac{y_j}{r_j}$, which means the maximization of the relay rate, r_j, strictly increases the utility of lower-tier devices.

Remark 1 *The social optimum for each subgame directly implies a social optimum across the complete game.*

5.1.4 System Evaluation

We consider a network cell with 1 base station, noting that the analysis so-far presented is independent of the spatial distribution of BS and MTCDs. We model the location of the BS as fixed, and the location of devices are distributed as independent Poisson Point Processes in a 1 km^2 circular area. In modeling the wireless channel, we use a path loss $L(d) = 37 + 33\log(d)$ for air interface channels, and the thermal noise power is $\sigma^2 = -104$ dBm. We suppose the power each device can transmit ranges from 0.01 mW to 100mW, and the sum of available resource blocks is normalized to 1.

The resulting association is shown in Fig. 5.2 from the simulation of Algorithm 1 for 200 devices. In Fig. 5.2, one base station and 200 randomly distributed MTCDs are located in the area, in which the red diamond in the center represents the BS, and MTCDs are denoted by blue circles. If a red circles is around a MTCD, it indicates that this particular MTCD is used as a relay. The blue lines between MTCDs indicate association links. From the scheme, it is shown in Fig. 2 that after each MTCD maximizes their utility in each subgame through deciding upon association, data is then aggregated into relays. Relays assisting these MTCDs in turn become players in the next subgame closer to the BS, determining the association to next tier relays, thereby maximizing data rate.

As a performance measure, Fig. 5.3 shows the cumulative distribution function (CDF) of long-term transmission rates in an MTC cellular network with different association schemes. In the network, the CDF for the proposed algorithm is compared with the state-of-art location-based algorithm from [174] and the popular

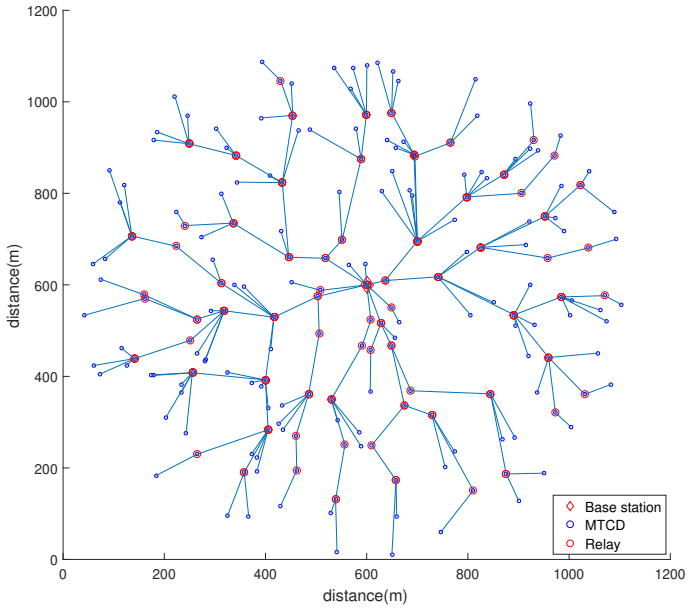


Figure 5.2: Association scheme in a M2M cellular network with 200 MTCDs

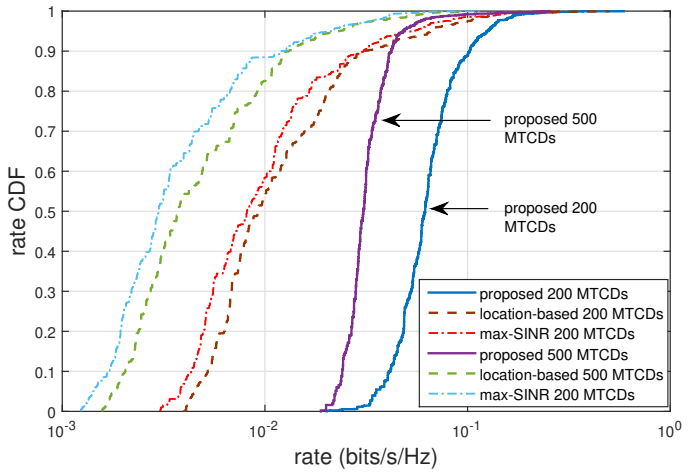


Figure 5.3: The cumulative distribution functions (CDFs) of rate per device in the network cell with 200 and 500 MTCDs

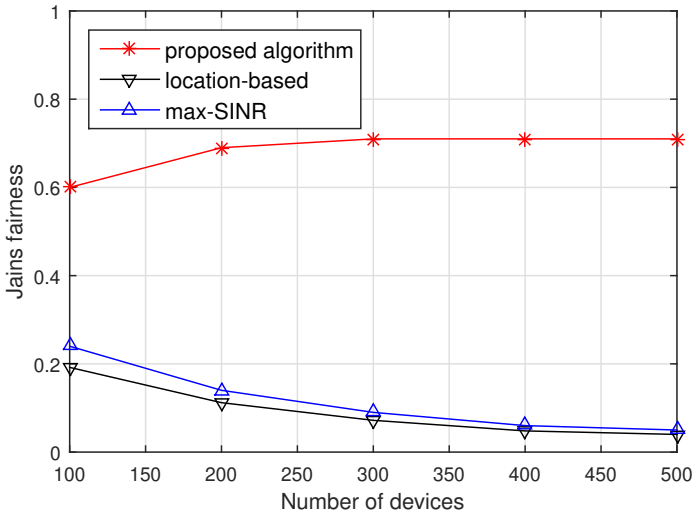


Figure 5.4: Jain's fairness index for rate of all MTCDs

max-SINR association scheme [28]. The CDF for the proposed distributed association shows large improvements. Fig. 5.3 shows the simulation for both 200 MTCDs and 500 MTCDs, 10% of all devices achieve a transmission rate larger than 0.02 bits/s/Hz and 0.07 bits/s/Hz respectively, while half of these achieve rates larger than 0.2 bits/s/Hz and 0.45 bits/s/Hz, respectively. Only 10% of all MTCDs have rate below 0.033 bits/s/Hz and 0.042 bits/s/Hz. From Fig. 5.3, it can be seen that the middle 80% of devices achieve a 6× gain and a 10× gain vs location-based and max-SINR schemes, respectively.

Jain's fairness index is a metric for evaluating the fairness of the rate of all MTCDs $J = \frac{(\sum_{i=1}^{K_{all}} \eta_i)^2}{K_{all} \sum_{i=1}^{K_{all}} \eta_i^2}$. Fig. 5.4 shows that in the proposed algorithm, Jain's index increases and converges to 0.71 as the number of devices approaches 500. By contrast, in location-based and max-SINR schemes, the Jain's index decreases to 0.05. Thus, overall system fairness is significantly improved, and stability is increased globally.

Finally, even though relay nodes needs to transmit at a higher rate to assist others transmissions, potentially consuming more power, each device aims to bet-

ter utilize wireless channels and associate with closer relays, so the power saved compensates for the higher data rate. The total power consumed in the proposed algorithm is 15.7 W, 12% lower than that of the max-SINR algorithm.

5.2 Improving Fairness for Spectrally-Efficient Downlink Scheduling

Instead of reusing existing UE as relays, as presented in Section 5.1, here we deploy relay access points (APs) in a downlink mmWave cellular network to help alleviate blockage and shorten transmission distances in Section 5.2. A cross-tier scheduling scheme which ensures every UE is served by AP while optimising sum throughput is presented.

5.2.1 Introduction

One promising solution to overcrowded microwave bands spectrum is to use mmWave bands (30–300 GHz), which provides abundant spectrum to support high data rates. However, traditional cellular network architectures are not suited to the unprecedented traffic. Characteristics of mmWave communication such as directional beamforming, susceptibility to blockage [175], and resultant smaller radio interference, are not considered in traditional architectures. Denser BS deployment [176] and heterogeneous networks [177] have both been proposed to compensate for high path loss and blockage, but at a heavy deployment cost. This has led to several proposals for multi-tier network architectures, comprised of one BS, densely deployed APs and many user equipments (UEs), e.g., [4, 178]. The BS communicates with APs and manages the whole network. APs serve as relays to forward data to UEs from the BS in downlink. UEs have the flexibility to asso-

ciate with any of the APs. Such multi-tier architecture solves coverage issues and blockage by leveraging multi-hop diversity.

Coverage performance of a naive 2-tier mmWave network, in which UEs associate with the nearest relay, is studied in [122]. The throughput capacity of a multi-tier mmWave network, considering various blockage probabilities, is characterized in [179]. A multi-hop obstacle avoidance algorithm, by steering blocked flows around obstacles is proposed in [178], with demonstrated superior performance in network throughput and connection robustness in small cells . An efficient association and multi-hop relaying scheme, with rate control between users and access points is proposed in [120], whereby overall transmission rate is maximized and power consumption is minimized, while ensuring load balancing and proportional fairness among users. The user association and backhaul problem is jointly considered in [4], regulating the transmissions in all tiers to achieve optimal sum throughput. However, service coverage is not taken into consideration in [4], leaving many UEs un-associated with APs and resulting in poor fairness. *Thus, here we propose a cross-tier scheduling algorithm for a multi-tier downlink mmWave network that optimises the overall downlink throughput while ensuring every user is served by APs.* The main contributions of this chapter are:

1. We formulate downlink cross-tier scheduling as a non-linear binary integer programming problem, which maximizes sum throughput from BS to UEs via APs.

2. We design an iterative scheduling scheme utilizing AP to AP links, which ensures all UEs are served with polynomial computational complexity.

3. We derive a sum throughput upper bound for the multi-tier downlink mmWave network.

4. The proposed scheme demonstrates far superior fairness to the state-of-art, and similar spectral efficiency.

5.2.2 System Model

Network Model

We now describe the downlink 2-tier mmWave cellular network, as shown in Fig. 5.5. We assume a single cell with a single BS at the center, m APs and k UEs. APs are relay nodes controlled by BS, which store the data transmitted to them and forward it to other APs or UEs. We model the spatial location of APs and UEs as 2D homogeneous Poisson point processes (PPPs). A PPP defined in $\mathbb{R}$ is a random process with number of points Φ in a bounded Borel set $\mathcal{B} \subset \mathbb{R}^2$:

$$\mathbb{P}(\Phi(\mathcal{B}) = b) = \frac{\Lambda^b}{b!} e^{-\Lambda}, b = 0, 1, 2, \cdots \tag{5.24}$$

where $\Lambda = \lambda v_2(\mathcal{B})$ is the expectation of the Poisson random variable for some intensity λ, and $v_2(\cdot)$ denotes the Lebesgue measure in $\mathbb{R}^2$.

In this chapter we focus on the scheduling of downlink transmissions from BS to UEs through APs. APs receiving data directly from BS are defined as direct APs, and they forward the data to other APs or UEs. APs receiving data from direct APs are defined as secondary APs, who can only forward the data to associated UEs meaning the maximal hop number is limited to 3. We assume the APs have full-duplex transmissions that allows them to receive and forward at the same time. We assume that UEs seek to continually receive data from the BS, but UEs can only associate with APs relaying data to UEs from the BS.

We label the BS as device 0. The APs are labeled as device $1, 2, 3 \cdots m$ and UEs are labeled as device $m+1, m+2, m+3 \cdots m+k$. Thus $\mathcal{D} = \{0, 1, 2, \cdots m+k\}$ represents the set of all devices in the network. We define the set of all APs as $\mathcal{M}$,

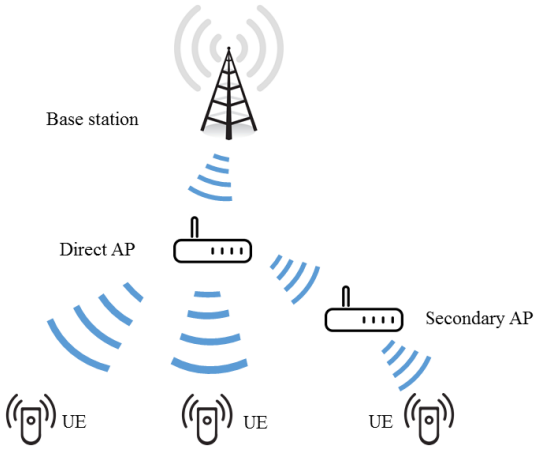


Figure 5.5: The multi-tier mmWave network with AP-AP communications.

and the set of all UEs as $\mathcal{K}$. The transmission link from device $i \subset \mathcal{D}$ to device $j \subset \mathcal{D}$ is denoted as (i, j).

The whole data transmission duration in a frame is denoted by T, which can be divided into N mini-slots, each of length $T_s = \frac{T}{N}$. A schedule defines which links are allowed to transmit in which mini-slot. We define a binary variable $x_{ij}(n)$ to represent the link transmission (i.e., $x_{ij}(n) = 1$ if link (i, j) is scheduled to transmit in mini-slot n, otherwise $x_{ij}(n) = 0$). Denote by $X = \{x_{ij}(n) : \forall i, j \in \mathcal{D}, n \in 1, 2 \cdots N\}$.

Channel Model

mmWave systems leverage directional beam forming to compensate for high path loss and minimize interference. According to the typical cone-sphere model, a beam pattern has a conical main lobe and a spherical side lobe. We denote by G^m and G^s the main lobe gain and side lobe gain, respectively. For a link (i, j), if the receiver j is located in the main lobe of the signal pattern, the beamforming gain at transmitter j equals to G_i^m; otherwise the beamforming gain at transmitter is G_i^s. Similarly, we define G_j^m and G_j^s as the main lobe gain and side lobe gain

respectively at receiver side. Hence in 2D, the beamforming gains are

$$G^m = \varrho\frac{2\pi}{\theta} \quad and \quad G^s = (1-\varrho)\frac{2\pi}{2\pi-\theta}, \tag{5.25}$$

where ϱ is the ratio of main lobe power to total power, θ denotes the beamwidth of the main lobe and is determined by properties of the antenna array, e.g., larger antenna arrays have higher ϱ and a narrower beam.

We assume the BS has L radio frequency (RF) chains, while each AP has ℓ RF chains. Each RF chain can serve 1 data stream. We also assume that each device serves multiple simultaneous transmission links in a non-orthogonal mode without interference mitigation, hence there exists inter-beam interference between links simultaneously served by the same device. We define $I_{ij}(X,n)$ as the interference experienced by link (i,j) in scheduling X at mini-slot n.

We consider the following path loss model without fading effects. Due to high attenuation in mmWave, multipath effects are negligible. For all scheduled links (i,j), i and j direct their main lobes to each other, which gives channel gain:

$$h_{ij} = \gamma G_i^m G_j^m d_{ij}^{-\alpha} B_{ij},\ \forall i,j \in \mathcal{D}, \tag{5.26}$$

where γ denotes the frequency dependent coefficient, G_i^m and G_j^m are transmit and receive main lobe antenna gains respectively, d_{ij} denotes the distance between i and j and B_{ij} is a binary variable denoting blockage (i.e., $B_{ij}=1$ if there is line of sight (LOS) link between i and j, otherwise $B_{ij}=0$) The probability of a LOS link existing depends on obstacle density and transmission distance:

$$Pr(B_{ij}=1) = P_{LOS}(d) = e^{-\beta d_{ij}}, i,j \in \mathcal{D}, \tag{5.27}$$

where β denotes linear obstacle density. We assume the links to be either LOS or blocked because NLOS links at mmWave usually have such high path loss that they can not be used.

The signal-to-interference-and-noise-ratio (SINR) of link (i, j) at mini-slot n is thus

$$SINR_{ij}(X, n) = \frac{P_i h_{ij} x_{ij}(n)}{N_0 W + I_{ij}(X, n)}, \tag{5.28}$$

where P_i denotes the transmit power of i, W denotes the bandwidth and N_0 denotes the noise spectral density. According to Shannon capacity, at mini-slot n, the achievable rate of link (i, j) is

$$R_{ij}(X, n) = W \log_2(1 + SINR_{ij}(X, n)). \tag{5.29}$$

Problem Formulation

Here, we aim to design a cross-tier scheduling algorithm that optimises the overall downlink throughput while ensuring every UE is served by APs. Existing scheduling schemes either focus on resource allocation between tiers with limited flexibility in link association [180, 181], or purely focus on throughput, leaving some UEs unserved [4]. Thus, a cross-tier scheduler in a flexible association scenario regarding service coverage is needed, leading to our following definition of the scheduling problem,

$$\max_{X} \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{K}} \sum_{n=1}^{N} R_{ij}(X, n) x_{ij}(n) T_s \tag{5.30}$$

$$s.t.$$

$$(\sum_{j \in \mathcal{M}} \sum_{n=1} x_{ij}(n) T_s)(\sum_{j \in \mathcal{M}} \sum_{n=1} x_{ji}(n) T_s) = 0, \ \forall i \in \mathcal{M}, j \neq i \tag{5.30a}$$

$$\sum_{n=1}^{N} R_{0i}(X, n) x_{ij}(n) T_s + \sum_{j \in \mathcal{M}} \sum_{n=1}^{N} R_{ji}(X, n) x_{ji}(n) T_s \leqslant \sum_{k \in \mathcal{K}} R_{ik}(X, n) x_{ik}(n) T_s, \ \forall i \in \mathcal{M} \tag{5.30b}$$

$$\sum_{n=1}^{N} x_{0i}(n) = \frac{NL}{m'}, \ \forall i \in \mathcal{M}' \tag{5.30c}$$

$$\sum_{j \in \mathcal{M}} x_{0i}(n) \leqslant L, \tag{5.30d}$$

$$\sum_{j \in \mathcal{M} \cup \mathcal{K}} x_{ij}(n) \leqslant \ell, \ \forall i \in \mathcal{M} \tag{5.30e}$$

$$\sum_{i \in \mathcal{M}} x_{ij}(X, n) = 1, \ \forall j \in \mathcal{K} \tag{5.30f}$$

$$x_{ii}(n) = x_{i0}(n) = x_{0j}(n) = x_{ji}(n) = 0, \quad \forall i \in \mathcal{D}, j \in \mathcal{K}, n \leqslant N \tag{5.30g}$$

$$x_{ij}(n) \in \{0, 1\}, \ \forall i, j \in \mathcal{D}, n \leqslant N, \tag{5.30h}$$

where (5.30) is the objective function, which maximizes the sum throughput from BS to UEs via APs. Constraint (5.30a) implies that an AP can not be at both transmit and receive side of an AP to AP link in each frame. Constraint (5.30b) guarantees that AP output throughput is lower than input throughput. In Constraint (5.30c), m' denotes the number of direct AP, and $\mathcal{M}'$ denotes the set of all direct APs, and (5.30c) implies that each direct AP shares the same resource allocation from the BS. Constraints (5.30d) and (5.30e) imply that the association number in any mini-slot is less than its number of RF chains. Constraint (5.30f) implies that every UE is served by one AP. Constraint (5.30g) enforces the downlink multi-tier architecture, such that the BS can not transmit directly to UEs, but only through APs. Constraint (5.30h) denotes the schedule for each link as a binary variable.

Formulation (5.30) is a non-linear binary integer programming problem, which is NP-hard in the number of network devices. Due to such impractical computational complexity, in Section 5.2.3, we propose a practical scheduling scheme for the multi-tier mmWave network.

5.2.3 Proposed Practical Scheduling Scheme

To address the impractical computational complexity of problem formulation (5.30), we design an iterative scheduling algorithm to approximately solve (5.30) that has polynomial computational complexity. The algorithm has 4 steps:

Step 1: AP Selection

In this step, the BS selects the APs to serve and determines the transmission duration for each AP. Define $\mathcal{M}^*$ as a relabeling of all APs in descending order of the achievable rate from BS. The first L APs in $\mathcal{M}^*$ are selected as direct APs and put in the set $\mathcal{M}'$, each allocated with the same transmission duration T.

The transmission duration of link $(0, i)$ in a frame for any $i \in \mathcal{M}^*$ is

$$t_{0i}^* = \begin{cases} T, & 0 < i < L, \forall i \in \mathcal{M}^* \\ 0, & i > L, \forall i \in \mathcal{M}^*. \end{cases} \tag{5.31}$$

All the other APs with 0 time allocation are put into indirect APs set $\mathcal{M}^0$

Step 2: AP To UE Matching-1

Next, we determine the link schedules that maximize the sum output throughput from direct APs in $\mathcal{M}'$ to indirect APs $\mathcal{M}^0$ and UEs in $\mathcal{K}$. Such scheduling requires the throughput knowledge of every potential link $(i, j), i \in \mathcal{M}', j \in \mathcal{M}^0 \cup \mathcal{K}$. With directional beamforming, we can effectively reduce the interference. However, although the interference between independent links (links with both different transmitter and receivers) is negligible due to rare probability, inter-beam interference (links with the same transmitter but a different receiver) is non-negligible. Therefore, for each AP i, we divide the devices $j \in \mathcal{M}^0 \cup \mathcal{K}$ into groups such that there does not exist inter-beam main-lobe interference between groups. Denote by $\mathcal{I}_{ij}$

the non main-lobe interference group that device j belongs to. We define

$$\delta_{ij} = \begin{cases} 1, & \text{if } j = \arg\max_{k \in \mathcal{I}_{ij}} R_{ik} \\ 0, & \text{if } j \neq \arg\max_{k \in \mathcal{I}_{ij}} R_{ik}. \end{cases} \tag{5.32}$$

If j has the highest rate link from i among devices in $\mathcal{I}_{ij}$, $\delta_{ij} = 1$, else $\delta_{ij} = 0$. We give a weight w_{ij} to each link (i, j)

$$w_{ij} = \delta_{ij} \frac{R_{0i} R_{ij}}{\eta_i}, \tag{5.33}$$

where η_i is the averaged sum-rate of concurrent scheduled links from AP i to other secondary APs or UEs.

This AP-UE scheduling process can then be abstracted as a weighted matching problem with weights w_{ij} in a bipartite graph. By using the Hungarian algorithm that finds the perfect matching in a weighted bipartite graph [182], the first matching problem is solved with computational complexity $O(m(m+k)^2)$, due to at most $m + k$ nodes on one side of the bipartite graph and at most $m(m + k)$ edges.

Step 3: AP to UE Matching-2

After matching from direct APs to indirect APs and the UE combined set, we define the set of APs that are matched to direct APs as secondary AP set $\mathcal{S}$. We then can solve the problem of matching secondary APs to UEs. For each direct and secondary AP $i \in \mathcal{M}' \cup \mathcal{S}$, we divide UEs $j \in \mathcal{K}$ into groups without main-lobe interference the same way as for step 2. Denote by $\mathcal{I}'_{ij}$ the non main-lobe interference group that device $j \in \mathcal{K}$ belongs to. We define

$$\delta'_{ij} = \begin{cases} 1, & \text{if } j = \arg\max_{j \in \mathcal{I}'_{ij}} R_{ij} \\ 0, & \text{if } j \neq \arg\max_{j \in \mathcal{I}'_{ij}} R_{ij}. \end{cases} \tag{5.34}$$

The weight of each link $(i, j), i \in \mathcal{M}' \cup \mathcal{S}, j \in \mathcal{K}$ is defined:

$$w'_{ij} = \begin{cases} \delta'_{ij} \frac{R_{0i} R_{ij}}{\eta_i}, & \text{if } i \in \mathcal{M}' \\ \delta'_{ij} \frac{R_{0h} R_{hi} R_{ij}}{\eta_h \eta_i}, & \text{if } i \in \mathcal{S}, h \in \mathcal{M}', \end{cases} \tag{5.35}$$

where h is the associated direct AP in step 2 if $i \in S$, and η_h and η_i is the averaged sum-rate of concurrent scheduled links from direct AP h and from secondary AP i to associated devices, respectively. This AP-UE scheduling process can also then be abstracted as a weighted matching problem with weights w'_{ij} in a bipartite graph, solved by the Hungarian algorithm with computational complexity $O(mk^2)$, due to at most k nodes on one side of the bipartite graph and at most mk edges.

Step 4: Iterative Scheduling

We repeat Step 2 and Step 3 to iteratively improve the scheduling. After each iteration, we select the AP i from the indirect AP set $\mathcal{M}^0$ with the highest achievable rate from BS, $i = \arg\max_{i \in \mathcal{M}^0} R_{0i}$. The selected AP then becomes a direct AP, meaning it is moved from $\mathcal{M}^0$ to the direct AP set $\mathcal{M}'$. As the number of direct APs increases, the transmit duration allocation in each frame is rescheduled to satisfy an equal resource allocation constraint:

$$t^*_{0i} = \begin{cases} \frac{LT}{m'}, & \forall i \in \mathcal{M}' \\ 0, & \forall i \in \mathcal{M}^0. \end{cases} \tag{5.36}$$

where m' denotes the number of direct APs. As Steps 2 and 3 are iterated, we increase the number of direct APs, and as a result, the total number of served UEs by either direct APs or secondary APs. The iterations stop when every UE in $\mathcal{K}$ is matched with an AP.

5.2.4 Performance Analysis

Computational complexity

Running the proposed scheduling algorithm, combining the computational complexity of steps 2 and 3, and the number of iterations in step 4, the overall computational complexity is $O\big(m(m-L)((m+k)^2+k^2))\big)$.

Throughput upper-bound analysis

The optimal schedule solved from (5.30) should be compared with the proposed algorithm to measure the performance. However, the impractical computational complexity of (5.30) implies that determining a throughput upper bound is a more feasible way to evaluate performance.

Assumption 1: The BS and AP have very narrow beamwidths, thus main-lobe interference is negligible.

Each link (i,j) suffers from side-lobe inter-beam interference caused by other concurrent links associated with transmitter i, so

$$R_{ij} = W \log_2 \left(1 + \frac{P_i h_{ij} x_{ij}(n)}{N_0 W + \sum_{j' \in \mathcal{D}} x_{ij'}(n) P_i h_{ij'}} \right). \tag{5.37}$$

Assumption 2: When the network has enough density, even with blockage, direct APs are associated with secondary APs and/or UEs whose combined downlink achievable throughput is higher than the associated direct AP's downlink throughput.

Thus, with the restriction of equal resource allocation among direct APs, sum throughput is upper bounded by the maximum throughput from BS to direct APs. To serve k UEs, at least $\frac{k}{L\ell}$ direct APs are needed.

Based on the system model in Section 5.2.3, the expected distance of ith nearest

LOS AP to the BS is

$$\mathrm{E}[d_{LOS}(i)] = \int_0^\infty \frac{2(\pi\lambda_{LOS}(d))^i}{(i-1)!} d^{2i} e^{-\pi\lambda_{LOS}(d)d^2} \mathrm{d}d, i \in \mathcal{M}, \tag{5.38}$$

where $\lambda_{LOS}(d)$ is the LOS AP intensity within distance d,

$$\lambda_{LOS}(d) = \int_0^d 2\pi d\lambda_m P_L(d) \mathrm{d}d \tag{5.39}$$

$$= 2\pi\lambda_m \left(\left(-\frac{d}{\beta} - \frac{1}{\beta^2} \right) e^{-\beta d} + \frac{1}{\beta^2} \right), \tag{5.40}$$

where $\lambda_m = \frac{s}{m}$ denotes the AP density in the area and s is the cell size. Therefore, the network sum throughput upper bound is given as

$$G = \sum_{i=1}^{\lceil \frac{k}{L\ell} \rceil} W \log_2 \left(1 + \frac{P_i \gamma G_0^m G_i^m E[d_{LOS}(i)]^{-\alpha}}{N_0 W + L P_i \gamma G_0^s G_i^m E[d_{LOS}(i)]^{-\alpha}} \right) \frac{L T_s}{\lceil \frac{k}{L\ell} \rceil}, \tag{5.41}$$

where i denotes the ith nearest LOS AP to BS, and operator $\lceil n \rceil$ represent the closest integer above n.

5.2.5 System Evaluation

In this section, we numerically evaluate our proposed algorithm's performance against 3 benchmark schedulers. We will measure the performance of schedulers in terms of spectral efficiency, service coverage and fairness.

Benchmark schedulers

The first benchmark scheduler is the direct access scheme that does not have APs. The BS directly transmits to L UEs with the highest achievable rates, each allocated the same transmission duration. The second benchmark is a greedy two-tier scheduler [3], where each AP independently picks at most ℓ UEs with the highest achievable rate, and the BS serves APs in descending order of achievable rates

$R_{0i}, \forall i \in \mathcal{M}$. The third benchmark is the cross-tier scheduler in [4], which jointly considers BS-AP, AP-AP and AP-UE transmissions in scheduling, aiming at maximizing sum throughput. These three benchmarks all seek to maximize network overall throughput, but give little regard to fairness and service coverage.

Simulation Setup

In simulation, the BS is located in the center of a cell of size $200m \times 200m$. APs and UEs are distributed in the cell according to a PPP. For all simulations, we assume that there are 24 APs. We assume that the beamwidth of the BS is 1°, the beamwidth of APs are 2° and the beamwidth of UEs are 18°, due to having much smaller antenna arrays. Usually BS and APs have limited RF chains, thus we assume the BS has $L = 5$ RF chains, while APs have $\ell = 3$ RF chains. Each UE has only 1 RF chain due to limited hardware. The transmitting power of each device is $P_i = 0.5$mW, and the bandwidth is $W = 1$GHz. Frequency dependent coefficient $\gamma = -51$dB, and the path loss exponent $\alpha = 2$. Noise spectral density is $N_0 = -90$dBm/MHz. Ratio of main lobe power to total power $\varrho = 0.9$. The linear density of obstacles $\beta_1 = 0.01/m$, $\beta_2 = 0.005/m$ and $\beta_3 = 0.0025/m$. The duration of a mini-slot $T_s = 100$ μs, and the data transmission duration in a frame $T = 50$ ms, thus each frame is comprised of $N = 500$ mini-slots.

Results and discussion

Fig. 5.6 shows the spectral efficiency of the network as a function of number of UEs. The spectral efficiency of our proposed algorithm first increases with number of UEs, approaching the upper bound in (5.41). The peak spectral efficiency of 35 bits/s/Hz appears at $k = 30$ UEs, slightly lower than theoretical upper bond of 38 bits/s/Hz. As the number of UEs further increases in Fig. 5.6, the spectral

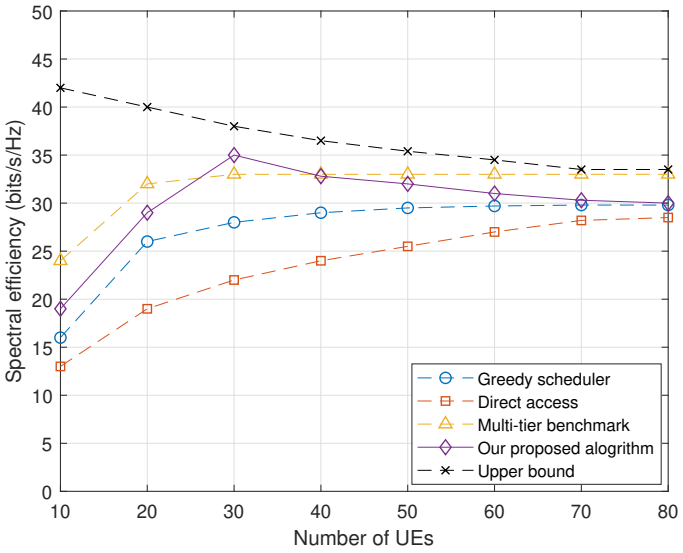


Figure 5.6: The network spectral efficiency with respect to number of UEs (compared with greedy scheduler [3] and multi-tier benchmark [4]).

efficiency decreases slowly due to the fact that to serve every UE, more direct APs are employed, which increases the probability of using poorer BS to AP links. However, for $k = 30$ UEs, the spectral efficiency of our proposed algorithm is superior to the multi-tier benchmark algorithm introduced in [4], which does not guarantee every UE's association with an AP. The improvement of our algorithm over the greedy scheduler in Fig. 5.6 shows the benefit of utilising AP-AP links and joint scheduling of BS-AP, AP-AP and AP-UE links. The further degradation of direct access to all other schemes in Fig. 5.6 demonstrates performance benefits of a multi-tier network architecture.

In Fig. 5.7, we evaluate the fairness of our algorithm, and compared with the other two benchmarks. Jain's fairness index is defined as

$$J = \frac{(\sum_{i=1}^{k} r_i)^2}{k \sum_{i=1}^{k} r_i^2}, \ \forall i \in \mathcal{K}. \tag{5.42}$$

where r_i represents the actual UE throughput. This metric measures whether the algorithm serves all UEs fairly. As k increases in Fig. 5.7, Jain's fairness inevitably

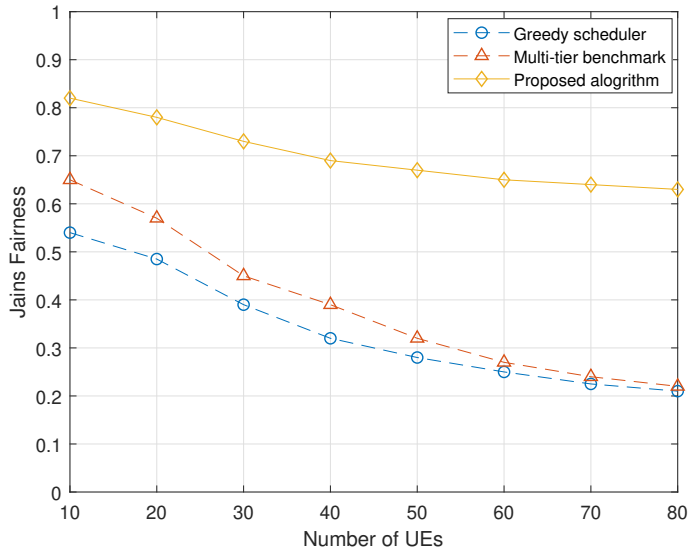


Figure 5.7: The UE throughput Jain's fairness with respect to number of UEs (compared with greedy scheduler [3] and multi-tier benchmark [4]).

decreases for all schemes as they are throughput-centric. However, our algorithm serves all UEs, while the multi-tier benchmark in [4] allows UEs to not be served. E.g, for 60 and 80 UEs in total, only 42.8 and 33.2 UEs are respectively served on average applying the solution in [4]. This gives our algorithm a large fairness advantage over [4], which is clearly shown in Fig. 5.7. The greedy scheduler performs worst in throughput fairness. As the name indicates, greedy scheduler let APs pick UEs with highest achievable rate, leaving many further UEs unserved which results in unfair services. By comparing Fig. 5.6 and Fig. 5.7, it can be seen that the fairness performance of our proposed algorithm significantly outperforms multi-tier benchmark algorithm, without sacrificing much spectral efficiency for a larger number of UEs.

5.3 Concluding Remarks

This chapter provided solutions for throughput and fairness oriented relaying in uplink and downlink cellular networks. The research assumes large number of devices in the network, and utilises multi-hop relaying to efficiently solve the congestion problem.

More specifically, in Section 5.1, we proposed a relaying and association scheme maximizing overall rate while achieve rate balancing in M2M uplink cellular networks. A logarithmic utility maximization for the overall network was decomposed into subgames. In each subgame, proportional resource allocation was optimal, using a distributed algorithm with dual decomposition. The distributed algorithm converged to a social optimum in a small number of iterations. We demonstrated large improvements of the proposed association algorithm over a location-based and max-SINR based relaying approach with respect to rate per device, Jain's fairness index and power consumption. Thus, for non-critical machine-type communication data, it is preferable to use such multi-hop communications incorporating distributed optimisation, also feasible as it introduces little extra overhead.

In section 5.2, we analysed the cross-tier mmWave downlink network scheduling problem, and formulated a multi-tier sum throughput optimisation problem where every UE is associated. Although the problem formulation is NP-hard, we developed an iterative scheduling algorithm, which has polynomial computational complexity. By relaxing the formulation according to suitable assumptions, we derived a sum throughput upper bound. Numerical results showed that the proposed algorithm approaches the sum throughput upper bound, significantly outperforming benchmark schedulers in terms of fairness while maintaining good spectral efficiency.

Chapter 6

Conclusion and Future Work

This chapter makes some concluding remarks about the contributions made by this thesis and discusses potential future directions of this work.

6.1 Conclusion

Here we provided some concluding remarks regarding our key contributions with respect to random access channel (RACH) optimisation, and multi-hop relaying optimisation, in cellular IoT

6.1.1 Deep Reinforcement Learning Based Optimisation of Random Access

To optimise RACH efficiency for M2M communications, we developed a DRL based controlling scheme, which adjusts the access class barring (ACB) factor and the back-off (BO) window for machine-type communication devices of different priority. We found that even though by applying an ACB scheme only we can significantly improve the efficiency of RACH, when the BO window is dynamically adjusted

in conjunction with an ACB scheme the efficiency can be further optimised. In comparison, exponential BO, which is widely used in WLANs, can not achieve as good efficiency as a DRL based ACB-BO scheme.

However, due to the fact that LTE was originally designed for human-to-human (H2H) communications, the current RACH has intrinsic constraints on simultaneous access number, even in the case of LTE-M, which was designed to reuse large of LTE technology to support IoT. Thus, we believe fundamental work needs to be done in the RACH in next generation cellular networks, in order to support the massive connection of cellular IoT devices, which require services beyond what can be provided by NB-IoT. For example, for small data packets, some researchers have proposed uplink grant-free RA solutions which piggyback small payloads to control messages in handshake procedure. However, such scheme still faces collision problem.

As we have shown that DRL can be utilised as a tool to optimise network parameters better than other prediction-based controlling schemes for next generation cellular IoT, we think that DRL-based algorithms could be applied to other key aspects of cellular networks optimisation.

6.1.2 Minimum-Delay Relaying in mmWave Uplink

We proposed a mmWave communications uplink relaying scheme called "Minimum-Delay" relaying, i.e., a delay-oriented relaying scheme, which minimises each UEs end-to-end delay and improves the overall network uplink efficiency. In the simulation based on a real-world urban map, we have gotten some key insights into mmWave relaying. First of all, mmWave radio links are very vulnerable to obstacles. Thus two path loss models – one for line-of-sight links, one for non-line-of-sight links – are widely used in academia, and are also applied in our analysis and sim-

ulation. We found that relaying in mmWave has large advantages over traditional direct-associate communications in terms of delay and coverage.

Secondly, we found that the use of beamforming in mmWave relaying is very beneficial, such that interference among devices is so weak such that it can be ignored, according to simulations based on a real-world urban map.

Lastly, we examined our algorithm for both two-tier and three-tier architectures. Surprisingly, neither architecture is always superior to the other according to key metrics. A 3-tier architecture has relatively lower average delay in low traffic load situations due to shorter transmission distances and higher probability of LOS links provided by the additional relay tier, while 2-tier architectures perform better under heavy traffic pressure because queueing delays dominate end-to-end delay and end users need to queue twice in 3-tier architectures, at both the first and second tier relays.

6.1.3 Socially Optimal Distributed User Association for multi-hop Machine-Type Communications

There are a large number of IoT applications that are delay-tolerant, such as smart metering and environment monitoring, which fall into the mMTC service category. Even though the devices in this category each carry only a small amount of data, it is expected that the large number of them together will generate significantly moredata more than for all human-to-human communications in the future. Therefore, optimising system capacity is of utmost importance.

We used existing MTCDs as relays, forming an uplink relaying network employing D2D links. For optimisation, we defined a non-cooperative game for data rate and user association across MTCDs, and then decomposed the game into subgames, each consisting of a subset of MTCDs aiming to maximise the sum of utilities

for user association. A distributed algorithm applying Lagrangian decomposition allowed each subgame to thus reach a socially optimal data rate. Unlike other data aggregation schemes, such as location-based relaying or SNR-based relaying, our proposed throughput-oriented relaying approach not only reduced direct accesses to the BS and reduced transmission distances, it also balanced traffic loads, among relays, caused by unevenly distributed MTCDs. The results have shown that the fairness of rate of all MTCDs does not get worse as device density increases. As opposed to network densification (multiple relay nodes are deployed along the source-destination transmission path), traffic dispersion (decompose the original heavy arrival traffic into sub-stream and deliver them through independent paths) is also investigated in [128], and the advantages of each scheme is compared. A hybrid scheme combining both schemes is proposed to further reduce latency in certain scenarios.

6.1.4 Improving Fairness for Spectrally-Efficient Downlink Scheduling

In this work, instead of reusing existing UEs as relays, we deployed relay access points in a downlink mmWave cellular network to help alleviate blockage and shorten transmission distances. However, due to the challenge of the scheduling of mixed BS-AP, AP-AP, AP-UE links, we developed a cross-tier scheduling scheme, which ensured every UE is served by an AP while optimising sum throughput. From the numerical results, we found that by using additional APs in the network, spectral efficiency is significantly improved.

In the design of the relaying network, pursuing maximised overall throughput tend to cause unfair services to devices with poor channel quality, as more resource blocks are given to links that are already good in order to exploit spectral efficiency.

This kind of design is represented by a greedy scheduler [3], which gives little to non considerations to rate fairness. We designed the scheduler of the relaying network with more focus on providing fairer services to every device. The results have shown that we significantly outperformed benchmark schedulers in terms of fairness, while only a small portion of spectral efficiency is sacrificed.

6.2 Future Directions

The work presented in this thesis opens up several new research directions, outlined in the following:

6.2.1 DRL-based algorithm deployed inside end user

So far, we have presented our research in DRL-based algorithms for RACH control. However, the DRL agent was deployed inside the BS, limiting its potential to deliver even more customised services to each individual device. More customised services could be achieved by deploying a variant of our DRL algorithm, in end devices. Taking the RACH for example, multiple methods have been proposed for efficient power ramping — stepping up transmission power due to the outage condition caused by low preamble SINR — during random access in LTE. DRL-based algorithms can be deployed inside UEs to control transmission power of preambles, which is an efficient means to improve the detection rate. Recent research in multi-agent cooperation reinforcement learning gives inspiration as to how end users can use DRL to cooperate together in order to solve the interference problem [183]. However, there remain many challenges, including, e.g., size and capability of processing units, deployment price, and power consumption. A new collaborative machine learning approach known as federated learning [184]

gives a good direction to tackle these challenges. By contrast to standard machine learning approaches, in federated learning devices download the current model and improve it by learning from data on the device, and then summarize the changes as a small focused update. Only this update to the model is sent to the cloud, where it is averaged with other user updates to improve the shared model. According to Google [185], federated learning allows for smarter models, lower latency and less power consumption, while in addition to providing an update to the shared model, the improved model can also be used to provide customised services.

6.2.2 Delay Oriented Relaying

We have presented a multi-hop relaying scheme for mmWave cellular networks, which optimises uplink latency. However, we only compared the performance of the proposed algorithm in 2-tier and 3-tier architectures. Our architectures could be extended to a 4-tier architecture, or even architectures without a hop constraint. However, with more hop numbers, there are other problems to be overcome. The first problem is overhead. This is because, as packets will be forwarded multiple times, control information for D2D communications will inevitably affect efficiency, causing an undesirable waste of resources. Besides optimisation of overhead for many-tier architectures, for IoT devices with constraint battery resources, extra energy consumption is a major challenge. Whether the improved spectral efficiency, extended coverage and power saved by reduced transmission distances, due to more hops, could bring more advantage than the cost of energy consumed by further relaying, remains an open issue.

6.2.3 Throughput and Fairness Oriented Relaying

In the work of both "Socially Optimal Distributed User Association for MTC" and "Improving Fairness for Spectrally-Efficient Downlink Scheduling", the relaying and scheduling schemes are designed to be served by one macro base station. However, the emerging trend in cellular networks is to deploy many small cells to form a heterogeneous network, where macro base stations provide wide area coverage, while small base stations act as aids to provide better services and fill in areas not covered by the macro cell. In such architectures, different cells may overlap, giving devices in the overlap area freedom to choose base stations. One research area could be extending the proposed schemes to optimise relaying in a heterogeneous network. Coordinated multi-point (ComP) technology [186], which coordinates the transmission and reception for a UE to and from more than one base station, brings many advantages in a heterogeneous network. By providing connections to multiple base stations, data can be pass through less loaded base stations for better resource allocation. Furthermore, using several base stations for each connection means that both uplink and downlink signal quality can be improved. Therefore, ComP technology brings new ways to to extend our relaying design for heterogeneous networks.

Appendices

Appendix A

Proof of chapter 4 - Lemma 1

For each base station, the probability distribution function (PDF) of the distance d of the i_{th} nearest LOS MTCD $f(d,i)$ is given as

$$f(d,i) = \frac{2(\pi\lambda_{LOS}(d))^i}{(i-1)!} d^{2i-1} e^{-\pi\lambda_{LOS}(d)d^2}, i \in \mathcal{I}, \tag{A.1}$$

where LOS MTCD intensity $\lambda_{LOS}(d)$ within distance d is

$$\lambda_{LOS}(d) = \int_0^d 2\pi d\lambda_m P_L(d) \mathrm{d}d. \tag{A.2}$$

Thus, the expectation of $d_{LOS}(i)$ according to probability theory gives

$$\mathrm{E}[d_{LOS}(i)] = \int_0^\infty df(d,i) \mathrm{d}d, \tag{A.3}$$

and by substituting (A.1) and (A.2) into (A.3), the expected distance of i_{th} nearest LOS MTCD to BS is

$$\mathrm{E}[d_{LOS}(i)] = \int_0^\infty \frac{2(\pi\lambda_{LOS}(d))^i}{(i-1)!} d^{2i} e^{-\pi\lambda_{LOS}(d)d^2} \mathrm{d}d, i \in \{I\}, \tag{A.4}$$

where $\lambda_{LOS}(d)$ is the LOS MTCD intensity within distance d

$$\lambda_{LOS}(d) = 2\pi\lambda_m q \left(\left(-\frac{d}{\beta} - \frac{1}{\beta^2} \right) e^{-\beta d} + \frac{1}{\beta^2} \right). \tag{A.5}$$

Appendix B

Proof of chapter 4 - Theorem 1

Let m be the number of first tier relays per each BS's coverage, and $\mathcal{M}$ be the set of first tier relays. The goal for association between first tier MTCDs and a BS is to minimize the average time to upload all generated data, in which the link delay times data arrival rate per link is estimated from (4.18):

$$\min_{f_{i,b}} \frac{\sum_{i=1}^{\mathcal{M}}(s/(c_{i,b}-f_{i,b}))}{m}, \quad i \in \mathcal{I} \tag{B.1}$$

$$s.t. \sum_{i=1}^{m} f(i,b) = \Gamma, \tag{B.2}$$

where $f(i,b)$ consists of the traffic generated by itself and the packets received from assisted second tier MTCDs. Using a lagrange multiplier for the above equation, we get

$$\begin{cases} \frac{c_{i,b}}{m(c_{i,b}-f_{i,b})^2} = \lambda, \forall i \in \mathcal{M} \\ \sum_{i=1}^{m} \frac{c_i\sqrt{\lambda m}-\sqrt{c_i}}{\sqrt{\lambda m}} = \Gamma, \end{cases} \tag{B.3}$$

where the lagrange multiplier λ can be expressed as,

$$\lambda = \left(\frac{\sum_{i=1}^{m}\sqrt{c_{i,b}}}{\sqrt{m}(\sum_{i=1}^{m} c_{i,b} - \Gamma)} \right)^2, \quad i \in \mathcal{M}. \tag{B.4}$$

Since expected delay increases with λ, optimal m is derived to minimise λ,

$$m = \arg\min_{m} \left(\frac{\sum_{i=1}^{m} \sqrt{c_{i,b}}}{\sqrt{m}(\sum_{i=1}^{m} c_{i,b} - \Gamma)} \right)^2, \quad m = 1, 2 \dots u. \tag{B.5}$$

Thus, the data arrival rate of MTCD i is

$$f_{i,b} = \frac{\sqrt{\lambda m} c_{i,b} - \sqrt{c_{i,b}}}{\sqrt{\lambda m}}, \quad i \in \mathcal{M}. \tag{B.6}$$

Then, using link delay equation defined in (4.18), the average delay of the i_{th} LOS MTCD relay can be easily computed.

Appendix C

Proof of chapter 4 - Theorem 4

$$
\begin{aligned}
& p(D_{T1} + D_{T2} > \tau) \\
& = 1 - p(D_{T1} + D_{T2} < \tau) \\
& = 1 - \int_0^{\infty} \int_0^{\tau - D_{T1}} f_{T1}(D_{T1}) f_{T2}(D_{T2}) \mathrm{d}D_{T2} \mathrm{d}D_{T1} \\
& = 1 - \int_0^{\infty} f_{T1}(D_{T1}) F_{T2}(\tau - D_{T1}) \mathrm{d}D_{T1}, \qquad \text{(C.1)}
\end{aligned}
$$

where f_{T1} and f_{T2} are PDF of first tier and second tier MTCD expected delay, respectively. $F_{T2}(D_{T2})$ is the CDF of second tier MTCD expected delay,

$$
\begin{aligned}
F_{T2}(D_{T2}) = & \int_0^{D_{T2}} f_{LOS,T2}(D_{T2}) \mathrm{d}D_{T2} \qquad \text{(C.2)} \\
& + \int_0^{D_{T2}} f_{NLOS,T2}(D_{T2}) \mathrm{d}D_{T2} \qquad \text{(C.3)}
\end{aligned}
$$

Here $f_{LOS,T2}$ and $f_{NLOS,T2}$ are the PDF of expected second tier MTCD transmission time to first tier relays, with LOS or NLOS links to first tier relays respectively.

$$
f_{T2,LOS}(D_{T2}) = 2\pi\lambda_{LOS,T2}\phi e^{-\pi\lambda_{LOS,T2}\phi^2} \qquad \text{(C.4)}
$$

$$
f_{T2,NLOS}(D_{T2}) = 2\pi\lambda_{NLOS,T2}\phi e^{-\pi\lambda_{NLOS,T2}\phi^2}, \qquad \text{(C.5)}
$$

where

$$\lambda_{LOS,T2} = \frac{(k-m)}{k}\lambda_m P_L(\phi) \tag{C.6}$$

$$\lambda_{NLOS,T2} = \frac{(k-m)}{k}\lambda_m (1 - P_L(\phi)) \tag{C.7}$$

ϕ denotes the distance at which D_{T2} can be satisfied. From combining (4.4), (4.5) and (4.10) with (4.18), ϕ can be easily derived as

$$\phi = 10^{\left(\left(\frac{P_{t_x} G_{m_t} G_{m_r}}{(2^{c_{T2}/B}-1)\sigma^2} - A_1 - A_3 \log_2(f_c)\right)/A_2\right)}, \tag{C.8}$$

where $c_{T2} = \frac{s}{D_{T2}} + \rho$.

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